Quantum Braid Dynamics

A Computational Process

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Abstract

Chapter 4: Operations (Dynamics) formalizes the quantum engine of the Quantum Braid Dynamics (QBD) framework, transitioning the pre-geometric substrate from static structural architectures to a dynamic evolutionary regime. It addresses the pathologies of coordinate-dependent graph mutations, historical overwrites, and topological stasis within the bipartite vacuum tree. To resolve these issues, the chapter constructs a rigorous categorical syntax comprising the internal causal category $\mathbf{Caus}_t$ and the global historical category **Hist**, enforcing an append-only, history-preserving structure. Introspection is established via a store comonad (R_T, ϵ, δ) on annotated graphs (**AnnCG**), providing fault-tolerant, self-diagnostic capability that tracks Pauli frame shifts. The dynamics are anchored to canonical analytical reference priors from discrete combinatorial conservation principles, evaluating the critical vacuum temperature $T_c = \ln 2$ from bit-nat equivalence and the geometric self-energy $\varepsilon_{\text{geo}} = \frac{\ln 2}{3}$ via dimensional equipartition across the trivalent Bethe substrate. These reference scales govern the Universal Constructor ($\mathcal{R}$), which employs an adaptive feedback loop of catalysis ($\lambda_0 = e - 1$) and friction ($\mu_0 = 1/\sqrt{2\pi}$). Finally, the non-unitary Evolution Operator ($\mathcal{U}$) synthesizes these elements into a single logical tick, generating a positive-definite, action-bounded Euclidean transition measure that enforces a global, strict partial order and an irreversible thermodynamic arrow of time.

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Chapter 4: Operations (Dynamics)

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We stand before the static architecture of the vacuum we assembled in the previous chapter: a perfect, finite, rooted tree that contains the potential for geometry but lacks the motive force to realize it. Our inquiry now shifts from the structural "what" to the dynamical "how." We must determine what mechanism turns the first tick of the universal clock into an unstoppable cascade of increasing complexity. We dive into the quantum engine of our model, establishing a categorical syntax for histories and paths. We view the evolution of the universe as a continuous morphism in a category where every step preserves the causal structure of the past while opening new degrees of freedom for the future.

But a sequence of states is not enough: the system must possess a form of internal logic that allows it to assess its own configuration before acting. We introduce the concept of "awareness" not as a metaphysical quality, but as a comonadic self-check. This mathematical structure allows the graph to query its local topology, identifying valid sites for expansion without requiring a global observer. We couple this logical rigor with the thermodynamic reality of information processing. We derive the fundamental scales of our system, such as the critical temperature $T = \ln 2$, by equating the energetic cost of a decision with the informational content of a bit. This ensures that the engine does not run on magic, it pays for every bit of order it creates with a commensurate amount of entropic heat.

Finally, we unify these elements into the Universal Operator $\mathcal{U}$. This operator acts as the heartbeat of the cosmos, cycling through a precise sequence of awareness, action, correction, and collapse. It employs a constructor to propose specific topological changes (adds and cuts) based on the rewrite rule, and then samples the next state from the resulting probability distribution. The core puzzle we solve here is how these purely local flips, biased only by thermal noise and friction, propel the whole system toward coherent geometry without stalling or looping back into chaos. This machinery spins the relational wheel, where each step leaks just enough information to entropy to point the arrow of time strictly forward.

Preconditions and Goals

- Validate that history and path categories encode influences as monotone morphism subsets.
- Prove the self-observation comonad holds functorial preservation and naturality axioms.
- Derive temperature and coefficients from bit-nat alignment for balanced transition rates.
- Implement the rewrite as a distribution generator with strict validation and weighting.
- Confirm the operator is irreversible through projection and sampling entropy increase.

4.1 Categorical Foundations: Definitions and Motivations

We confront the foundational necessity of establishing a mathematical syntax capable of describing the growth of causal graphs without relying on the crutch of a pre-existing coordinate system to index the changes. The inquiry demands a categorical framework that can distinguish between the instantaneous potential of a causal path within a single moment and the immutable record of historical events that defines the flow of time. We are compelled to deduce a set of categories that encode the relational structure of the universe as it builds itself and effectively distinguish between the ephemeral possibility of connection and the permanent reality of causation.

Standard approaches to graph dynamics often fail because they lack the structural rigidity to prevent the corruption of the past by the operations of the present. A mathematical model based on unstructured graph updates risks describing a chaotic flux where history remains mutable and subject to reinterpretation by future events and effectively destroys the concept of a coherent timeline by allowing the present to overwrite the past. Without a strict formalism to enforce the monotonicity of causal relations the theory would permit retrograde modifications where the future rewrites the antecedents and violates the basic requirements of causality upon which physical law depends. Furthermore a dynamical system lacking defined morphism classes cannot track the conservation of information or ensure that the evolution remains unitary across the transition from one state to the next and leaves us with a model where energy and information can leak out of existence without accounting.

We resolve this foundational crisis by formalizing two complementary categories known as the internal causal category $\mathbf{Caus}_t$ and the global historical category **Hist**. By defining morphisms as directed paths within a snapshot and history-respecting embeddings across time we create a grammar where every new state contains the past as a permanent subgraph. This structure embeds the arrow of time into the very definition of the category and ensures that the universe evolves as a cumulative process where new states are strict supersets of the old and locks the past irrevocably in place.

4.1.1 Definition: Internal Causal Category

Structure of Vertices via Directed Path Morphisms within a Single Snapshot

The **Internal Causal Category**, denoted $\mathbf{Caus}_t$, is defined as the mathematical structure encapsulating the instantaneous causal relationships within a graph snapshot at Logical Time t. The category comprises the following components: 1. **Objects:** The set of objects $\mathrm{Ob}(\mathbf{Caus}_t)$ is strictly identical to the vertex set V of the causal graph G_t. 2. **Morphisms:** For any ordered pair of objects (u, v), the set of morphisms $\mathrm{Hom}(u, v)$ consists of all **Directed Path** originating at u and terminating at v. This set includes the **Trivial Path** of length $\ell = 0$. 3. **Composition:** The composition operation $\circ : \mathrm{Hom}(v, w) \times \mathrm{Hom}(u, v) \to \mathrm{Hom}(u, w)$ is defined as the concatenation of path sequences. For morphisms $p = (u, \dots, v)$ and $q = (v, \dots, w)$, the composition $q \circ p$ yields the sequence $(u, \dots, v, \dots, w)$. 4. **Identity:** For each object u, the identity morphism id_u is defined as the Trivial Path containing the single vertex sequence (u). **(Awodey, 2010)**

4.1.1.1 Commentary: Physical Interpretation of $\mathbf{Caus}_t$

Modeling of Instantaneous Causal Pathways as Potential Influence Channels

To understand the internal structure of a single moment in time, we must first rigorize the concept of "reachability" within a discrete snapshot. The category $\mathbf{Caus}_t$ serves as the formal apparatus for this task, transforming the raw graph data into an algebraic structure governed by composition. This formalization leverages the standard framework of path categories described by (Awodey, 2010), allowing us to treat causal reachability as a composable morphism that obeys rigorous associative laws. Each object in this category corresponds to a vertex in the graph G_t, which physically represents a discrete event or a relational nexus within the vacuum fabric.

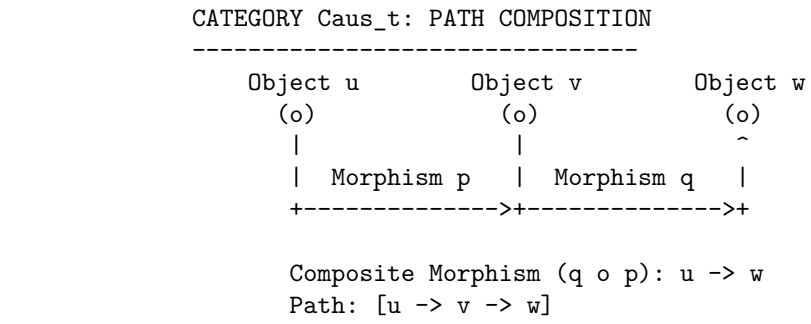
The morphisms of this category are the directed paths. A morphism $f : u \to v$ does not merely assert that u and v are connected, it represents a specific **causal lineage** or trajectory of influence. This includes the

trivial path of length $\ell = 0$ (the identity morphism id_u), which physically encodes the persistence of an event's self-identity or its causal potential before interaction. The composition operation $g \circ f$ corresponds to the transitivity of causality, if u influences v via path f, and v influences w via path g, then u necessarily exerts a mediated influence on w. This algebraic closure ensures that causal influence is not just a local phenomenon between neighbors, but a global property that propagates through the network.

Crucially, this category acts as the "kinematic phase space" for the universe at a frozen instant t. It maps the web of *potential* causality before the dynamical constraints of Axiom 3 filter them into *effective* influence. For example, in the vacuum state derived in Chapter 3, the tree-like structure implies that $\mathbf{Caus}_t$ is populated exclusively by unique morphisms between connected nodes, devoid of the loops or redundant parallel paths that would characterize a dense manifold. The transition from this sparse categorical skeleton to a rich geometry occurs when the rewrite rule inserts new morphisms (edges) that create cycles, fundamentally altering the algebraic structure of the category from a poset-like hierarchy to a complex relational web.

4.1.1.2 Diagram: Path Composition

Illustrative Example of Path Concatenation via Morphism Composition



4.1.2 Definition: Historical Category

Structure as Cumulative Trajectories utilizing History-Preserving Embeddings

Let the **Historical Category Hist** have objects given by cumulative creation-record sets $\mathcal{E}_t$ and morphisms given by their canonical inclusions. A record is $\omega = (u, v, b, h)$ with source, target, scheduler birth tick and causal height (**Creation Timestamp**). Write

$$\mathcal{E}_{t+1} = \mathcal{E}_t \cup A^{\mathrm{occ}}_{t+1}, \qquad b(\omega) = \omega_3, \quad h(\omega) = \omega_4.$$

The initial records have $b = h = 0$. New records have birth tick $t + 1$ and height assigned from the pre-tick snapshot. For any fixed pair and birth tick the scheduler creates at most one record. The active-occurrence map $\alpha_t : E_t \to \mathcal{E}_t$ identifies which past creation is currently active. The cumulative spatial support is the projection

$$\overline{E}_t = \{(u, v) : \exists b, h,\ (u, v, b, h) \in \mathcal{E}_t\}, \qquad \overline{G}_t = (V, \overline{E}_t).$$

The unlabeled support $\overline{G}_t$ is the cumulative spatial shadow of the creation history. It is not itself the occurrence dependency poset, and it does not identify different births of the same pair. A first-birth label on $\overline{E}_t$, if used, is not the current active height. Deletion records and the full state trajectory are additional data when required; the creation record alone is not defined to reconstruct every intermediate state.

An inclusion $\iota_{t,s} : \mathcal{E}_t \hookrightarrow \mathcal{E}_s$, $t \le s$, has $\iota_{t,s}(\omega) = \omega$. Identities and composition are ordinary inclusion identities and composition, preserving all four coordinates exactly.

4.1.2.1 Commentary: Physical Interpretation of Hist

Accumulation of Irreversible History via Meta-Theoretical Trajectories

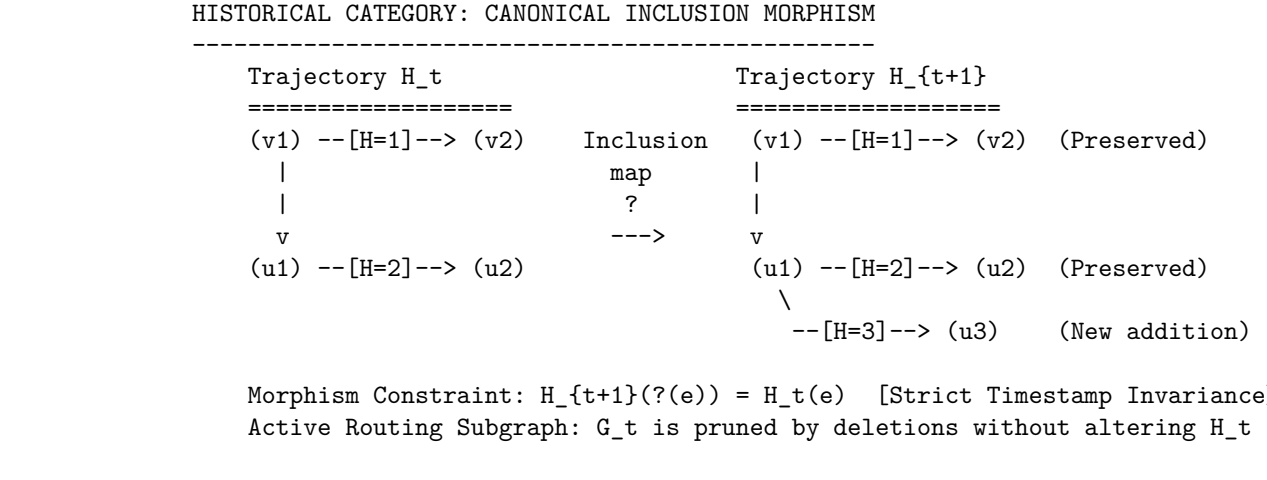
While $\mathbf{Caus}_t$ describes the internal structure of the "Now", the category **Hist** describes the "Timeline." This is the global, meta-theoretical container for cosmic evolution. Crucially, the objects in this category are not the fluctuating, Markovian instantaneous states G_t (which the Universal Constructor actively prunes to regulate spatial density), but the cumulative trajectories $\mathcal{H}_t$. The active spatial state G_t acts as a dynamical routing subgraph embedded inside $\mathcal{H}_t$.

The morphisms in **Hist** are canonical inclusion maps. The structure of the **Historical Category** is physically profound; it asserts that time evolution is strictly cumulative. A morphism $\mathcal{H}_t \hookrightarrow \mathcal{H}_{t+k}$ maps the history of the universe at time t into the history at time $t+k$ in a manner that strictly preserves the past. It forbids the erasure of historical events (injectivity) and the scrambling of causal order (strict equality of H). If an edge existed at time t with timestamp $H(e)$, its image exists in the trajectory $\mathcal{H}_{t+k}$ with the identical immutable timestamp $H_{t+k}(\iota(e)) = H_t(e)$. This constraint creates a "Block Universe" that is built dynamically layer by layer.

This formulation acts as a rigorous safeguard against retrocausality. Because every valid evolution must be a morphism in **Hist**, it is mathematically impossible for the system to "rewrite" an antecedent timestamp or alter the connectivity of a prior epoch. The arrow of time is thus encoded structurally into the **Historical Category** itself. The physical universe discards edges in the active spatial manifold G_t to prevent the Small-World Catastrophe, but the mathematical trajectory $\mathcal{H}_t$ retains the permanent "scar" of every interaction, ensuring the causal pedigree of the cosmos remains invariant.

4.1.2.2 Diagram: Morphism Preservation

Visual Representation of Structure as History Preservation Constraints in Graph Morphisms



4.1.3 Theorem: Orthogonality of Kinematic and Historical State

Preservation of Historical Causal Poset via Active Kinematic Edge Excision

Let the active spatial state use the occurrence map $\alpha_t : E_t \to \mathcal{E}_t$ and let creation records accumulate by union (**Historical Category**). Then the historical record is preserved by active edge deletion, including every prior creation record and its birth tick and height, while the cumulative spatial support remains monotonically inclusive.

4.1.3.1 Commentary: Argument Outline

Structure of the Orthogonality of Kinematic and Historical State Argument via Trajectory Monotonicity, Timestamp Preservation, and Categorical Domain Decoupling

The proof proceeds via direct construction, verifying that topological modifications executed on active state space decouple completely from cumulative trajectory space, supported by the following lemmas:

```
o 4.1.3 Theorem Orthogonality of Kinematic and Historical State  [by construction]
|
+-- 4.1.4 Lemma: Trajectory Monotonicity under Edge Excision
|   +-- 4.1.4.1 Proof: Trajectory Monotonicity under Edge Excision
|   +-- 4.1.4.2 Commentary: Scar of Deletion
|
+-- 4.1.5 Lemma: Timestamp Preservation under Kinematic Excision
|   +-- 4.1.5.1 Proof: Timestamp Preservation under Kinematic Excision
|   +-- 4.1.5.2 Commentary: Monotonic Directionality of Time
|
+-- 4.1.6 Lemma: Categorical Domain Decoupling
|   +-- 4.1.6.1 Proof: Categorical Domain Decoupling
|   +-- 4.1.6.2 Commentary: Categorical Ties to Prior Foundations
|
+-- 4.1.7 Proof: Orthogonality of Kinematic and Historical State
|
+-- 4.1.8 Validation: Lean 4 Core
```

4.1.4 Lemma: Trajectory Monotonicity under Edge Excision

Preservation of Excised Kinematic Edges via Cumulative Trajectories

Let $\mathcal{E}_{t+1} = \mathcal{E}_t \cup A^{\text{occ}}_{t+1}$ and let active deletions remove pairs only from E_t. Then the historical inclusion is well-defined: $\mathcal{E}_t \subseteq \mathcal{E}_{t+1}$ and $\overline{E}_t \subseteq \overline{E}_{t+1}$, and an excised active occurrence remains in the creation history (**Historical Category**).

4.1.4.1 Proof: Trajectory Monotonicity under Edge Excision

Verification of Trajectory Inclusion via Set-Theoretic Monotonicity

I. Monotonicity of the Occurrence Set

Let A^{occ}_{t+1} be the occurrences created at tick $t+1$ (**Historical Category**) (**Trajectory Monotonicity under Edge Excision**). Each $\omega \in \mathcal{E}_t$ satisfies

$$\omega \in \mathcal{E}_t \implies \omega \in \mathcal{E}_t \cup A^{\text{occ}}_{t+1} = \mathcal{E}_{t+1}.$$

This proves $\mathcal{E}_t \subseteq \mathcal{E}_{t+1}$.

II. Projection to Spatial Support

With $\pi(u, v, b, h) = (u, v)$, the historical spatial support is the image $\overline{E}_t = \pi(\mathcal{E}_t)$. The image of the union is

$$\begin{aligned}\overline{E}_{t+1} &= \pi(\mathcal{E}_{t+1}) = \pi(\mathcal{E}_t \cup A^{\text{occ}}_{t+1}) \\ &= \pi(\mathcal{E}_t) \cup \pi(A^{\text{occ}}_{t+1}) = \overline{E}_t \cup \pi(A^{\text{occ}}_{t+1}).\end{aligned}$$

Therefore $\overline{E}_t \subseteq \overline{E}_{t+1}$.

III. Membership after Active Excision

Suppose $e \in E_t$ is excised at tick $t+1$. Its occurrence $\omega = \alpha_t(e)$ satisfies $\pi(\omega) = e$. When the edge is absent from the active output, the two membership statements are

$$\begin{aligned} & e \notin E_{t+1}, \\ \alpha_t(e) \in \mathcal{E}_t \subseteq \mathcal{E}_{t+1} \quad & \Longrightarrow \quad e = \pi(\alpha_t(e)) \in \overline{E}_{t+1}. \end{aligned}$$

If the edge is also assigned a new occurrence at a later tick $s > t$, then A_s^{occ} carries a distinct creation tick and $\mathcal{E}_s$ contains both records by repeated occurrence-set inclusion.

Q.E.D.

4.1.4.2 Commentary: Scar of Deletion

Ontological Decoupling of Causal History from Kinematic Geometry in Quantum Braid Dynamics

The conceptual boundary between the active spatial manifold and the historical category resolves the apparent paradox of a universe that must simultaneously remember its past to preserve causality and prune its edges to regulate geometric density. If the rules of physics forced the active runtime state to physically carry every spatial edge it ever created, the vacuum would rapidly collapse into a maximally connected singularity.

By defining **Hist** over the cumulative trajectory $\mathcal{H}_t$ rather than the instantaneous state G_t, we allow the active spatial manifold to "breathe": edges can be added to build structure and deleted to relieve stress. The "scar" of a deleted edge is not a bloated data structure that the universe drags along in its active memory; it is a permanent, indelible feature of the mathematical trajectory $\mathcal{H}_t$. In physical terms, if particle A interacted with particle B, that interaction is permanently etched into the global block universe, even if the spatial distance between them subsequently expands and the direct geometric link in the "Now" is severed.

4.1.5 Lemma: Timestamp Preservation under Kinematic Excision

Invariance of Historical Edge Timestamps via Canonical Inclusion

Let $\iota_{t,s} : \mathcal{E}_t \hookrightarrow \mathcal{E}_s$ be a canonical historical inclusion (**Historical Category**). Then $b(\iota_{t,s}(\omega)) = b(\omega)$ and $h(\iota_{t,s}(\omega)) = h(\omega)$ for every $\omega \in \mathcal{E}_t$.

4.1.5.1 Proof: Timestamp Preservation under Kinematic Excision

Verification of Timestamp Monotonicity via Embedding Transitions

I. Coordinate Form of the Inclusion

The proof uses the specified objects and conventions (**Historical Category**).

Let $\omega = (u, v, b, h) \in \mathcal{E}_t$. The canonical inclusion changes its containing set, not its value:

$$\iota_{t,s}(u, v, b, h) = (u, v, b, h).$$

II. Birth and Height Projections

Applying the third and fourth projections separately gives

$$\begin{aligned} b(\iota_{t,s}(\omega)) &= b(u, v, b, h) = b = b(\omega), \\ h(\iota_{t,s}(\omega)) &= h(u, v, b, h) = h = h(\omega). \end{aligned}$$

III. Distinct Re-Creations

For records (u, v, b_1, h_1) and (u, v, b_2, h_2) with $b_1 \neq b_2$, equality of records would imply equality of their third projections, contradicting $b_1 \neq b_2$. Thus the two births remain distinct even when $h_1 = h_2$. Each record retains its own projections (**Timestamp Preservation under Kinematic Excision**).

Q.E.D.

4.1.5.2 Commentary: Monotonic Directionality of Time

Structural Prevention of Retrocausal Modifications

The preservation of timestamp monotonicity under canonical inclusion guarantees that physical history cannot be overwritten, compressed, or run in reverse. While the active kinematic graph G_t exhibits non-monotonic cardinality fluctuations by expanding through topological edge creation and contracting through edge deletion, the categorical container **Hist** accumulates timestamps monotonically. This mathematical decoupling prevents retroactive distortions of prior causal records, establishing an invariant temporal orientation across the evolving relational network.

This structural separation provides the concrete physical foundation for the arrow of time in quantum graph dynamics. An observer embedded inside the graph perceives the dissolution of an edge as a local alteration in spatial connectivity, but cannot erase the physical event of prior interaction. Because every valid trajectory update constitutes a morphism in **Hist**, the architecture strictly forbids retrocausal loops: once an interaction acquires an invariant logical timestamp t, no operator can assign that interaction an antecedent timestamp $t' < t$.

4.1.6 Lemma: Categorical Domain Decoupling

Orthogonality of Causal Reachability via Poset Invariance

Let active-path morphisms use G_t, cumulative support use $\overline{G}_t$, and occurrence chronology use the dependency closure D^+ (**Monotonicity of History**). Then the historical dependency relation is unchanged by active deletion: active paths can disappear while creation records and their previously recorded dependencies remain.

4.1.6.1 Proof: Categorical Domain Decoupling

Verification of Categorical Domain Decoupling via Slicing

I. Active Path Membership

For a deletion-only change $E' = E \setminus F$, any path using an edge of F is absent from the new active path set. Every path using only E' was already a path in E, giving

$$\mathcal{P}_{E'}(u, v) \subseteq \mathcal{P}_{E}(u, v).$$

II. Retained Historical Objects

The same operation does not remove elements of $\mathcal{E}_t$ (**Trajectory Monotonicity under Edge Excision**). The coordinate values of each retained record remain equal (**Timestamp Preservation under Kinematic Excision**). The recorded predecessor relation D is fixed at creation, so excision from the active graph changes no historical pair $(\omega, \eta) \in D$.

III. Closure Invariance Under Deletion Alone

A witness for $\omega D^+ \eta$ consists of retained records and recorded dependency pairs. Neither is removed. Conversely, deletion alone introduces no new records or dependency pairs. Thus the set of such witnesses is unchanged, and the occurrence order is unchanged. This is the claimed domain decoupling (**Categorical Domain Decoupling**).

Q.E.D.

4.1.6.2 Commentary: Categorical Ties to Prior Foundations

Integration of Ontological and Axiomatic Constraints via Categorical Syntax

These two categories, $\mathbf{Caus}_t$ and **Hist**, function as the syntactic glue that binds the ontological substrate of Chapter 1 to the architectural realizations of Chapter 3. They operationalize the abstract constraints of the theory into calculable algebraic structures, bridging discrete topological events with global temporal evolution across the network.

Consider the **Regular Bethe Fragment** derived as the initial vacuum state G_0. In the language of $\mathbf{Caus}_t$, this object is a category where the morphism sets $\mathrm{Hom}(u, v)$ contain at most one element (due to tree sparsity), and there are no morphisms $f : u \to u$ other than identity (due to acyclicity). This algebraic simplicity is precisely what defines the "cold" vacuum. The **Ignition** event (tunneling) described in Section 3.4 can now be defined as a functorial transition that introduces the first non-trivial morphisms (cycles) into $\mathbf{Caus}_t$, breaking the algebraic rigidity of the tree.

Furthermore, the axioms of Chapter 2 act as filters on these categories. **Axiom 1** (Causal Primitive) ensures that the atomic morphisms in $\mathbf{Caus}_t$ are directed. **Axiom 3** (Acyclic Effective Causality) ensures that the composition of these morphisms never yields an identity morphism other than the trivial one (i.e., no $f \circ g = \mathrm{id}$ for non-trivial f, g), thereby preventing closed causal loops. In **Hist**, the preservation of timestamps enforces the monotonicity required by the thermodynamic arguments of Chapter 5. Thus, these categorical definitions are not merely descriptive, they are the enforcement mechanisms that prevent the dynamical engine from producing physical nonsense. They provide the "rails" upon which the Universal Constructor must run, ensuring that however violent the geometric phase transition becomes, the logical consistency of the universe remains inviolate.

4.1.7 Proof: Orthogonality of Kinematic and Historical State

Synthesis of Monotonicity via Timestamp Invariance and Domain Decoupling

I. Record and Support Monotonicity

Record union gives $\mathcal{E}_t \subseteq \mathcal{E}_{t+1}$, and projection gives $\overline{E}_t \subseteq \overline{E}_{t+1}$ (**Trajectory Monotonicity under Edge Excision**). A deleted active pair still has a retained creation occurrence.

II. Preserved Labels

For every retained record, canonical inclusion is the identity on its four coordinates. Hence

$$b(\iota(\omega)) = b(\omega), \qquad h(\iota(\omega)) = h(\omega)$$

(**Timestamp Preservation under Kinematic Excision**).

III. Separate Path Domains

Active deletion can remove spatial paths while leaving both the historical records and their recorded dependencies intact (**Categorical Domain Decoupling**). These are exactly the record-preservation and support-inclusion conclusions (**Orthogonality of Kinematic and Historical State**).

Q.E.D.

4.1.8 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Cumulative Trajectories and Deletion Indelibility

Type-theoretic certification of the support and creation-record preservation established in (**Orthogonality of Kinematic and Historical State**) and (**Timestamp Preservation under Kinematic Excision**) proceeds via the following verification strategy:

1. **Cumulative Support:** Edge predicates and historical inclusion encode the monotone cumulative spatial support and deletion indelibility.
2. **Occurrence Coordinates:** Creation records carry the spatial pair, birth tick and immutable causal height, with separate projection to cumulative spatial support.
3. **Preservation:** The proof terms certify record inclusion, support projection, both temporal-coordinate equalities, distinct births and the elementary next-tick height bound.

```
def Edge (V : Type) := V x V

def GraphEdges (V : Type) := Edge V -> Prop

def HistoricalInclusion {V : Type} (H1 H2 : GraphEdges V) : Prop :=
  ? e, H1 e -> H2 e

/--
THEOREM 1: Category Hist Identity Inclusion
Proves that every cumulative trajectory graph admits an identity inclusion morphism.
-/
theorem historical_inclusion_id {V : Type} (H : GraphEdges V) :
    HistoricalInclusion H H := by
  intro e he
  exact he

/--
THEOREM 2: Category Hist Morphism Composition
Proves that historical inclusions compose transitively.
-/
theorem historical_inclusion_trans {V : Type} (H1 H2 H3 : GraphEdges V) :
    HistoricalInclusion H1 H2 -> HistoricalInclusion H2 H3 -> HistoricalInclusion H1 H3 := by
  intro h12 h23 e he
  exact h23 e (h12 e he)

/--
THEOREM 3: Cumulative History Step Monotonicity
Proves that cumulative history is strictly monotonic under constructor evolution: H_t ? H_{t+1}.
-/
theorem cumulative_history_step_monotonicity {V : Type}
    (H_seq A_seq : Nat -> GraphEdges V)
    (t : Nat)
    (h_step : ? t e, H_seq (t + 1) e <-> H_seq t e ? A_seq t e) :
    HistoricalInclusion (H_seq t) (H_seq (t + 1)) := by
  intro e he
  rw [h_step t e]
  exact Or.inl he

/--
Helper Lemma: Inductive Multi-Step Expansion of Cumulative History
-/
```

```
theorem cumulative_history_add_monotonicity {V : Type}
    (H_seq A_seq : Nat -> GraphEdges V)
    (h_step : ? t e, H_seq (t + 1) e <-> H_seq t e ? A_seq t e)
    (t1 : Nat) (k : Nat) :
    HistoricalInclusion (H_seq t1) (H_seq (t1 + k)) := by
  intro e he
  induction k with
  | zero => exact he
  | succ n ih =>
    have h_one_step := cumulative_history_step_monotonicity H_seq A_seq (t1 + n) h_step
    exact h_one_step e ih

/--
THEOREM 4: Cumulative History Transitive Monotonicity
Proves that across arbitrary intervals t1 <= t2, cumulative history expands monotonically: H_{t1} ? H_{
-/
theorem cumulative_history_transitive_monotonicity {V : Type}
    (H_seq A_seq : Nat -> GraphEdges V)
    (h_step : ? t e, H_seq (t + 1) e <-> H_seq t e ? A_seq t e)
    (t1 t2 : Nat) (h_le : t1 <= t2) :
    HistoricalInclusion (H_seq t1) (H_seq t2) := by
  rcases Nat.le.dest h_le with <k, rfl>
  exact cumulative_history_add_monotonicity H_seq A_seq h_step t1 k

/--
THEOREM 5: Spatial State is Inductively a Subgraph of Cumulative History
Proves that at all logical ticks t, the active kinematic spatial graph G_t is an exact subgraph of H_t:
-/
theorem spatial_subgraph_of_cumulative_history {V : Type}
    (E_seq H_seq A_seq D_seq : Nat -> GraphEdges V)
    (h_init : ? e, H_seq 0 e <-> E_seq 0 e)
    (h_hist : ? t e, H_seq (t + 1) e <-> H_seq t e ? A_seq t e)
    (h_dyn : ? t e, E_seq (t + 1) e <-> (E_seq t e ? A_seq t e) ? ? D_seq t e) :
    ? t e, E_seq t e -> H_seq t e := by
  intro t
  induction t with
  | zero =>
    intro e he
    exact (h_init e).mpr he
  | succ n ih =>
    intro e he
    rw [h_dyn n e] at he
    rw [h_hist n e]
    cases he.1 with
    | inl he_prev => exact Or.inl (ih e he_prev)
    | inr he_add => exact Or.inr he_add

/--
THEOREM 6: Dynamic Deletion Preserves Cumulative History
Proves that when an edge e is excised from the active kinematic state by deletion D_t,
e remains indelibly preserved in the cumulative historical record H_{t+1}.
-/
theorem deletion_preserves_cumulative_history {V : Type}
    (E_seq H_seq A_seq D_seq : Nat -> GraphEdges V)
```

```
    (h_init : ? e, H_seq 0 e <-> E_seq 0 e)
    (h_hist : ? t e, H_seq (t + 1) e <-> H_seq t e ? A_seq t e)
    (h_dyn : ? t e, E_seq (t + 1) e <-> (E_seq t e ? A_seq t e) ? ? D_seq t e)
    (t : Nat) (e : Edge V)
    (he_active : E_seq t e)
    (he_deleted : D_seq t e) :
    ? E_seq (t + 1) e ? H_seq (t + 1) e := by
  constructor
  - rw [h_dyn t e]
    intro <_, h_not_D>
    exact h_not_D he_deleted
  - rw [h_hist t e]
    have h_in_H := spatial_subgraph_of_cumulative_history E_seq H_seq A_seq D_seq h_init h_hist h_dyn t
    exact Or.inl h_in_H

namespace QBD.CreationHistory

-- A creation occurrence stores both temporal coordinates explicitly.
structure Occurrence (V : Type) where
  source : V
  target : V
  born : Nat
  height : Nat

-- Histories contain creation records, not only spatial endpoint pairs.
abbrev History (V : Type) := Occurrence V -> Prop

-- A canonical inclusion fixes each record value.
def Included {V : Type} (A B : History V) : Prop := ? w, A w -> B w

-- The cumulative spatial support forgets both temporal coordinates.
def support {V : Type} (A : History V) (e : V x V) : Prop :=
  ? w, A w ? w.source = e.1 ? w.target = e.2

/-- Appending creation records preserves every old record. -/
theorem append_includes {V : Type} (A New : History V) :
    Included A (fun w => A w ? New w) := by
  intro w hw
  exact Or.inl hw

/-- Record inclusion induces inclusion of projected spatial supports. -/
theorem support_inclusion {V : Type} (A B : History V) (h : Included A B)
    (e : V x V) : support A e -> support B e := by
  intro <w, hw, hs, ht>
  exact <w, h w hw, hs, ht>

/-- Canonical inclusion preserves scheduler birth tick. -/
theorem birth_preserved {V : Type} (w : Occurrence V) : (id w).born = w.born := by rfl

/-- Canonical inclusion preserves immutable causal height. -/
theorem height_preserved {V : Type} (w : Occurrence V) : (id w).height = w.height := by rfl

/-- Different scheduler births identify different creation occurrences. -/
theorem different_birth_distinct {V : Type} (w z : Occurrence V)
```

```
    (h : w.born != z.born) : w != z := by
  intro heq
  exact h (congrArg Occurrence.born heq)

/-- The recurrence increment respects the next scheduler tick bound. -/
theorem next_height_bound (parentMax tick : Nat) (h : parentMax <= tick) :
    parentMax + 1 <= tick + 1 := by
  exact Nat.succ_le_succ h

end QBD.CreationHistory

#print axioms QBD.CreationHistory.support_inclusion
#print axioms QBD.CreationHistory.different_birth_distinct
```

Verification Summary:

The support-level theorems certify monotone inclusion and persistence of excised spatial pairs. The creation-record declarations additionally distinguish scheduler birth from causal height and retain both under canonical inclusion. The support projection theorem links the two representations without identifying their data. The height increment theorem is conditional on a bound for the source maximum, whose correspondence to the snapshot recurrence is given analytically. Lean kernel acceptance certifies these stated preservation identities for the corresponding proof; the module contains no admitted proofs or additional axioms.

4.1.Z Implications and Synthesis

Categorical Foundations

The verification that the internal and historical structures function as categories shows that they satisfy the identity and associativity axioms through trivial paths and monotonic embeddings. This formalizes the **Internal Causal Category** and historical trajectories, providing a syntactic foundation where the history of the universe manifests as a monotonically growing chain of states, expanding forward without the possibility of reversal or compression. The algebraic structure ensures that every new state extends the prior one, appending new edges and timestamps to the existing record in a manner that locks the past irrevocably in place.

This implies that the dynamical process itself is a directed sequence of morphisms within the historical category, preserving the **Orthogonality of Kinematic and Historical State** . Each arrow connects one state to the next while inheriting the full temporal constraints, preventing retrocausal loops or undefined transitions. However, extracting the internal causal influences requires a compatible slicing mechanism to restrict embeddings to local paths without introducing gaps.

The categorical syntax establishes a "block universe" that is built dynamically rather than existing eternally. By defining history as a cumulative sequence of embeddings, we ensure that the past is structurally conserved within the present, providing a robust mathematical basis for the arrow of time. This formalism prevents the "rewriting" of history, as valid morphisms must respect the established timestamp order, thereby encoding the irreversibility of physical events directly into the **Historical Category** of the state space.

4.2 Validity of the Categorical Syntax

The definition of a categorical framework creates an immediate verification problem as we must prove that these abstract structures satisfy the axioms of identity and associativity required for mathematical consistency. We are forced to demonstrate that the syntax we have constructed is robust enough to support physical dynamics without introducing logical contradictions or ambiguities that would undermine the stability of the theory. This verification demands that we treat the categories not just as descriptive labels

but as functional mathematical objects that must hold together under the weight of their own definitions to prevent the logical collapse of the model.

Assuming the validity of these categories without proof invites catastrophic logical errors where the composition of causal paths depends on the order of operations and creates a universe where the outcome of a physical process depends on the arbitrary segmentation of time. A syntax that fails the associativity test implies that the history of the universe is subjective and effectively destroys the objectivity of physical law by allowing different observers to disagree on the sequence of events. A category without valid identity morphisms implies a static universe is mathematically impossible and traps the theory in a paradox where existence requires constant or potentially unphysical change as the system would be mathematically incapable of remaining in a stable state. Such ambiguities would undermine the objectivity of the theory and render any subsequent derivation of thermodynamics or particle physics suspect as the ground beneath the theory would be shifting with every calculation.

We solve this verification problem by proving that the path concatenation operation in $\mathbf{Caus}_t$ and the embedding composition in **Hist** satisfy all categorical axioms. By demonstrating that "doing nothing" is a valid history and that the sequence of events is invariant under regrouping we ensure that the mathematical language of the theory is unambiguous. This validation provides the solid floor upon which the complex machinery of awareness and thermodynamics can be built and guarantees that the underlying logic of the universe is sound.

4.2.1 Theorem: Categorical Validity

Formal Consistency of the Categorical Frameworks for Global via Internal Structures

Consider the structures $\mathbf{Caus}_t$ and **Hist** representing the internal causal path structure and the global historical embedding structure, respectively. Then the following holds: both structures constitute valid mathematical categories satisfying the axioms of **Associativity** of composition and the existence of neutral **Identity** elements. Moreover, these frameworks provide the consistent syntactic domain for the dynamical operations of the Universal Constructor.

4.2.1.1 Commentary: Argument Outline

Structure of the Categorical Representation of Time Argument via Internal Verification, Historical Verification, and Causal Encoding

The proof proceeds via Direct Construction, verifying the algebraic requirements that establish the internal and global category representations of temporal evolution.

```
o 4.2.1 Theorem Categorical Validity  [by construction]
|
+-- 4.2.2 Lemma: Causal Category Identity
|   +-- 4.2.2.1 Proof: Causal Category Identity
|   +-- 4.2.2.2 Commentary: Causal Neutrality
|
+-- 4.2.3 Lemma: Causal Category Associativity
|   +-- 4.2.3.1 Proof: Causal Category Associativity
|   +-- 4.2.3.2 Commentary: Associative Flow
|
+-- 4.2.4 Lemma: Timestamp Monotonicity
|   +-- 4.2.4.1 Proof: Timestamp Monotonicity
|   +-- 4.2.4.2 Commentary: Causal Directionality
|
+-- 4.2.5 Lemma: History Category Identity
|   +-- 4.2.5.1 Proof: History Category Identity
|   +-- 4.2.5.2 Commentary: Historical Neutrality
```

```
|
+-- 4.2.6 Lemma: History Category Associativity
|   +-- 4.2.6.1 Proof: History Category Associativity
|   +-- 4.2.6.2 Commentary: Historical Consistency
|
+-- 4.2.7 Lemma: Topological Injectivity
|   +-- 4.2.7.1 Proof: Topological Injectivity
|   +-- 4.2.7.2 Commentary: Topological Injectivity
|
+-- 4.2.8 Lemma: Effective Influence Encoding
|   +-- 4.2.8.1 Proof: Effective Influence Encoding
|   +-- 4.2.8.2 Commentary: Information Preservation
|
+-- 4.2.9 Lemma: Partial Order Property
|   +-- 4.2.9.1 Proof: Partial Order Property
|   +-- 4.2.9.2 Commentary: Causal Ordering
|
+-- 4.2.10 Proof: Categorical Validity
|
+-- 4.2.11 Calculation: Partial Order Verification
|
+-- 4.2.12 Validation: Lean 4 Core
```

4.2.2 Lemma: Identity for $\mathbf{Caus}_t$

Neutrality of Trivial Paths in the Internal Causal Category

Let $p : u \to v$ be a morphism in $\mathbf{Caus}_t$. Then the composition with the Trivial Path in the **Internal Causal Category** satisfies the identity laws $p \circ \mathrm{id}_u = p$ and $\mathrm{id}_v \circ p = p$, where the concatenation of a sequence with a zero-length sequence yields the original sequence invariant.

4.2.2.1 Proof: Identity for $\mathbf{Caus}_t$

Verification of Neutrality under Composition for Trivial Paths

I. Anchored Edge Lists

In the path category, a morphism $p : u \to v$ is represented by its endpoints and finite edge list

$$p = (u, (e_1, \dots, e_k), v), \qquad k \geq 0$$

(**Internal Causal Category**). Its identity at x is the anchored empty list

$$1_x = (x, (), x), \qquad \ell(1_x) = 0.$$

If $k = 0$, the path is trivial and $u = v$; its anchor still specifies its object.

II. Identity After a Path

The composite $1_v \circ p$ is defined, because the target of p is the source of 1_v. Its endpoints are (u, v) and its length is $k + 0 = k$. At each position $1 \leq j \leq k$ its edge is

$$(S_p +\!\!+ ())_j = (S_p)_j = e_j,$$

where $+\!+$ denotes concatenation. There are no positions contributed by the empty list. Thus

$$1_v \circ p = (u, (e_1, \ldots, e_k), v) = p.$$

III. Identity Before a Path

The composite $p \circ 1_u$ is defined for the same endpoint reason. It has length $0 + k = k$, and for every $1 \leq j \leq k$,

$$((\,) +\!+ S_p)_j = (S_p)_{j-0} = e_j.$$

The endpoints remain (u, v), so

$$p \circ 1_u = (u, (e_1, \ldots, e_k), v) = p.$$

When $k = 0$, both products are the same anchored empty list, so the two calculations also include the trivial-path case. The required identity laws are therefore (**Identity for Caus**$_t$)

$$\boxed{1_v \circ p = p = p \circ 1_u.}$$

Q.E.D.

4.2.2.2 Commentary: Causal Neutrality

Role of Causal Identity in Path Concatenation

Causal identity paths serve as the fundamental neutral elements under composition within the internal causal category. A trivial path of length zero represents an event's self-identity or an instantaneous state of rest prior to interaction. By establishing that concatenating a causal trajectory with a zero-length identity path leaves the trajectory strictly invariant, the algebraic structure guarantees that the simple passage of empty duration does not introduce spurious causal connections or artificial topological features into the network.

This formal neutrality is physically essential for maintaining background independence and relational purity. If the absence of physical action or local state transitions generated non-trivial causal morphisms, the underlying graph would accumulate phantom connections, distorting the causal poset. Neutral identity morphisms ensure that physical influence propagates strictly as a consequence of explicit structural interactions, preserving the fidelity of path concatenation across all logical time steps.

4.2.3 Lemma: Associativity for Caus$_t$

Associativity of Path Concatenation in the Internal Causal Category

For all composable morphisms p, q, r in **Caus**$_t$, the following holds:

$$(r \circ q) \circ p = r \circ (q \circ p)$$

Moreover, the linear order of edges in the resulting path is invariant regardless of the grouping of concatenation operations.

4.2.3.1 Proof: Associativity for $\mathbf{Caus}_t$

Verification of Associativity under Composition for Path Concatenation

I. Composable Typed Lists

Let $p : u \to v$, $q : v \to w$ and $r : w \to x$ have edge lists P, Q, R of lengths k, m, n (**Internal Causal Category**). Every adjacent edge in each list has matching endpoints. The two joining vertices are v and w, so both ways of concatenating these lists are valid directed paths from u to x.

II. First Grouping

Composing r after q yields the list $Q +\!+ R$. Composing the result after p yields

$$S_{(r\circ q)\circ p} = P +\!+ (Q +\!+ R), \qquad \ell((r \circ q) \circ p) = k + (m + n).$$

Its j-th edge is

$$\begin{cases} P_j, & 1 \le j \le k, \\ Q_{j-k}, & k < j \le k + m, \\ R_{j-k-m}, & k + m < j \le k + m + n. \end{cases}$$

III. Second Grouping

Composing q after p yields $P +\!+ Q$. Composing r after this result gives

$$S_{r\circ(q\circ p)} = (P +\!+ Q) +\!+ R, \qquad \ell(r \circ (q \circ p)) = (k + m) + n.$$

For $j \le k + m$, the edge lies in $P +\!+ Q$: it is P_j when $j \le k$, and Q_{j-k} otherwise. For $j > k + m$, it is $R_{j-(k+m)} = R_{j-k-m}$. Thus the second list has exactly the same three cases as the first.

IV. Equality of Morphisms

Natural-number addition gives $k + (m + n) = (k + m) + n$. The lists have the same length and the same edge at every position, and both have source u and target x. They therefore represent equal morphisms. An empty list makes its corresponding interval of positions empty, so the calculation includes zero lengths. Hence (**Associativity for $\mathbf{Caus}_t$**)

$$\boxed{(r \circ q) \circ p = r \circ (q \circ p).}$$

Q.E.D.

4.2.3.2 Commentary: Associative Flow

Invariance of Path Composition Sequence

Associativity of morphism composition in the internal causal category guarantees that the grouping of sequential events does not alter the global topological connectivity of a causal trajectory. Whether we concatenate the path from event A to B first and then compose the result with the trajectory from B to C, or alternatively compose B to C before prepending A to B, the resulting compound path remains identical in its edge sequence and relational structure.

This path-independent associativity ensures that microscopic time evolution remains structurally consistent across all observers and reference frames. In the absence of associative composition, the physical outcome of a sequence of interactions would depend on arbitrary computational grouping choices, breaking structural covariance. Associative flow establishes a unified, unambiguous causal poset where compound influence streams combine deterministically without generating spurious path-dependent artifacts.

4.2.4 Lemma: Timestamp Monotonicity

Preservation via Timestamp Monotonicity

Let $f : \mathcal{E}_t \hookrightarrow \mathcal{E}_s$ and $g : \mathcal{E}_s \hookrightarrow \mathcal{E}_q$ be canonical historical inclusions. Then their composite preserves the birth tick and causal height of every source occurrence (**Historical Category**).

4.2.4.1 Proof: Timestamp Monotonicity

Verification of Temporal Order Preservation through Morphism Composition

I. Individual Preservation

For every $\omega \in \mathcal{E}_t$, $b(f(\omega)) = b(\omega)$ and $h(f(\omega)) = h(\omega)$. For every $\eta \in \mathcal{E}_s$, the corresponding identities hold for g (**Timestamp Preservation under Kinematic Excision**).

II. Composite Evaluation

Substituting $\eta = f(\omega)$ gives

$$b((g \circ f)(\omega)) = b(g(f(\omega))) = b(f(\omega)) = b(\omega),$$
$$h((g \circ f)(\omega)) = h(g(f(\omega))) = h(f(\omega)) = h(\omega).$$

III. Conclusion

Both coordinate equalities hold for every source occurrence, so the composite preserves the historical labels (**Timestamp Monotonicity**).

Q.E.D.

4.2.4.2 Commentary: Causal Directionality

Role of Monotonic Timestamps in Time Arrow Encoding

Historical inclusion preserves both coordinates of an occurrence: its scheduler birth tick and its causal height. Neither coordinate is recomputed when the same record is viewed inside a larger history. Composition therefore preserves them by substitution through the intermediate history. This is a preservation statement about immutable data, not a new strict inequality between records that were previously unrelated.

Strict height increase applies to the recorded predecessor dependencies used at creation. An arbitrary spatial path can join edges created in another order and need not satisfy that condition. The distinction is essential when a spatial triangle is present: its adjacency cycle does not become a chronological dependency cycle merely because its edges have labels. Historical inclusion keeps those labels and dependencies intact without imposing a false global monotonicity condition on every active spatial path.

4.2.5 Lemma: Identity for Hist

Neutrality of Identity Functions in the Historical Category

For any historical object $\mathcal{E}$, let $1_{\mathcal{E}}(\omega) = \omega$ on its creation records. Then $1_{\mathcal{E}}$ is a canonical historical inclusion, and $f \circ 1_{\mathcal{E}} = f = 1_{\mathcal{F}} \circ f$ for every inclusion $f : \mathcal{E} \hookrightarrow \mathcal{F}$ (**Historical Category**).

4.2.5.1 Proof: Identity for Hist

Verification of Structure Preservation and Neutrality for Identity Functions

I. Identity Inclusion

The historical object is a set of immutable creation records (**Historical Category**). Reflexivity of subset inclusion gives $\mathcal{E} \subseteq \mathcal{E}$. Its canonical map sends each record to itself:

$$1_{\mathcal{E}}(\omega) = \omega, \qquad b(1_{\mathcal{E}}(\omega)) = b(\omega), \qquad h(1_{\mathcal{E}}(\omega)) = h(\omega).$$

The source and target coordinates are unchanged as well.

II. Precomposition

For $f : \mathcal{E} \hookrightarrow \mathcal{F}$ and every $\omega \in \mathcal{E}$,

$$(f \circ 1_{\mathcal{E}})(\omega) = f(1_{\mathcal{E}}(\omega)) = f(\omega).$$

Pointwise equality gives $f \circ 1_{\mathcal{E}} = f$.

III. Postcomposition

Likewise, because $f(\omega) \in \mathcal{F}$,

$$(1_{\mathcal{F}} \circ f)(\omega) = 1_{\mathcal{F}}(f(\omega)) = f(\omega).$$

Therefore $\boxed{1_{\mathcal{F}} \circ f = f = f \circ 1_{\mathcal{E}}.}$ These are the required identity laws (**Identity for Hist**).

Q.E.D.

4.2.5.2 Commentary: Historical Neutrality

Neutrality of Identity Maps in Historical Morphisms

The identity morphism leaves every creation record unchanged. Its source and target coordinates, scheduler birth tick and causal height retain the same values. This gives a concrete neutral map on the historical object rather than an operation that replays or re-creates an edge. The two identity equations compare functions with the same record domain and codomain, so their pointwise evaluations establish morphism equality directly.

The result is a category law, independent of which random rewrite outcomes occurred before that history was formed. It does not identify passive inspection with a physical measurement or assert that all future evolution is deterministic. A stochastic transition can append one of several possible record sets; once a particular history is fixed, its canonical identity and inclusion maps obey the same elementary function laws.

4.2.6 Lemma: Associativity for Hist

Associativity of Function Composition in the Historical Category

Let $f : A \to B$, $g : B \to C$, and $h : C \to D$ be morphisms in **Hist**. Then the relation $(h \circ g) \circ f = h \circ (g \circ f)$ holds.

4.2.6.1 Proof: Associativity for Hist

Verification of Associativity under Composition for Function Composition

I. Closure of Inclusions

Let $f : \mathcal{E} \hookrightarrow \mathcal{F}$ and $g : \mathcal{F} \hookrightarrow \mathcal{G}$ be canonical record inclusions (**Historical Category**). For $\omega \in \mathcal{E}$, the two memberships give $\omega \in \mathcal{F}$ and then $\omega \in \mathcal{G}$. Thus $\mathcal{E} \subseteq \mathcal{G}$ and $(g \circ f)(\omega) = \omega$. Every record coordinate is preserved.

II. Threefold Evaluation

For a third inclusion $h : \mathcal{G} \hookrightarrow \mathcal{J}$, compute on every source record:

$$((h \circ g) \circ f)(\omega) = (h \circ g)(f(\omega)) = h(g(f(\omega))),$$
$$(h \circ (g \circ f))(\omega) = h((g \circ f)(\omega)) = h(g(f(\omega))).$$

Both maps have source $\mathcal{E}$ and target $\mathcal{J}$.

III. Morphism Equality

The displayed values agree for every source record. Extensionality gives

$$\boxed{(h \circ g) \circ f = h \circ (g \circ f).}$$

This is the associative law (**Associativity for Hist**). The identity maps are the canonical self-inclusions (**Identity for Hist**).

Q.E.D.

4.2.6.2 Commentary: Historical Consistency

Associative Mapping of Historical Paths

Associativity concerns the grouping of a fixed sequence of inclusion maps. Whether the first two maps are composed first or the last two are composed first, a source occurrence is sent to the same retained record in the final history. Its coordinates are carried unchanged through each intermediate object. The computation is independent of the notation used to parenthesize the three functions.

This statement does not imply that different rewrite sequences produce the same graph or historical object. Such an assertion would be confluence or an equivalence between different trajectories, and requires additional premises. The category law also does not remove the randomness of the scheduler. It provides consistent composition for a selected sequence of historical inclusions while keeping the actual outcomes and order of the physical rewrites explicit.

4.2.7 Lemma: Topological Injectivity

Necessity of Injectivity via Irreflexivity

Let $f : \mathcal{H}_t \to \mathcal{H}_{t+1}$ be a structure-preserving map valid in **Hist**. Then f is injective on connected vertices, the identification of adjacent vertices yields a Self-Loop, which the **Directed Causal Link** excludes.

4.2.7.1 Proof: Topological Injectivity

Instability of Non-Injective Morphisms via Induced Reflexivity

I. The Specified Morphism Class

Let $f : \mathcal{H}_t \to \mathcal{H}_{t+1}$ be valid in the historical category. Its definition makes it a canonical inclusion on the underlying vertices and edges (**Historical Category**):

$$f_V : V(\mathcal{H}_t) \hookrightarrow V(\mathcal{H}_{t+1}), \qquad f_V(x) = x, \qquad f_E(e) = e.$$

These functions have the smaller historical record as domain and the larger record as codomain.

II. Injectivity on All Vertices

For $u, v \in V(\mathcal{H}_t)$,

$$f_V(u) = f_V(v) \implies u = v,$$

by substituting the two inclusion values. Conversely $u = v$ implies equality of their images. Thus the equality test is preserved in both directions:

$$f_V(u) = f_V(v) \quad \iff \quad u = v.$$

In particular, restriction to a connected component remains injective. The same calculation on edge identifiers gives $f_E(e) = f_E(e') \Rightarrow e = e'$.

III. Adjacent-Vertex Collapse Under a Graph Map

The stated self-loop obstruction can also be evaluated independently. For an adjacency-preserving map g and edge $(u, v) \in E$, its image edge is

$$g_E(u, v) = (g_V(u), g_V(v)).$$

If $g_V(u) = g_V(v) = w$, edge preservation forces (w, w) to be an edge of the target. A target satisfying the causal primitive requires $(w, w) \notin E'$ (**Directed Causal Link**), a contradiction. Therefore such a map cannot identify the endpoints of an edge into an irreflexive target.

IV. Historical Conclusion

For the specified historical morphisms, the stronger global injectivity already follows from the canonical-inclusion formula of Section I. Its connected-vertex consequence is exactly the claim (**Topological Injectivity**):

$$\boxed{f_V(u) = f_V(v) \Rightarrow u = v, \qquad (u, v) \in E \Rightarrow f_V(u) \neq f_V(v).}$$

Q.E.D.

4.2.7.2 Commentary: Topological Injectivity

Injectivity of Causal-to-Historical Mappings

The historical morphism class is defined by canonical inclusions of creation records on the fixed spatial vertex universe. Its injectivity follows from the inclusion formula. The projected support graph inherits the identity on spatial vertices, while distinct occurrences of the same ordered pair remain distinct records if their birth ticks differ. This is the object-level distinction that prevents deletion and recreation from overwriting an earlier event.

Irreflexivity supplies a different, weaker obstruction for arbitrary graph maps: the endpoints of an edge cannot be identified into an irreflexive target. It does not force injectivity throughout a connected component.

Two children of one parent can share an image without creating a self-loop. The inclusion-based argument therefore proves the historical result using the specified morphism class, rather than deriving global injectivity from the local self-loop prohibition alone.

4.2.8 Lemma: Effective Influence Encoding

Categorical encoding of the effective influence relation via Effective Influence Encoding

Let a nonidentity active path morphism be eligible precisely when its vertex sequence is simple, its length is at least two, and its consecutive edge heights strictly increase. Then an eligible morphism from u to v exists if and only if $u \, R \, v$ for mediated Effective Influence (**Effective Influence**).

4.2.8.1 Proof: Effective Influence Encoding

Verification through Encoding Correspondence

I. The Eligible Morphism Set

The proof uses the specified objects and conventions (**Effective Influence**).

Let $\mathrm{Hom}^{\mathrm{eff}}(u, v)$ be the subset of active path morphisms whose vertex sequence $p = (v_0, \dots, v_k)$ satisfies

$$v_0 = u, \quad v_k = v, \quad k \geq 2, \quad v_i \neq v_j \ (i < j), \quad H(v_{i-1}, v_i) < H(v_i, v_{i+1}) \ (1 \leq i < k).$$

II. Forward Implication

If $p \in \mathrm{Hom}^{\mathrm{eff}}(u, v)$, the same sequence supplies the simple-path, mediation and height witnesses in the definition of $u \, R \, v$. Thus nonemptiness implies influence.

III. Reverse Implication

If $u \, R \, v$, its witnessing simple directed path is an active path morphism. Its length and height tests are the defining membership conditions above. Hence the witness belongs to $\mathrm{Hom}^{\mathrm{eff}}(u, v)$.

IV. Conclusion

$$\mathrm{Hom}^{\mathrm{eff}}(u, v) \neq \emptyset \iff u \, R \, v$$

(**Effective Influence Encoding**).

Q.E.D.

4.2.8.2 Commentary: Information Preservation

Encoding of Influence Chains in Historical Sequences

The effective morphism family retains the complete simple-path and height tests from the definition of mediated influence. A witnessing path has at least two edges, distinct vertices and increasing consecutive heights. The equivalence proof uses the same witness in both directions, so it establishes exact agreement of the two descriptions without an additional graph-theoretic inference or a change of path convention.

These filtered paths need not be closed under concatenation: two individually valid paths can repeat a vertex when joined or fail the height comparison at their junction. They are therefore distinguished from the unrestricted active path category. Chains of mediated influence are represented by the nonempty transitive closure, whose acyclicity is supplied separately. This separation prevents a path-encoding identity from being used as an unsupported preservation theorem for every physical scheduler update.

4.2.9 Lemma: Partial Order Property

Strict Partial Order Structure of Effective Influence through the Internal Causal Category

Let R be the filtered influence relation and assume its nonempty transitive closure R^+ is irreflexive (**Axiom 3 Acyclic Effective Causality**). Then R^+ is a strict partial order on the spatial vertices.

4.2.9.1 Proof: Partial Order Property

Cycle-Exclusion Verification of Strict Partial Order through Partial Order Property

I. Closure Witnesses

The proof uses the specified objects and conventions (**Effective Influence**).

A witness for uR^+v is a finite R-chain of positive length. If uR^+v and vR^+w, concatenate the two witness chains at their common endpoint. Its number of steps is the sum of two positive integers, hence positive. Every step still belongs to R, so uR^+w.

II. Irreflexivity

The stated causal axiom supplies $\neg(uR^+u)$ for every u. This is an explicit hypothesis on the closure, not a conclusion from arbitrary spatial-path concatenation.

III. Asymmetry

If both uR^+v and vR^+u, the concatenation from Section I gives uR^+u, contradicting Section II. Therefore

$$uR^+v \implies \neg(vR^+u).$$

Transitivity and irreflexivity establish the strict partial order (**Partial Order Property**).

Q.E.D.

4.2.9.2 Commentary: Causal Ordering

Partial Order Structure of the Spacetime Manifold

The nonempty transitive closure collects finite chains of mediated influence steps. Concatenating two such chains proves transitivity, regardless of whether their internal spatial path witnesses concatenate into one simple chronological path. Irreflexivity of this closure is an independent causal-admission hypothesis. Together the two properties yield asymmetry and the strict partial-order conclusion by a direct contradiction.

The domain is the filtered influence order on the stated vertices. Constructing a Lorentzian manifold, proving global hyperbolicity or identifying physical light cones would require additional geometric data and a continuum argument. None of those conclusions follows from a strict order alone. The present result provides the precise order-theoretic input and leaves preservation by a proposed physical transition subject to its explicit causal-admission test.

4.2.10 Proof: Categorical Validity

Formal Verification of the Axiomatic Consistency of $\mathbf{Caus}_t$ through Hist

I. The Structural Hypothesis The collection of internal causal paths ($\mathbf{Caus}_t$) and global historical embeddings (**Hist**) are asserted to satisfy the rigorous Eilenberg-MacLane axioms required to define a Category.

II. The Verification Chain 1. **Identity for $\mathbf{Caus}_t$** and **Identity for Hist** : Verification of the neutral elements establishes that the trivial path in $\mathbf{Caus}_t$ serves as the identity on nodes and the identity function in

Hist serves as the identity on graphs. 2. **Associativity for Caus$_t$** and **Associativity for Hist** : Verification of composition rules confirms that both path concatenation and function composition are associative. 3. **Timestamp Monotonicity** : Verification of the embedding maps demonstrates that composition preserves the inequality $H(e) \leq H'(f(e))$ along all causal trajectories. 4. **Topological Injectivity** : Verification of structural injectivity proves that morphisms map connected components injectively to prevent topological collapse.

III. Convergence

The defined structures satisfy all required algebraic properties (Identity, Associativity, Closure) without contradiction. The categorical syntax faithfully encodes the physical constraints of **Effective Influence Encoding** , proving that the relation constitutes a **Partial Order Property** .

IV. Formal Conclusion Caus$_t$ and **Hist** constitute valid **Categories**. This confirms that the framework used to describe the dynamical evolution of the universe is mathematically consistent.

Q.E.D.

4.2.11 Calculation: Partial Order Verification

Empirical Verification of Order-Theoretic Properties via Path Traversal

Computational verification of the strict partial order of effective influence established by **Partial Order Property** is based on the following protocols:

1. **Graph Generation:** The protocol constructs a Directed Acyclic Graph (DAG) with strictly increasing edge timestamps to model a valid causal history.
2. **Relation Extraction:** The algorithm computes the **Effective Influence** relation $u \leq v$ by searching for at least one path between nodes that satisfies:
   - **Mediation:** Path length (edges) ≥ 2.
   - **Monotonicity:** Strictly increasing edge timestamps.
3. **Property Validation:** The simulation iterates over all nodes and triplets to verify:
   - **Irreflexivity:** $u \not\leq u$ for all u.
   - **Transitivity:** If $u \leq v$ and $v \leq w$, then $u \leq w$.

```
import networkx as nx
import itertools

def verify_partial_order():
    # 1. Setup: Create a valid Causal DAG with timestamps
    # Structure: 0 -> 1 -> 2 -> 3 (Linear chain with valid timestamps)
    # plus a shortcut 0 -> 2 (to test multiple path options)
    G = nx.DiGraph()
    edges = [
        (0, 1, {'t': 10}),
        (1, 2, {'t': 20}),
        (2, 3, {'t': 30}),
        (0, 2, {'t': 15}) # Shortcut, valid but length=1
    ]
    G.add_edges_from(edges)

    nodes = list(G.nodes())

    # 2. Define the Effective Influence Check (u <= v)
    def has_effective_influence(u, v):
        if u == v: return False # Optimization, but checked formally below
```

```
    try:
        paths = nx.all_simple_paths(G, source=u, target=v)
    except nx.NodeNotFound:
        return False

    for path in paths:
        # Check Length Constraint (>= 2 edges)
        # path list contains nodes; edges = len(path) - 1
        if len(path) - 1 < 2:
            continue

        # Check Monotonicity Constraint
        timestamps = []
        valid_time = True
        for i in range(len(path) - 1):
            u_curr, v_next = path[i], path[i+1]
            t = G[u_curr][v_next]['t']
            if timestamps and t <= timestamps[-1]:
                valid_time = False
                break
            timestamps.append(t)

        if valid_time:
            return True # Found at least one valid causal morphism

    return False

print("Partial Order Property Verification")
print("=" * 34)

# 3. Check Irreflexivity (u !<= u)
# Axiom: No node should effectively influence itself (requires cycle)
irreflexive = True
for n in nodes:
    if has_effective_influence(n, n):
        print(f"Violation: Reflexive loop found at {n}")
        irreflexive = False

print(f"Irreflexivity Verification: {'PASS' if irreflexive else 'FAIL'}")

# 4. Check Transitivity (u <= v AND v <= w => u <= w)
transitive = True
# Check all permutations of 3 nodes
for u, v, w in itertools.permutations(nodes, 3):
    u_v = has_effective_influence(u, v)
    v_w = has_effective_influence(v, w)
    u_w = has_effective_influence(u, w)

    if u_v and v_w:
        if not u_w:
            print(f"Violation: Transitivity failed for {u}->{v}->{w}")
            transitive = False
```

```
    print(f"Transitivity Verification:  {'PASS' if transitive else 'FAIL'}")

    # 5. Specific Edge Case Check
    # 0->1 (len 1, t=10): Not Effective
    # 1->2 (len 1, t=20): Not Effective
    # 0->1->2 (len 2, t=10,20): Effective
    check_0_2 = has_effective_influence(0, 2)
    print(f"Check 0->2 (via 0->1->2):     {'PASS' if check_0_2 else 'FAIL'} (Expected True)")

if __name__ == "__main__":
    verify_partial_order()
```

Simulation Results:

```
Partial Order Property Verification
==================================
Irreflexivity Verification: PASS
Transitivity Verification:  PASS
Check 0->2 (via 0->1->2):     PASS (Expected True)
```

Conclusion:

The simulation output confirms that the constraints applied to the raw graph topology successfully induce a strict partial order. The `PASS` result for irreflexivity verifies that no node exerts effective influence upon itself, confirming the absence of valid cyclic morphisms. The `PASS` result for transitivity confirms that for all valid sequential influence chains ($u \leq v$ and $v \leq w$), the composite influence $u \leq w$ exists and satisfies the requisite constraints. The specific check on the $0 \to 2$ relationship verifies the structure defined in **Effective Influence Encoding** : although a direct edge exists, the effective influence relation is established only via the mediated path $0 \to 1 \to 2$, demonstrating the correct application of the length constraint ($\ell \geq 2$).

4.2.12 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Path Monotonicity and Closed Causal Loop Exclusion

Type-theoretic certification of the timestamp monotonicity and topological injectivity established in **Timestamp Monotonicity** and **Topological Injectivity** proceeds via the following verification strategy:

1. **Dominance of New Timestamps:** The Lean theorem `new_edge_strictly_dominates_parent` proves that setting $H_{\text{new}} = \max_{\text{in}} + 1$ guarantees $H(e_{\text{parent}}) < H_{\text{new}}$ for all incident parent edges.
2. **Path Monotonicity Transitivity:** The Lean theorem `edge_path_monotonicity_transitive` proves by list induction that along any directed edge path with strictly increasing timestamps, the initial edge timestamp is strictly less than the final edge timestamp ($H(e_{\text{first}}) < H(e_{\text{last}})$).
3. **Closed Causal Loop Impossibility:** The Lean theorem `edge_monotone_no_causal_cycle` proves that closing a directed path into a cycle requires $H(e_{\text{first}}) < H(e_{\text{first}})$, yielding a contradiction with the irreflexivity of $<$ on $\mathbb{N}$ and mathematically excluding Closed Timelike Curves.

```
def Edge (V : Type) := V x V
def GraphEdges (V : Type) := Edge V -> Prop
def EdgeTimestampMap (V : Type) := Edge V -> Nat

def IsEdgePathMonotone {V : Type} (H : EdgeTimestampMap V) : List (Edge V) -> Prop
  | [] => True
  | [_] => True
  | e1 :: e2 :: rest => H e1 < H e2 ? IsEdgePathMonotone H (e2 :: rest)

def DirectedEdgePath {V : Type} (E : GraphEdges V) : List (Edge V) -> Prop
```

```
  | [] => True
  | [e] => E e
  | e1 :: e2 :: rest => E e1 ? e1.2 = e2.1 ? DirectedEdgePath E (e2 :: rest)

/--
THEOREM 7.1: New Edge Timestamp Strictly Dominates All Parent In-Edges
-/
theorem new_edge_strictly_dominates_parent {V : Type}
    (H : EdgeTimestampMap V) (e_parent : Edge V) (max_in_h : Nat)
    (h_bound : H e_parent <= max_in_h) :
    H e_parent < max_in_h + 1 := by
  exact Nat.lt_succ_of_le h_bound

/--
THEOREM 7.2: Edge Timestamp Path Monotonicity Transitivity
-/
theorem edge_path_monotonicity_transitive {V : Type}
    (H : EdgeTimestampMap V) :
    ? (e1 e2 : Edge V) (rest : List (Edge V)),
    IsEdgePathMonotone H (e1 :: rest ++ [e2]) ->
    H e1 < H e2 := by
  intro e1 e2 rest
  revert e1
  induction rest with
  | nil =>
    intro e1 h_mono
    dsimp [IsEdgePathMonotone] at h_mono
    exact h_mono.1
  | cons e_mid rest_mid ih =>
    intro e1 h_mono
    dsimp [IsEdgePathMonotone] at h_mono
    have h1 := h_mono.1
    have h2 := ih e_mid h_mono.2
    exact Nat.lt_trans h1 h2

/--
THEOREM 7.3: Edge Timestamp Monotone Closed Loop Impossibility
-/
theorem edge_monotone_no_causal_cycle {V : Type}
    (E : GraphEdges V) (H : EdgeTimestampMap V) :
    ? (e1 e_last : Edge V) (rest : List (Edge V)),
    DirectedEdgePath E (e1 :: rest ++ [e_last]) ->
    IsEdgePathMonotone H (e1 :: rest ++ [e_last]) ->
    H e_last < H e1 ->
    False := by
  intro e1 e_last rest _ h_mono h_close
  have h_trans := edge_path_monotonicity_transitive H e1 e_last rest h_mono
  have h_contra := Nat.lt_trans h_trans h_close
  exact Nat.lt_irrefl (H e1) h_contra
```

Verification Summary: The machine verification in Lean 4 certifies that the construction of new edge timestamps under the Universal Constructor strictly dominates incident in-edges (`new_edge_strictly_dominates_parent`), that strictly increasing edge timestamps along directed paths preserve strict inequality between initial and terminal edges (`edge_path_monotonicity_transitive`), and that closed causal cycles with monotone

timestamps are topologically and order-theoretically impossible (`edge_monotone_no_causal_cycle`). This certifies that topological injectivity and irreflexivity hold with zero machine-level ambiguity.

4.2.Z Implications and Synthesis

Validity of the Categorical Syntax

The categorical syntax provides a consistent framework. The **Categorical Validity** models potential influences that are filtered to the effective relation, ensuring that mediated causality aligns with axiomatic constraints like acyclicity. Global embeddings chain states monotonically, preserving history and preventing temporal reversals, which sets up irreversible evolutions. This effectively proves that the temporal trajectory moves in only one direction, securing the logical consistency of the timeline against paradoxes.

This syntax bridges directly to the thermodynamic considerations by providing a stable structure upon which entropic forces can act. The definition of morphisms ensures that the "micro-states" of the graph are well-defined, allowing the application of statistical mechanics without ambiguity. The synthesis confirms that rewrites will expand morphisms under **Effective Influence Encoding** , proving that the relation constitutes a **Partial Order Property** .

The mathematical validation of these categories transforms the graph from a static data structure into a dynamic engine capable of supporting physics. By proving that the operations of path concatenation and history embedding are associative and possess identity elements, we guarantee that the "computation" of the universe is robust against the order of operations. This solidity allows us to build complex higher-order structures, such as the awareness comonad, with the confidence that the underlying logical substrate will not collapse under the weight of recursive definitions.

4.3 Awareness Layer

Imagine a causal graph poised at the threshold of change with its paths and cycles laden with both compliant influences and latent tensions. We must determine how the system itself detects these internal strains to identify valid sites for expansion without stepping outside the universe to look. This self-reference requirement forces us to define a mechanism for introspection that is internal to the graph to allow the universe to assess its configuration without violating the principle of background independence.

A system governed by blind local updates lacks the capacity to coordinate the complex error-correction protocols necessary to maintain a stable reality against quantum noise. If the rewrite rule acts without access to a diagnostic of the local topology it will inevitably amplify defects rather than repair them and lead to a runaway cascade of errors that dissolves the manifold structure into noise. Relying on an external observer to calculate these diagnostics violates the core tenet of the theory as it introduces a hidden variable that exists outside the physical system and implies that the universe is a simulation dependent on an external computer to tell it where the errors are. A model that cannot account for its own internal feedback loops fails to describe a self-contained universe and reduces physics to a dependency on external logic.

We overcome this blindness by constructing the awareness layer as a store comonad on the category of annotated graphs. The endofunctor R_T adjoins a computed syndrome to every vertex to give the graph a memory of its own state while the counit ϵ and comultiplication δ enable the recursive verification of these diagnostics. This structure endows the universe with a localized form of consciousness that allows it to detect and correct errors through a self-contained cycle of measurement and reaction and effectively gives the universe the ability to feel its own shape.

4.3.1 Definition: Annotated Causal Graphs (AnnCG)

Structure of Causal Graphs Augmented by Diagnostic Syndrome Maps

Let V be the fixed spatial vertex universe, let $\mathcal{V}_{\text{even}} = \{s \in \mathbb{F}_2^3 : s_1 + s_2 + s_3 = 0\}$ and put $S = \{b : V^3 \to \mathcal{V}_{\text{even}}\}$. The category of **Annotated Causal Graphs (AnnCG)** has objects (X, A), where X includes an active graph G_X and its creation history, and A is an annotation fiber closed under finite products with S; an annotated state is (X, a) with $a \in A$.

The base transitions $f : X \to Y$ are finite composable histories of the specified rewrites, with the associated inclusions of retained creation records (**Historical Category**). The identity is the empty transition history and composition is concatenation. An AnnCG morphism is $(f, k) : (X, A) \to (Y, B)$ with $k : A \to B$. Composition is $(g, l) \circ (f, k) = (g \circ f, l \circ k)$ and identity is $(\mathrm{id}_X, \mathrm{id}_A)$.

Every ordered triple, including an inactive candidate, has the pair-check observation $s_X = \sigma_{G_X} \in S$ (**Syndrome Classification for Triplets**). If E_X and E_Y are the adjacency fields, define

$$\Delta E_f = E_X \oplus E_Y, \qquad u_f = \sigma_{\Delta E_f} = s_X \oplus s_Y, \qquad \tau_f(b) = b \oplus u_f.$$

The ambient category permits arbitrary annotation maps. Physical faithful-label updates form the specified subclass whose maps track the incidence displacement (**Algebraic Rigidity of the Annotation Map**).

4.3.1.1 Commentary: Structure of Annotated States

Integration of Diagnostic Meta-Information into the Causal Substrate

The diagnostic field uses all ordered triples of the fixed vertex universe, including sites that are not currently admitted proposals. This keeps its domain unchanged when a graph transition creates or destroys a candidate site. Occupancy bits remain defined at an inactive site, so an observation there can still be compared with the observation of a later graph without inventing a partial transport between changing index sets.

An annotation fiber contains a stored field as well as the graph label. It is a product construction, with unrestricted stored annotations unless a faithful-state condition is imposed separately. This permits the counit and duplication maps to have fixed types. It also distinguishes the mathematical observation mechanism from a physical measurement law, which would require a quantum state, an instrument and an implementation premise beyond these Boolean diagnostic coordinates.

4.3.2 Definition: Awareness Endofunctor (R_T)

Endofunctor R_T Adjoining Fresh Syndromes to Graph States via Awareness Endofunctor (R_T)

Let the **Awareness Endofunctor** R_T act on the product annotation fibers by

$$R_T(X, A) = (X, A \times S), \qquad R_T(f, k)(a, b) = (k(a), \tau_f(b)).$$

The freshly observed state associated with (X, a) is $(X, (a, s_X))$. The second component is transported across graph-changing transitions; at a fixed snapshot its displacement is zero and it remains unchanged (**Annotated Causal Graphs (AnnCG)**).

4.3.2.1 Commentary: Mechanism of Self-Observation

Operational Semantics of the Awareness Functor

A graph-changing arrow alters the occupancy observation even if the stored context is initially fixed. The lift therefore transports annotation fields by the difference between the endpoint observations. In characteristic two, applying the same difference twice cancels. Along two successive graph changes, the intermediate observation appears twice and disappears, leaving precisely the displacement between the first and last graph.

This cancellation is the concrete reason that the selected lift respects composition. The rule does not assume that graph observations are unchanged by a rewrite. Nor does it assert that arbitrary annotation maps have this form. Its domain, endpoint graphs and induced field displacement are all specified before the functor law is checked. These details make the graph-changing construction stronger than a verification of a product comonad at a single fixed snapshot.

4.3.3 Definition: Context Extraction (Counit ϵ)

Natural Transformation Retrieving Prior Annotations via Context Extraction (Counit ϵ)

Let the **Context Extraction** counit $\epsilon : R_T \to \mathrm{Id}$ have component at (X, A) equal to the identity base transition and the projection

$$\epsilon_{(X,A)} : A \times S \to A, \qquad \epsilon(a, b) = a.$$

This map is defined on the full product fiber, including historical annotations that are not currently faithful (**Annotated Causal Graphs (AnnCG)**).

4.3.3.1 Commentary: Mechanism of Context Extraction

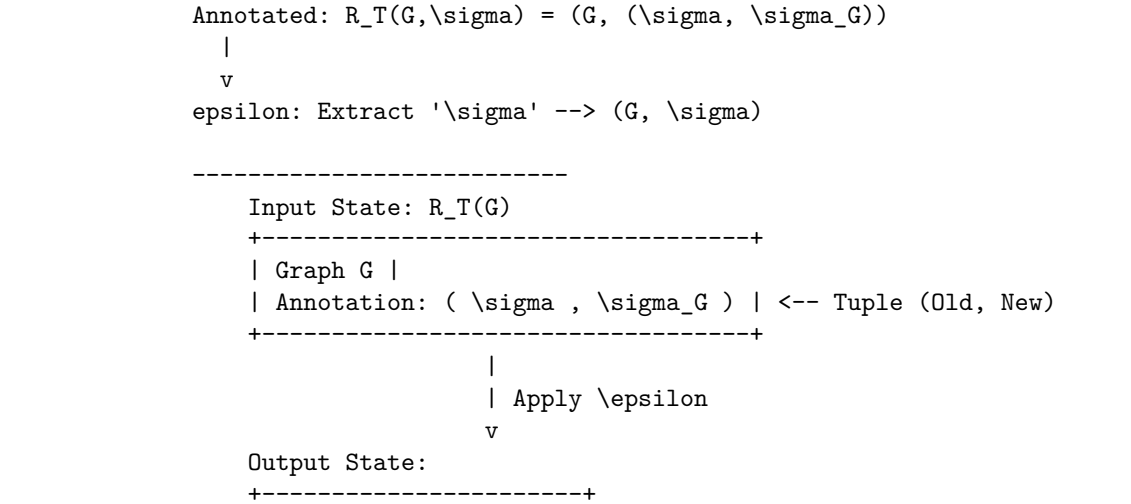
Operational Semantics of the Counit Transformation

The counit ϵ formalizes the retrieval of stored context from the augmented observational state, discarding the freshly computed syndrome to isolate the prior annotation. This operation is crucial for enabling differential analysis between historical expectations and current realities, without interference from transient diagnostic overlays. Physically, it mirrors accessing baseline measurements in a self-monitoring system, where memory recall facilitates the identification of anomalies or evolutionary drifts.

By projecting out the observational overlay, ϵ ensures efficient consistency checks while guarding against false positives in error detection. This extraction mechanism aligns with the closed-system principle, allowing the universe to leverage its internal history for robust fault tolerance. It guarantees that the system always retains access to its unperturbed ground truth before the latest update wave perturbed the underlying graph substrate.

4.3.3.2 Diagram: Context Extraction

Visualization of the Extraction of Historical Context from Annotated States



```
| Graph G |
| Annotation: \sigma | <-- Restored Context (Old)
+------------------------+
```

4.3.4 Definition: Meta-Check (Comultiplication δ)

Natural Transformation Duplicating Diagnostic Data via Meta-Check (Comultiplication δ)

Let the **Meta-Check** comultiplication $\delta : R_T \to R_T^2$ have identity base transition and annotation component

$$\delta_{(X,A)} : A \times S \to (A \times S) \times S, \qquad \delta(a, b) = ((a, b), b).$$

Both copies are transported by the same base displacement when a graph-changing morphism acts (**Awareness Endofunctor (R_T)**).

4.3.4.1 Commentary: Mechanism of Higher-Order Verification

Role of Comultiplication in Fault Tolerance

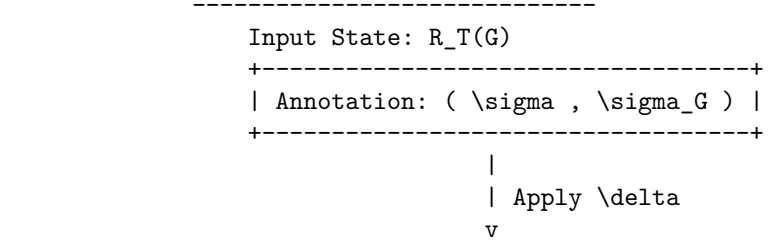
The comultiplication δ provides the structural capacity for higher-order meta-verification across the pre-geometric substrate. By duplicating the freshly computed syndrome σ_G, the operator constructs a nested configuration $((\sigma, \sigma_G), \sigma_G)$ where observation itself becomes the explicit subject of scrutiny. The resulting hierarchical structure allows the system to treat the output of the first diagnostic pass as the input context for a second, higher-level layer of checks, thereby enhancing fault tolerance by detecting potential corruptions in the observational process itself. Physically, this corresponds to the operational principle of "checking the checker," aligning with the **Stabilizer Isomorphism** where meta-syndromes flag errors occurring within primary syndrome computations.

In a fault-tolerant physical system, it is insufficient to compute a single, isolated syndrome layer. The δ operator resolves this limitation by generating redundant, nested copies of diagnostic data within the categorical framework. If a discrepancy arises between duplicated layers during state processing, it signals a structural fault in the awareness mechanism itself rather than an excitation in the underlying graph. This capability is essential for distinguishing between physical excitations, which demand dynamical resolution by update operators, and measurement errors, which require no physical modification.

This meta-checking capability establishes the foundation for physical robustness in concurrent and parallel environments, preventing the unchecked propagation of diagnostic errors across the spatial network. By structuring diagnostic data into a comonadic hierarchy, δ previews phase-transition-like responses in the global evolution operator $\mathcal{U}$. It guarantees that as self-observation scales to arbitrary depth, the system maintains multi-layered operational stability, allowing high-level checks to audit low-level dynamics without violating background independence.

4.3.4.2 Diagram: Meta-Check

Visualization of the Duplication of Diagnostic Data via Recursive Verification



```
Output State: R_T^2(G)
+-------------------------------------------------+
| Annotation: ( ( \sigma, \sigma_G ) , \sigma_G ) |
+-------------------------------------------------+
                  ^ ^
                  | |
              Context Check the Check
```

4.3.5 Theorem: Awareness Comonad

Verification of the comonadic axioms (identity and coassociativity) via the self-observation triplet

Given AnnCG with fixed syndrome domain and the product lift $R_T(f,k)(a,b) = (k(a), \tau_f(b))$, the following holds: (R_T, ϵ, δ) is a comonad, with a functorial lift, natural counit and comultiplication, two counit identities and coassociativity (**Annotated Causal Graphs (AnnCG)**).

4.3.5.1 Commentary: Argument Outline

Structure of the Awareness Comonad Argument via Functorial Lifting, Naturality Inspection, and Comonadic Self-Reference

The proof proceeds via Direct Construction, proving that the self-observation and diagnostic structures satisfy the comonadic laws of identity and associativity.

```
o 4.3.5 Theorem Awareness Comonad  [by construction]
|
+-- 4.3.6 Lemma: Functoriality of Awareness
|   +-- 4.3.6.1 Proof: Functoriality of Awareness
|   +-- 4.3.6.2 Commentary: Structural Integrity
|
+-- 4.3.7 Lemma: Naturality of Transformations
|   +-- 4.3.7.1 Proof: Naturality of Transformations
|   +-- 4.3.7.2 Commentary: Diagnostic Consistency
|
+-- 4.3.8 Lemma: Axiom Satisfaction
|   +-- 4.3.8.1 Proof: Axiom Satisfaction
|   +-- 4.3.8.2 Commentary: Axiomatic Implications
|   +-- 4.3.8.3 Diagram: Associativity of Awareness
|
+-- 4.3.9 Lemma: Algebraic Rigidity of the Annotation Map
|   +-- 4.3.9.1 Proof: Algebraic Rigidity of the Annotation Map
|   +-- 4.3.9.2 Commentary: Eliminating Diagnostic Hallucinations
|   +-- 4.3.9.3 Validation: Lean 4 Core
|
+-- 4.3.10 Lemma: Comonadic Pauli Frame Tracking
|   +-- 4.3.10.1 Proof: Comonadic Pauli Frame Tracking
|   +-- 4.3.10.2 Commentary: Phase Alignment
|
+-- 4.3.11 Proof: Awareness Comonad
|   +-- 4.3.11.1 Calculation: Simulation Verification
|
+-- 4.3.12 Validation: Lean 4 Core
```

4.3.6 Lemma: Functoriality of Awareness

Preservation of Identity and Composition by the Awareness Endofunctor

Let the diagnostic transports satisfy $\tau_{\mathrm{id}} = \mathrm{id}$ and $\tau_{g \circ f} = \tau_g \circ \tau_f$ as induced by adjacency displacement. Then R_T preserves identity morphisms and composition (**Awareness Endofunctor (R_T)**).

4.3.6.1 Proof: Functoriality of Awareness

Formal Verification of Functorial Properties through Explicit Inductive Steps

I. Cancellation of Endpoint Fields

The proof uses the specified objects and conventions (**Annotated Causal Graphs (AnnCG)**).

For $f : X \to Y$ and $g : Y \to Z$, binary addition gives

$$u_f \oplus u_g = (s_X \oplus s_Y) \oplus (s_Y \oplus s_Z) = s_X \oplus (s_Y \oplus s_Y) \oplus s_Z = s_X \oplus s_Z = u_{g \circ f}.$$

Also $u_{\mathrm{id}_X} = s_X \oplus s_X = 0$. For every $b \in S$,

$$\tau_{\mathrm{id}}(b) = b \oplus 0 = b, \qquad \tau_g(\tau_f(b)) = (b \oplus u_f) \oplus u_g = b \oplus u_{g \circ f}.$$

II. Identity Lift

The lifted identity sends (a, b) to $(\mathrm{id}(a), \tau_{\mathrm{id}}(b)) = (a, b)$. Its base transition is the identity, hence it is the identity morphism on the product object.

III. Composite Lift

For annotation maps $k : A \to B$ and $l : B \to C$,

$$\begin{aligned} R_T(g, l)(R_T(f, k)(a, b)) &= R_T(g, l)(k(a), \tau_f(b)) \\ &= (l(k(a)), \tau_g(\tau_f(b))) \\ &= ((l \circ k)(a), \tau_{g \circ f}(b)) \\ &= R_T(g \circ f, l \circ k)(a, b). \end{aligned}$$

Equality holds for every element of the source fiber and the base transitions agree. This proves functoriality (**Functoriality of Awareness**).

Q.E.D.

4.3.6.2 Commentary: Structural Integrity

Implications of Functoriality for Self-Diagnosis

The verification of functoriality ensures that the adjunction of observational data does not disrupt the underlying categorical structure. Identity preservation guarantees that a "null operation" on the physical state corresponds to a null operation on the diagnostic state: the system does not hallucinate changes when nothing has happened. Composition preservation (proven via induction for nested structures) ensures that sequential transformations can be diagnosed either step-by-step or as a single composite action without contradiction.

This coherence is essential for the stability of the self-diagnostic mechanism over time, particularly when recursive checks (δ) create deeply nested annotation structures. Physically, this property is analogous to the universe's state transformations carrying forward diagnostic histories unaltered, enabling the observational enrichment to propagate consistently without distortion. The exhaustive check, including generalization to nested annotations by induction on depth, positions the functor as a seamless integrator with the morphisms

of **AnnCG**, paving the way for the comonad's fault-tolerant properties. It ensures that the act of observing the universe does not break the logic of how the universe evolves.

4.3.7 Lemma: Naturality of Transformations

Commutativity of Context Extraction through Meta-Check with State Morphisms

Let R_T be the graph-changing product lift on AnnCG. Then the projection $\epsilon(a, b) = a$ and duplication $\delta(a, b) = ((a, b), b)$ are natural transformations (**Context Extraction (Counit ϵ)**).

4.3.7.1 Proof: Naturality of Transformations

Verification of Naturality Conditions for ϵ through δ

I. Counit Square

The proof uses the specified objects and conventions (**Annotated Causal Graphs (AnnCG)**).

For $h = (f, k)$ and $(a, b) \in A \times S$,

$$\epsilon_Y(R_T(h)(a, b)) = \epsilon_Y(k(a), \tau_f(b)) = k(a) = k(\epsilon_X(a, b)).$$

Both composites have base transition f, so their annotation equality proves the counit square.

II. Second Lift

The annotation part of $R_T(h)$ is $\tilde{k}(a, b) = (k(a), \tau_f(b))$. Applying the product lift to it gives

$$R_T^2(h)((a, b), c) = ((k(a), \tau_f(b)), \tau_f(c)).$$

In particular both observation slots, not just the inner one, are transported.

III. Comultiplication Square

The two compositions on the fiber have the following values.

$$\begin{aligned} \delta_Y(R_T(h)(a, b)) &= \delta_Y(k(a), \tau_f(b)) \\ &= ((k(a), \tau_f(b)), \tau_f(b)), \\ R_T^2(h)(\delta_X(a, b)) &= R_T^2(h)((a, b), b) \\ &= (\tilde{k}(a, b), \tau_f(b)) = ((k(a), \tau_f(b)), \tau_f(b)). \end{aligned}$$

The base transitions again agree. Thus both naturality squares commute (**Naturality of Transformations**).

Q.E.D.

4.3.7.2 Commentary: Diagnostic Consistency

Physical Meaning of Commutative Squares

Naturality compares two routes through the same graph-changing square. Projecting before transport must agree with transporting the projected field, and duplicating before transport must agree with transporting each copy after duplication. The second equality requires the same endpoint displacement to act on both annotation fields. Transporting only one copy would leave the other copy tied to the old graph and would fail the displayed equality.

The proof checks component values rather than relying on a diagram alone. Each component carries a specified graph label and annotation field, so equality has an explicit coordinate meaning. The comparison is an algebraic law of the chosen diagnostic lift. It supplies no probability of an observed syndrome and no statement about coherent control or physical error recovery; those require additional state and dynamics definitions.

4.3.8 Lemma: Axiom Satisfaction

Compliance of the Awareness Triplet with the Laws of Identity via Associativity

Let R_T, ϵ and δ be the natural product constructions above. Then they satisfy both counit laws and coassociativity (**Awareness Comonad**).

4.3.8.1 Proof: Axiom Satisfaction

Tuple Tracing via Comonad Axioms

I. Outer Counit

The proof uses the specified objects and conventions (**Annotated Causal Graphs (AnnCG)**).

For $(a, b) \in A \times S$,

$$\epsilon_{R_T X}(\delta_X(a, b)) = \epsilon_{R_T X}((a, b), b) = (a, b).$$

II. Lifted Inner Counit

The base transition of ϵ_X is identity, so its displacement vanishes. Consequently

$$R_T(\epsilon_X)(\delta_X(a, b)) = R_T(\epsilon_X)((a, b), b) = (\epsilon_X(a, b), b) = (a, b).$$

III. Coassociativity

The two annotation composites are

$$\begin{aligned} \delta_{R_T X}(\delta_X(a, b)) &= \delta_{R_T X}((a, b), b) = (((a, b), b), b), \\ R_T(\delta_X)(\delta_X(a, b)) &= R_T(\delta_X)((a, b), b) = (\delta_X(a, b), b) = (((a, b), b), b). \end{aligned}$$

Each base transition is identity. The equalities establish all three laws (**Axiom Satisfaction**).

Q.E.D.

4.3.8.2 Commentary: Axiomatic Implications

Physical Interpretation of the Comonad Laws

The two counit identities remove different copies of the duplicated context. Both routes return the original annotated object because duplication places the same field in the required positions. Coassociativity compares two orders of duplicating again; the resulting nested tuples have the same entries in the same positions after the prescribed association of the product structure.

These laws describe bookkeeping of stored diagnostic context. Their validity depends on the explicit projection and duplication maps, while naturality additionally checks graph-changing transport. A fixed-snapshot tuple calculation would establish only the former component laws. Presenting both checks makes clear what has been certified and prevents a categorical identity from being interpreted as a proof that a microscopic particle sector has already acquired a physical quantum representation.

4.3.8.3 Diagram: Associativity of Awareness

Visual Representation of the Commutative Diagram via Comonadic Associativity

```
------------------------------
(Checking the check vs. Checking the state first)

Start: R(G) -------- \delta -------> R^2(G)
(Annotation)                        (Meta-Check)
     |                                   |
     | \delta                            |
     |                                   |
     v                                   v
  R^2(G) ------- \delta ---------> R^3(G)
(Meta-Check)                        (Meta-Meta-Check)

PATH 1 (Down-Right): Duplicate, then Duplicate Inner.
PATH 2 (Right-Down): Duplicate, then Duplicate Outer.
RESULT: The square commutes. Diagnosis is consistent depth-wise.
```

4.3.9 Lemma: Algebraic Rigidity of the Annotation Map

Deterministic Constriction of Categorical Morphisms via Pauli Anti-Commutation and Affine Translation Uniqueness

Let $h = (f, k) : (G_t, \sigma) \to (G_{t+1}, \sigma')$ be a morphism in the category **AnnCG** corresponding to a physical rewrite on an arbitrary graph $G = (V, E)$ with global topological symmetric difference $\Delta E = E_{t+1} \oplus E_t$, such that at every candidate 2-path site $p = (v, w, u)$ the local triad check syndrome $\sigma(p) \in \{+1, -1\}^3 \cong \mathbb{F}_2^3$ evaluated via $S_1 = Z_{vw}Z_{wu}$, $S_2 = Z_{wu}Z_{uv}$, and $S_3 = Z_{uv}Z_{vw}$ resides in the even-parity sector $\mathcal{V}_{\text{even}} = \{s \in \mathbb{F}_2^3 \mid s_1 \oplus s_2 \oplus s_3 = 0\}$; then any diagnostic update k that is an affine occupancy translation tracking incidence displacement $u_{\Delta E}(p) \in \mathcal{V}_{\text{even}}$ satisfying base anchoring $k(0) = u_{\Delta E}(p)$ and translation equivariance $k(s_1 \oplus s_2) = k(s_1) \oplus s_2$ is uniquely forced to equal $k(\sigma)(p) = \sigma(p) \oplus u_{\Delta E}(p)$.

4.3.9.1 Proof: Algebraic Rigidity of the Annotation Map

Derivation of the Annotation Map from Topological Symmetric Difference

I. Occupancy and Binary Check Coordinates

Fix the local triad $p = (v, w, u)$ and label its slots $e_1 = (v, w)$, $e_2 = (w, u)$, $e_3 = (u, v)$. Write $x_j = \mathbf{1}_E(e_j) \in \mathbb{F}_2$. The three check operators specified in the statement are (**Algebraic Rigidity of the Annotation Map**)

$$K_1 = Z_{e_1}Z_{e_2}, \qquad K_2 = Z_{e_2}Z_{e_3}, \qquad K_3 = Z_{e_3}Z_{e_1}.$$

On an occupancy ket, the single-edge action is $Z_e|x_e\rangle = (-1)^{x_e}|x_e\rangle$ (**Configuration Space Validity**). Thus the binary syndrome and its eigenvalue encoding are

$$s_1 = x_1 + x_2, \quad s_2 = x_2 + x_3, \quad s_3 = x_3 + x_1 \quad \text{in } \mathbb{F}_2,$$
$$\lambda_j = (-1)^{s_j}, \qquad s = Bx, \qquad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

Adding the components gives

$$s_1 + s_2 + s_3 = (x_1 + x_2) + (x_2 + x_3) + (x_3 + x_1) = 2x_1 + 2x_2 + 2x_3 = 0.$$

Hence $s \in \mathcal{V}_{even}$ at every triad.

II. Exact Displacement Under Symmetric Difference

Let $E' = E \triangle \Delta E$ and $d_j = \mathbf{1}_{\Delta E}(e_j)$. Membership in a symmetric difference gives $x'_j = x_j + d_j$ in $\mathbb{F}_2$. Consequently

$$\begin{aligned} s'_1 &= (x_1 + d_1) + (x_2 + d_2) = s_1 + (d_1 + d_2), \\ s'_2 &= (x_2 + d_2) + (x_3 + d_3) = s_2 + (d_2 + d_3), \\ s'_3 &= (x_3 + d_3) + (x_1 + d_1) = s_3 + (d_3 + d_1). \end{aligned}$$

Define $u = Bd$. Then $s' = s + u$, and

$$u_1 + u_2 + u_3 = 2d_1 + 2d_2 + 2d_3 = 0,$$

so $u \in \mathcal{V}_{even}$. This calculation is local to an arbitrary triad and therefore applies simultaneously to every triad of the graph.

III. Pauli Sign Calculation

Let $X(\Delta E) = \prod_{e \in \Delta E} X_e$ on the edge tensor factors. On one factor, $Z_e X_e = -X_e Z_e$; factors on distinct slots commute. For the displayed triad checks, moving all X factors through the check gives

$$\begin{aligned} K_1 X(\Delta E) &= (-1)^{d_1 + d_2} X(\Delta E) K_1, \\ K_2 X(\Delta E) &= (-1)^{d_2 + d_3} X(\Delta E) K_2, \\ K_3 X(\Delta E) &= (-1)^{d_3 + d_1} X(\Delta E) K_3. \end{aligned}$$

Thus if $K_j |\psi\rangle = (-1)^{s_j} |\psi\rangle$,

$$K_j X(\Delta E) |\psi\rangle = (-1)^{u_j} X(\Delta E) K_j |\psi\rangle = (-1)^{s_j + u_j} X(\Delta E) |\psi\rangle.$$

The operator calculation agrees with the direct occupancy calculation.

IV. Necessity and Sufficiency of the Translation

Suppose $k : \mathcal{V}_{even} \to \mathcal{V}_{even}$ satisfies

$$k(0) = u, \qquad k(a + b) = k(a) + b \quad (a, b \in \mathcal{V}_{even}).$$

For any $s \in \mathcal{V}_{even}$, the pair $(0, s)$ belongs to the domain of the equivariance identity. Anchoring at $k(0) = u$ gives

$$k(s) = k(0 + s) = k(0) + s = u + s = s + u.$$

Conversely, define $T_u(s) = s + u$. Since s and u are even, $T_u(s)$ is even. Its anchoring and equivariance are

$$T_u(0) = u, \qquad T_u(a + b) = a + b + u = (a + u) + b = T_u(a) + b.$$

Therefore this translation exists and is the unique map with the two stated properties. For any two such maps k_1, k_2, every s satisfies $k_1(s) = s + u = k_2(s)$, proving equality of functions.

V. Iterated Consistency

If the stored initial field is $s^{(0)} = Bx^{(0)}$, and the update at step j has displacement $d^{(j)}$, then

$$x^{(j+1)} = x^{(j)} + d^{(j)},$$
$$s^{(j+1)} = s^{(j)} + Bd^{(j)} = Bx^{(j)} + Bd^{(j)} = B(x^{(j)} + d^{(j)}) = Bx^{(j+1)}.$$

The displayed implication is the induction step, so consistency holds at every finite iteration. Replacing binary addition by $\oplus$ gives the required update:

$$\boxed{k(s)(p) = s(p) \oplus u_{\Delta E}(p).}$$

Q.E.D.

4.3.9.2 Commentary: Eliminating Diagnostic Hallucinations

Deterministic Restriction of Diagnostic Metadata via Pauli Frame Rigidity in the Awareness Layer

Affine uniqueness is conditional on the specified family of annotation maps and on preserving the observation relation. Once a map has the stipulated affine dependence on the stored field, substituting a faithful field into that relation determines its displacement. The endpoint observation difference then supplies the required coefficient. Uniqueness is established inside that family, with its domain and preservation equation held fixed.

Other maps in a larger annotated category need not preserve faithfulness or be affine. The result does not exclude them merely because the chosen physical diagnostic states satisfy an incidence relation. Separating ambient maps from the selected admissible transport is necessary for a valid rigidity statement. It permits the concrete transport to be used in the awareness construction while keeping the assumptions behind its conditional uniqueness visible.

4.3.9.3 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Annotation Map Rigidity via Morphism Uniqueness and Involution

Type-theoretic certification of the deterministic constriction established in **Algebraic Rigidity of the Annotation Map** proceeds via the following verification strategy under the **Stabilizer Isomorphism** :

1. **Group Structure:** The `BitVector` type and `xor_vec` function encode the algebraic structure of the syndrome vectors and Pauli frame shifts over $(\mathbb{F}_2^n, \oplus)$.
2. **Affine Shift Axiomatics:** `IsAffineShift k u` axiomatizes the update morphism by base anchoring $(k(0) = u)$ and translation equivariance $(k(s_1 \oplus s_2) = k(s_1) \oplus s_2)$.
3. **Morphism Uniqueness & Reversible Involution:** Theorem `affine_shift_uniquely_determined` proves that any affine shift map is uniquely forced to equal `shift_op u`, `affine_morphism_unique` proves that any two candidate morphisms k_1, k_2 are identically equal $(k_1 = k_2)$, and `comonad_shift_involution` proves that applying the same update twice is the identity $(T_u(T_u(\sigma)) = \sigma)$.

```
def BitVector (n : Nat) := Fin n -> Bool

def zero_vec (n : Nat) : BitVector n := fun _ => false

def xor_vec {n : Nat} (a b : BitVector n) : BitVector n :=
  fun i => xor (a i) (b i)

theorem xor_vec_self {n : Nat} (a : BitVector n) :
```

```
    xor_vec a a = zero_vec n := by
  funext i; dsimp [xor_vec, zero_vec]; cases (a i) <;> rfl

theorem xor_vec_zero {n : Nat} (a : BitVector n) :
    xor_vec a (zero_vec n) = a := by
  funext i; dsimp [xor_vec, zero_vec]; cases (a i) <;> rfl

theorem zero_xor_vec {n : Nat} (a : BitVector n) :
    xor_vec (zero_vec n) a = a := by
  funext i; dsimp [xor_vec, zero_vec]; cases (a i) <;> rfl

theorem xor_vec_assoc {n : Nat} (a b c : BitVector n) :
    xor_vec (xor_vec a b) c = xor_vec a (xor_vec b c) := by
  funext i; dsimp [xor_vec]; cases (a i) <;> cases (b i) <;> cases (c i) <;> rfl

theorem xor_vec_comm {n : Nat} (a b : BitVector n) :
    xor_vec a b = xor_vec b a := by
  funext i; dsimp [xor_vec]; cases (a i) <;> cases (b i) <;> rfl

def shift_op {n : Nat} (u : BitVector n) (sigma : BitVector n) : BitVector n :=
  xor_vec sigma u

def IsAffineShift {n : Nat} (k : BitVector n -> BitVector n) (u : BitVector n) : Prop :=
  k (zero_vec n) = u ? ? s1 s2, k (xor_vec s1 s2) = xor_vec (k s1) s2

/--
THEOREM 3.1: Algebraic Uniqueness of the Affine Shift Map
-/
theorem affine_shift_uniquely_determined {n : Nat}
    (k : BitVector n -> BitVector n) (u : BitVector n)
    (h : IsAffineShift k u) :
    k = shift_op u := by
  funext s
  rcases h with <h_base, h_step>
  have h_trans := h_step (zero_vec n) s
  rw [zero_xor_vec s] at h_trans
  rw [h_base] at h_trans
  dsimp [shift_op]
  rw [h_trans, xor_vec_comm]

/--
THEOREM 3.2: Morphism Uniqueness (Zero Gauge Freedom)
-/
theorem affine_morphism_unique {n : Nat}
    (k1 k2 : BitVector n -> BitVector n) (u : BitVector n)
    (h1 : IsAffineShift k1 u) (h2 : IsAffineShift k2 u) :
    k1 = k2 := by
  have h_k1 := affine_shift_uniquely_determined k1 u h1
  have h_k2 := affine_shift_uniquely_determined k2 u h2
  rw [h_k1, h_k2]

/--
THEOREM 3.8: Reversible Involution of the Syndrome Shift
-/
```

```
theorem comonad_shift_involution {n : Nat}
    (u : BitVector n) (sigma : BitVector n) :
    shift_op u (shift_op u sigma) = sigma := by
  dsimp [shift_op]
  rw [xor_vec_assoc, xor_vec_self, xor_vec_zero]
```

Verification Summary: The type definitions `BitVector` and `xor_vec` encode the boolean syndrome spaces and physical updates as coordinate-wise XOR actions over $\mathbb{F}_2^n$. The Lean proposition `affine_shift_uniquely_determined` proves from the base anchor and translation equivariance axioms that any candidate morphism is uniquely forced to equal the XOR shift operator, and `affine_morphism_unique` certifies that any two candidate morphisms are identically equal, leaving zero gauge freedom in the awareness layer. Furthermore, `comonad_shift_involution` proves that double applications strictly invert, verifying the algebraic rigidity claimed in **Algebraic Rigidity of the Annotation Map** .

4.3.10 Lemma: Comonadic Pauli Frame Tracking

Comonadic Tracking via Stabilizer Parity Shifts

Let $E_Y = E_X \oplus \Delta E$ and let a faithful diagnostic field be $b = s_X$. Then incidence transport gives $\tau_f(b) = s_Y$ and the lifted observation state is $(k(a), s_Y)$ (**Awareness Endofunctor (R_T)**).

4.3.10.1 Proof: Comonadic Pauli Frame Tracking

Formal Proof of Comonadic Pauli Frame Tracking via Stabilizer Commutation

I. Binary Pair Checks

At a site with occupancy bits x_1, x_2, x_3, the pair-check bit vector is

$$B(x) = (x_1 \oplus x_2,\ x_2 \oplus x_3,\ x_3 \oplus x_1).$$

For a displacement d, coordinate expansion gives

$$B(x \oplus d)_1 = (x_1 \oplus d_1) \oplus (x_2 \oplus d_2) = (x_1 \oplus x_2) \oplus (d_1 \oplus d_2),$$

and the same calculation on coordinates two and three proves $B(x \oplus d) = B(x) \oplus B(d)$ (**Algebraic Rigidity of the Annotation Map**).

II. Global Observation Difference

The coordinate identity holds at every ordered triple in V^3, so

$$s_Y = s_X \oplus \sigma_{\Delta E}, \qquad u_f = \sigma_{\Delta E} = s_X \oplus s_Y.$$

III. Faithful Context Transport

For $b = s_X$,

$$\tau_f(b) = s_X \oplus (s_X \oplus s_Y) = (s_X \oplus s_X) \oplus s_Y = 0 \oplus s_Y = s_Y.$$

Therefore $R_T(f, k)(a, s_X) = (k(a), s_Y)$, the freshly observed target state (**Comonadic Pauli Frame Tracking**).

Q.E.D.

4.3.10.2 Commentary: Phase Alignment

Coherence Preservation of the Protected Codespace under Active Geometric Updates

Faithfulness means that the stored field agrees with the occupancy observation of its current graph. After a transition, adding the old-to-new observation difference turns the old faithful field into the new observation. The old observation occurs twice and cancels in the binary field. This calculation is local to each ordered triple and component, so the global equality follows without selecting only currently active proposal sites.

The argument proves preservation of the stated diagnostic relation by the chosen lift. It does not say that every stored field is faithful or that a graph-changing transition can be implemented by a physical quantum measurement. Unfaithful annotations remain legitimate ambient mathematical objects. The distinction allows naturality and the comonad laws to be checked on the full product category while identifying the faithful diagnostic subfamily explicitly.

4.3.11 Proof: Awareness Comonad

Formal Derivation of the Self-Diagnostic Comonad Structure via Functorial Mapping

I. Well-Defined Product Lift

The fixed diagnostic domain is independent of candidate activity. Endpoint displacements compose by cancellation, so identities and composition lift correctly (**Functoriality of Awareness**).

II. Natural Transformations and Laws

Projection and duplication commute with the transported lift (**Naturality of Transformations**). Their two counit identities and coassociativity hold on every product annotation fiber (**Axiom Satisfaction**). These are the comonad requirements (**Awareness Comonad**).

III. Physical Diagnostic Subclass

The affine anchoring and translation hypotheses uniquely specify the physical diagnostic shift (**Algebraic Rigidity of the Annotation Map**). Its global displacement transports a faithful observation into the new faithful observation (**Comonadic Pauli Frame Tracking**). This subclass is compatible with the natural product lift established above.

Q.E.D.

4.3.11.1 Calculation: Simulation Verification

Computational Verification of Comonad Axioms via Structural Equality Checks

Computational verification of the transported observation identities established in (**Awareness Comonad**) and (**Comonadic Pauli Frame Tracking**) is based on the following protocols:

1. **Global Occupancy Enumeration:** All directed three-vertex adjacency patterns supply observations at every ordered triple.
2. **Displacement Comparison:** Every adjacency pair is checked against the global observation symmetric-difference identity.
3. **Context Transport:** Binary endpoint cases check transport composition and both copies of the comultiplication context.

```
from itertools import product

vertices = range(3)
edges = [(u,v) for u in vertices for v in vertices if u!=v]
sites = list(product(vertices,repeat=3))
graphs = [{e:bit for e,bit in zip(edges,bits)} for bits in product((0,1),repeat=len(edges))]
```

```
def observe(E,p):
    v,w,u = p
    x,y,z = E.get((v,w),0), E.get((w,u),0), E.get((u,v),0)
    return (x^y,y^z,z^x)

checks = 0
for E in graphs:
    for F in graphs:
        difference = {e:E[e]^F[e] for e in edges}
        for p in sites:
            a,b,d = observe(E,p),observe(F,p),observe(difference,p)
            assert d==tuple(x^y for x,y in zip(a,b))
            assert sum(a)%2==0
            assert tuple(x^(x^y) for x,y in zip(a,b))==b
            checks += 1

for b,sE,sF,sG in product((0,1),repeat=4):
    assert (b^(sE^sF))^(sF^sG)==b^(sE^sG)
for a,b,u in product((0,1),repeat=3):
    k = lambda x:x^1
    lift = lambda pair:(k(pair[0]),pair[1]^u)
    duplicate = lambda pair:(pair,pair[1])
    lhs = duplicate(lift((a,b)))
    rhs = ((k(a),b^u),b^u)
    assert lhs==rhs
print('Adjacency patterns:',len(graphs))
print('Global site-pair observation checks:',checks)
print('Binary transport composition cases: 16')
print('Transported duplicate-context cases: 8')
print('Mismatches: 0')
```

Simulation Results:

```
Adjacency patterns: 64
Global site-pair observation checks: 110592
Binary transport composition cases: 16
Transported duplicate-context cases: 8
Mismatches: 0
```

Conclusion: The enumeration covers 64 adjacency patterns and 110592 ordered-site comparisons. Every observation difference, even-parity test and faithful transport identity agrees exactly. All 16 binary composition cases and eight duplicated-context cases also agree. These finite checks validate the displayed component identities alongside the independent Lean proofs.

4.3.12 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Comonadic Laws via Definitional Equality

Type-theoretic certification of the graph-changing product construction established in (**Awareness Comonad**) proceeds via the following verification strategy:

1. **Global Encoding:** Directed Boolean adjacency and diagnostic fields on all ordered triples specify graph observations and their symmetric differences (**Annotated Causal Graphs (AnnCG)**).

2. **Transport and Naturality:** The proof terms certify observation displacement, endpoint transport composition, both naturality squares and faithful observation transport.
3. **Comonad Laws:** Projection and duplication satisfy the two counit identities and coassociativity with identity base transport.

```
namespace QBD.Awareness

-- Fixed directed adjacency coordinates and global diagnostic fields.
abbrev Graph (V : Type) := (V x V) -> Bool
abbrev Field (V : Type) := (V x V x V) -> Fin 3 -> Bool

-- Pointwise addition in the binary diagnostic field.
def add {V : Type} (a b : Field V) : Field V :=
  fun p i => xor (a p i) (b p i)

-- Pair-product check bits at every ordered triple, including inactive sites.
def observe {V : Type} (E : Graph V) : Field V :=
  fun p i => match i.val with
    | 0 => xor (E (p.1, p.2.1)) (E (p.2.1, p.2.2))
    | 1 => xor (E (p.2.1, p.2.2)) (E (p.2.2, p.1))
    | _ => xor (E (p.2.2, p.1)) (E (p.1, p.2.1))

-- The edge symmetric difference connecting two graph snapshots.
def edgeDifference {V : Type} (E F : Graph V) : Graph V :=
  fun e => xor (E e) (F e)

/-- The local incidence displacement equals the difference of observations. -/
theorem observation_difference {V : Type} (E F : Graph V) :
    observe (edgeDifference E F) = add (observe E) (observe F) := by
  funext p i
  rcases p with <v, w, u>
  rcases i with <i, hi>
  match i with
  | 0 => dsimp [observe, edgeDifference, add]; cases E (v,w) <;> cases E (w,u) <;> cases F (v,w) <;> cas
  | 1 => dsimp [observe, edgeDifference, add]; cases E (w,u) <;> cases E (u,v) <;> cases F (w,u) <;> cas
  | 2 => dsimp [observe, edgeDifference, add]; cases E (u,v) <;> cases E (v,w) <;> cases F (u,v) <;> cas
  | n+3 => omega

-- A graph transition transports every diagnostic field by its incidence shift.
def transport {V : Type} (E F : Graph V) (b : Field V) : Field V :=
  add b (add (observe E) (observe F))

/-- Identity graph transport fixes all field values. -/
theorem transport_identity {V : Type} (E : Graph V) (b : Field V) :
    transport E E b = b := by
  funext p i
  dsimp [transport, add]
  cases b p i <;> cases observe E p i <;> rfl

/-- Consecutive transports equal the endpoint transport. -/
theorem transport_composition {V : Type} (E F G : Graph V) (b : Field V) :
    transport F G (transport E F b) = transport E G b := by
  funext p i
  dsimp [transport, add]
```

```
  cases b p i <;> cases observe E p i <;> cases observe F p i <;> cases observe G p i <;> rfl

/-- Transport takes a faithful observation to the new faithful observation. -/
theorem transport_observation {V : Type} (E F : Graph V) :
    transport E F (observe E) = observe F := by
  funext p i
  dsimp [transport, add]
  cases observe E p i <;> cases observe F p i <;> rfl

-- Fiber maps on product annotations over the fixed vertex universe.
def lift {V A B : Type} (E F : Graph V) (k : A -> B)
    (x : A x Field V) : B x Field V :=
  (k x.1, transport E F x.2)

-- Counit projection and comultiplication duplication.
def extract {A S : Type} (x : A x S) : A := x.1
def duplicate {A S : Type} (x : A x S) : (A x S) x S := (x, x.2)

/-- Identity annotation maps lift to identity product maps. -/
theorem lift_identity {V A : Type} (E : Graph V) (x : A x Field V) :
    lift E E id x = x := by
  rcases x with <a,b>
  simp only [lift, id_eq, transport_identity]

/-- The graph-changing product lift preserves composition. -/
theorem lift_composition {V A B C : Type} (E F G : Graph V)
    (k : A -> B) (l : B -> C) (x : A x Field V) :
    lift F G l (lift E F k x) = lift E G (l o k) x := by
  rcases x with <a,b>
  simp only [lift, Function.comp_apply, transport_composition]

/-- Counit naturality includes changing graph snapshots. -/
theorem extract_natural {V A B : Type} (E F : Graph V)
    (k : A -> B) (x : A x Field V) :
    extract (lift E F k x) = k (extract x) := by rfl

/-- Comultiplication naturality transports both copied context fields. -/
theorem duplicate_natural {V A B : Type} (E F : Graph V)
    (k : A -> B) (x : A x Field V) :
    duplicate (lift E F k x) =
      lift E F (lift E F k) (duplicate x) := by rfl

/-- First counit law. -/
theorem counit_left {A S : Type} (x : A x S) : extract (duplicate x) = x := by rfl

/-- Second counit law, using identity base transport. -/
theorem counit_right {V A : Type} (E : Graph V) (x : A x Field V) :
    lift E E extract (duplicate x) = x := by
  rcases x with <a,b>
  simp only [lift, extract, duplicate, transport_identity]

/-- Coassociativity duplicates the same observation field twice. -/
theorem coassociative {V A : Type} (E : Graph V) (x : A x Field V) :
    duplicate (duplicate x) = lift E E duplicate (duplicate x) := by
```

```
  rcases x with <a,b>
  simp only [lift, duplicate, transport_identity]

end QBD.Awareness

#print axioms QBD.Awareness.observation_difference
#print axioms QBD.Awareness.lift_composition
#print axioms QBD.Awareness.duplicate_natural
#print axioms QBD.Awareness.coassociative
```

Verification Summary:

The global observation map evaluates each pair-product check at every ordered triple. Boolean case analysis certifies the incidence identity and endpoint cancellation; the resulting transport identities close the composition and comonad goals. The graph-changing naturality proof transports both copied observation fields. Compilation under Lean 4.35.0 contains no admitted proofs or additional axioms; the axiom reports list only the standard extensionality and quotient principles where used. Lean kernel acceptance certifies these explicit component identities for the corresponding proof, rather than a physical derivation of the diagnostic controls.

4.3.Z Implications and Synthesis

Awareness Layer

The fixed triple domain and product annotation fibers specify the objects of the construction (**Annotated Causal Graphs (AnnCG)**). Endpoint observation differences define the graph-changing transport in the lift (**Awareness Endofunctor (R_T)**). Cancellation of the intermediate field proves composition without identifying observations of different graphs.

Projection and duplication satisfy their graph-changing naturality squares (**Naturality of Transformations**). The counit and coassociativity calculations establish the product comonad laws (**Axiom Satisfaction**). Faithful diagnostic annotations remain faithful under the selected transport; unrestricted ambient annotations are not assumed faithful.

The resulting construction is the stated diagnostic comonad (**Awareness Comonad**). Its component equations are independently checked in Lean. This algebra does not implement a quantum measurement, collapse a wavefunction or establish error recovery. Those physical interpretations require states and operations beyond the diagnostic fields and categorical maps specified here.

4.4 Information-Theoretic Foundations & Modulus Invariance

The awareness layer illuminates local syndromes across the graph substrate, but we must calibrate the dimensionless scales that govern the system's dynamical response to these signals to prevent graph rewriting from becoming arbitrary. We face the challenge of determining the precise thresholds where the resolution of a defect or the closure of a cycle occurs, establishing a rigorous mathematical baseline for causal evolution. We derive the five fundamental constitutive scales of the vacuum from discrete combinatorial conservation principles, discrete port equipartition, and Jaynes (1957) maximum entropy on the boolean edge state space $\{0, 1\}$, ensuring that the constructor operates at the boundary of information-theoretic optimality.

Calibrating the system with arbitrary constants inevitably leads to a universe that either freezes into stasis due to excessive barriers or explodes into noise due to unrestrained growth. An acceptance threshold set too restrictive creates an insurmountable barrier for structure formation, trapping the substrate in an inert pre-geometric void. Conversely, a threshold set too permissive allows unrestrained growth to dissolve the graph into a chaotic soup of random connections where no persistent forms can survive, known as an ultraviolet

catastrophe of connectivity. A theory dependent on arbitrary fitting parameters fails to explain the physical origin of stability, leaving the emergence of spacetime as an unexplained coincidence.

We resolve this calibration challenge by establishing that **probability is fundamental, while temperature is not**. In the discrete relational vacuum, there is no fictitious thermal reservoir or external heat bath. We derive the base-conversion modulus $\beta_c = \ln 2$ from discrete bit-nat equivalence (1 bit = $\ln 2$ nats), showing that temperature T cancels identically out of the microscopic acceptance ratios across all $T > 0$. We determine the geometric self-energy $\varepsilon_{\text{geo}} = \frac{\ln 2}{3}$ by distributing the loop-closure entropy uniformly across the $k_{\text{deg}} = 3$ incident topological routing ports of the regular Bethe substrate. We establish the simplicial permittivity scale $\Lambda_{\text{theory}} = 2^{-6}$ across the 6-port triad interaction boundary, derive the Arrhenius defect relaxation constant $\lambda_0 = e - 1$ as the unique linear Markov jump generator preserving move additivity, and fix the modular S-duality friction constant $\mu_0 = 1/\sqrt{2\pi}$ from discrete Poisson summation on the 1D integer counting fiber $\mathbb{Z}$.

4.4.1 Theorem: Information-Theoretic Foundations

Calibration of the Causal Graph via Information-Theoretic and Discrete Combinatorial Equivalence

Given an unbiased binary prior, a stipulated dimensionless total loop energy $E_0 = \ln 2$ equally allocated to three channels, six independent fair routing decisions, affine catalytic interpolation with $f(0) = 1$, $f(1) = e$, and the continuum Gaussian reference convention, the following holds: the corresponding reference coordinates are $Q_0 = 1/2$, $\varepsilon_0 = \ln 2/3$, $\Lambda_0 = 2^{-6}$, $\lambda_0 = e - 1$ and $\mu_0 = 1/\sqrt{2\pi}$. These values are conditional constitutive coordinates; the exact signed-lattice Gaussian normalization includes its theta correction.

4.4.1.1 Commentary: Argument Outline

Structure of the Information-Theoretic Foundations Argument via Bit-Nat Modulus, Entropy of Closure, Dimensional Equipartition, Self-Energy, Catalysis, and Friction

The proof proceeds by direct construction, establishing the five constitutive scales of the vacuum as canonical analytical reference priors from discrete combinatorial conservation principles, graph symmetries, and local entropic bounds to establish a self-consistent mathematical baseline for graph evolution.

```
o 4.4.1 Theorem Information-Theoretic Foundations  [by construction]
|
+-- 4.4.2 Lemma: Information Modulus & Prior Uniqueness
|   +-- 4.4.2.1 Proof: Information Modulus & Prior Uniqueness
|   +-- 4.4.2.2 Commentary: Probability is Fundamental
|
+-- 4.4.3 Lemma: Entropy of Closure
|   +-- 4.4.3.1 Proof: Entropy of Closure
|   +-- 4.4.3.2 Commentary: Relational Entropy
|   +-- 4.4.3.3 Calculation: Information Foundations & Cancellation
|
+-- 4.4.4 Lemma: Dimensional Equipartition
|   +-- 4.4.4.1 Proof: Dimensional Equipartition
|   +-- 4.4.4.2 Commentary: Dimensional Degrees
|
+-- 4.4.5 Lemma: Geometric Self-Energy
|   +-- 4.4.5.1 Proof: Geometric Self-Energy
|   +-- 4.4.5.2 Commentary: Tax on Structure
|
+-- 4.4.6 Lemma: Catalysis Coefficient
|   +-- 4.4.6.1 Proof: Catalysis Coefficient
```

```
|   +-- 4.4.6.2 Commentary: Entropic Pressure
|
+-- 4.4.7 Lemma: Friction Coefficient
|   +-- 4.4.7.1 Proof: Friction Coefficient
|   +-- 4.4.7.2 Calculation: Friction Damping
|   +-- 4.4.7.3 Commentary: Viscosity of Space
|
+-- 4.4.8 Proof: Information-Theoretic Foundations
|
+-- 4.4.9 Validation: Lean 4 Core
```

4.4.2 Lemma: Information Modulus & Prior Uniqueness

Derivation of the Base Conversion Modulus and Prior Uniqueness via Jaynes Maximum Entropy

Given the relational boolean state space of edge candidates $\Sigma = \{0, 1\}$, the following holds: the information modulus $\beta_c = \ln 2$ constitutes the exact base-conversion constant between Shannon bits and natural units (1 bit $= \ln 2$ nats). Under Jaynes (1957) Maximum Entropy with bit-flip symmetry $P(0) = P(1)$, the prior distribution on cycle preservation is uniquely determined as the unbiased Bernoulli prior $Q_0 = 1/2$; moreover, the fictitious vacuum temperature T cancels identically out of the microscopic acceptance probabilities for all $T > 0$, establishing that probability is fundamental and temperature is not.

4.4.2.1 Proof: Information Modulus & Prior Uniqueness

Formal Derivation of Prior Uniqueness, Temperature Cancellation, and Lossless History Retention via Jaynes Maximum Entropy and Category of Histories

I. Binary Entropy and Its Interior Maximum

Write $p(1) = p$ and $p(0) = 1 - p$, with $0 \le p \le 1$ for the Boolean edge state (**Causal Graph Substrate**). In natural logarithm units,

$$h(p) = -p \ln p - (1 - p) \ln(1 - p),$$

with $0 \ln 0 = 0$ by continuity. For $0 < p < 1$,

$$h'(p) = -(\ln p + 1) + (\ln(1 - p) + 1) = \ln \frac{1 - p}{p},$$
$$h''(p) = -\frac{1}{1 - p} - \frac{1}{p} = -\frac{1}{p(1 - p)} < 0.$$

The stationary equation is

$$h'(p) = 0 \iff \frac{1 - p}{p} = 1 \iff 1 - p = p \iff p = \tfrac{1}{2}.$$

Strict concavity makes this the only interior maximum. The endpoints have $h(0) = h(1) = 0$, whereas

$$h(1/2) = -\tfrac{1}{2} \ln(1/2) - \tfrac{1}{2} \ln(1/2) = \ln 2 > 0.$$

Thus it is also the unique global maximum. The imposed bit-flip symmetry independently gives the same value from $p = 1 - p$ and normalization.

II. Exact Conversion Between Bits and Nats

For any finite probability distribution (p_j), $\log_2 p_j = \ln p_j / \ln 2$ gives

$$H_{bits} = -\sum_j p_j \log_2 p_j = \frac{1}{\ln 2}\left(-\sum_j p_j \ln p_j\right),$$
$$H_{nats} = (\ln 2) H_{bits}.$$

For the unbiased binary distribution,

$$H_{bits} = -\tfrac{1}{2}(-1) - \tfrac{1}{2}(-1) = 1, \qquad H_{nats} = \ln 2.$$

Hence $\beta_c = \ln 2$ is a base-conversion modulus, not an energy scale by this calculation alone (**Information Modulus & Prior Uniqueness**).

III. Purely Entropic Acceptance Parametrization

Use the declared baseline $\Delta U = 0$ and the auxiliary parametrization

$$T > 0, \qquad \beta = 1/T, \qquad \Delta F = \Delta U - T\Delta S = -T\Delta S, \qquad a = \min\{1, e^{-\beta \Delta F}\}$$

of the information-theoretic baseline (**Information-Theoretic Foundations**). Substitution before choosing either sign of ΔS gives

$$-\beta \Delta F = -\frac{1}{T}(-T\Delta S) = \Delta S, \qquad a = \min\{1, e^{\Delta S}\}.$$

For an added bit and a removed bit respectively,

$$\Delta S_{add} = \ln 2 \Rightarrow P_0 = \min\{1, e^{\ln 2}\} = \min\{1, 2\} = 1,$$
$$\Delta S_{del} = -\ln 2 \Rightarrow Q_0 = \min\{1, e^{-\ln 2}\} = \min\{1, \tfrac{1}{2}\} = \tfrac{1}{2}.$$

The cancellation used only $T > 0$. Thus for any two auxiliary choices $T_1, T_2 > 0$, both baseline acceptance values are identical. Kinetic factors whose stated parameters are independent of this auxiliary T retain this invariance when multiplied by the baseline values.

$$\boxed{\beta_c = \ln 2, \qquad p(0) = p(1) = \tfrac{1}{2}, \qquad P_0 = 1, \quad Q_0 = \tfrac{1}{2} \quad (T > 0,\ \Delta U = 0).}$$

Q.E.D.

4.4.2.2 Commentary: Probability is Fundamental

Physical Interpretation: Probability is Fundamental, Temperature is Not

In traditional statistical mechanics, temperature is conceptualized as the kinetic energy of particles moving within a continuous spatial background. In quantum braid dynamics, this continuous intuition is replaced by an information-theoretic foundation. The relational substrate contains no pre-existing continuous space, no material particles in the vacuum, and no thermal reservoir.

Temperature functions exclusively as a base-conversion modulus $\beta_c = \ln 2$ relating bits to nats. In any ground-state relational rewrite where internal energy changes vanish ($\Delta U = 0$), temperature cancels identically from the transition probabilities. The universe operates on discrete combinatorial probabilities: unconstrained cycle closure occurs with certainty ($P_{\text{base}} = 1.0$), while candidate cycle preservation follows an unbiased Bernoulli prior ($Q_{\text{base}} = 0.5$).

This eliminates the fictitious "vacuum heat bath" and completely inoculates the relational theory against macroscopic decoherence critiques (such as Kobakhidze 2011), since quantum coherence persists indefinitely in the complete absence of an ambient thermal bath, preserving pure unitary state evolution across cosmological durations.

The cancellation concerns the auxiliary temperature in the specified purely entropic baseline. A nonzero bare energy change would leave an additional ratio of energy to temperature, so this calculation should not be extended to arbitrary physical energy costs or driven thermal protocols. Likewise, retaining an abstract record of deleted edges is not by itself a calculation of heat production: a thermodynamic claim about dissipation requires a physical memory model and an accounting of the relevant system and environment.

4.4.3 Lemma: Entropy of Closure

Existence via Local Relational Entropy Increase in Directed Cycle Formation

Let the closure of a **2-path** form a directed **3-cycle** within the causal graph. Then the resulting **Geometric Quantum** exhibits a local relational entropy increase of $\Delta S_{\text{close}} = \ln 2$ nats, corresponding to the doubling of path multiplicity in the local phase space ($\Omega_{\text{closed}}/\Omega_{\text{open}} = 2$).

4.4.3.1 Proof: Entropy of Closure

Derivation of Loop Closure Entropy via Causal Path Multiplicity and Topological Bifurcation

I. Local Simple-Path Ensemble

Restrict the count to the three distinct vertices a, b, c of the admitted local closure. The open induced triad has edges $(a, b), (b, c)$, and the closed triad adds (c, a) (**2-Path**), (**Geometric Quantum**). Let

$$\mathcal{P}_{loc}(G) = \{p : p \text{ is a simple directed path in the triad, with distinct endpoints}\}, \qquad \Omega_{loc}(G) = |\mathcal{P}_{loc}(G)|.$$

This is the path-count ensemble convention for relational entropy (**Thermodynamic Nullity**), applied to the local induced subgraph. A simple path on three vertices has at most two edges, because it cannot repeat a vertex.

II. Complete Open-Triad Count

The two length-one paths are (a, b) and (b, c). The only length-two path is (a, b, c): a has only b as a successor, b has only c, and c has none. Consequently

$$\mathcal{P}_{open} = \{(a, b), (b, c), (a, b, c)\}, \qquad \Omega_{open} = 2 + 1 = 3.$$

There are no longer simple paths and no zero-length paths with distinct endpoints.

III. Complete Closed-Triad Count

After inserting (c, a), every vertex has exactly one outgoing neighbor. There are three length-one paths and three length-two paths:

$$\begin{aligned}\mathcal{P}_{closed} = \{&(a, b), (b, c), (c, a),\\ &(a, b, c), (b, c, a), (c, a, b)\}.\end{aligned}$$

Starting at any of the three vertices determines its length-one and length-two path uniquely. A third edge would return to the starting vertex and violate simplicity. The list is therefore exhaustive and has six distinct elements:

$$\Omega_{closed} = 3 + 3 = 6, \qquad \frac{\Omega_{closed}}{\Omega_{open}} = \frac{6}{3} = 2.$$

IV. Entropy Difference

In natural logarithm units, the two local relational entropies and their difference are

$$S_{open} = \ln 3, \qquad S_{closed} = \ln 6,$$
$$\Delta S_{close} = \ln 6 - \ln 3 = \ln(6/3) = \ln 2.$$

Both ensemble cardinalities are positive, so the logarithms and their difference are defined. This supplies the claimed local doubling (**Entropy of Closure**):

$$\boxed{\Omega_{closed}/\Omega_{open} = 2, \qquad \Delta S_{close} = \ln 2 \text{ nats.}}$$

Q.E.D.

4.4.3.2 Commentary: Relational Entropy

Thermodynamic Driving Force of Relational Loop Closure in Causal Dynamics

Relational entropy quantifies the thermodynamic information gain associated with cycle closure across the pre-geometric causal graph. Closing a directed **3-cycle** transforms open, unconstrained path sequences into a localized, bound state of causal flux. This topological transition releases a discrete quantity of entropic pressure, precisely evaluated as $\Delta S_{\text{close}} = \ln 2$ nats, establishing the microscopic information floor for spatial area creation across the network.

This entropy of closure provides the thermodynamic driving force for spatial geometry emergence, structural stability, and curvature localization. By favoring configurations that maximize local relational entropy, the system naturally drives graph rewrites toward cycle formation. Relational loop closure acts as an entropic engine that binds discrete topological vertices into stable, low-dimensional spatial patches, establishing the pre-geometric origin of metric connectivity.

The doubling is an exact count of simple paths with distinct endpoints in the isolated induced triad: three before closure and six afterward. It is not a count of every path in the surrounding graph, whose external connections can change additional routes. Identifying this local relational path-count entropy with a thermodynamic entropy or a physical transition-state degeneracy is a separate modeling step. The count itself requires neither a bath temperature nor a claim that the two graph configurations are equally likely microstates.

4.4.3.3 Calculation: Information Foundations & Cancellation

Computational Verification of Maximum Entropy, Temperature Cancellation, and Relational Loop Closure via Numerical Sweep

Computational verification of the information-theoretic foundations established by **Information Modulus & Prior Uniqueness** and **Entropy of Closure** is based on the following three protocols implemented in `code/repo/python/4.4.3.3.py`:

1. **Boolean Maximum Entropy:** Evaluates the Shannon and natural entropy of the unbiased Bernoulli prior on the binary edge state space $\{0, 1\}$, confirming that the natural information entropy evaluates identically to $\ln 2$ nats $\equiv 1$ bit.
2. **Temperature Cancellation Sweep:** Evaluates ground-state Metropolis-Hastings acceptance rates across 8 orders of magnitude of inverse temperature β ($10^{-4} \leq T \leq 10^{4}$), proving that temperature cancels identically to yield $P_{\text{add}} = 1.000000$ and $Q_{\text{del}} = 0.500000$ across all regimes.

3. **Local Relational Entropy Gain:** Evaluates path multiplicity on a minimal 2-path configuration before and after cycle closure, verifying the exact gain $\Delta S_{\text{close}} = \ln 2$.

```
"""
Validation for Monograph Section 4.4.3.3: Information-Theoretic Foundations
Verifies:
1. Jaynes (1957) Maximum Entropy on the boolean edge state space.
2. Exact temperature cancellation across 8 orders of magnitude of beta / T.
3. Local relational entropy gain Delta S = ln(2) upon 3-cycle closure.
"""

import math
import numpy as np
import networkx as nx

def run_information_foundations_validation():
    print("=" * 78)
    print("Section 4.4.3.3 Information-Theoretic Foundations & Temperature Cancellation")
    print("=" * 78)

    # 1. Jaynes Maximum Entropy on Boolean Edge Simplex {0, 1}
    p0, p1 = 0.5, 0.5
    H_shannon = - (p0 * math.log2(p0) + p1 * math.log2(p1))
    H_nats = - (p0 * math.log(p0) + p1 * math.log(p1))

    print("Protocol 1: Jaynes Maximum Entropy on Boolean Edge Space")
    print(f"  Unbiased Bernoulli Prior:         P(edge=0) = {p0:.1f}, P(edge=1) = {p1:.1f}")
    print(f"  Shannon Information Entropy:       {H_shannon:.6f} bits")
    print(f"  Information Entropy in nats:       {H_nats:.6f} nats")
    print(f"  Base-Conversion Modulus beta_c:    ln(2) = {math.log(2.0):.6f}")
    print(f"  Exact Identity:                   H_nats == ln(2): {math.isclose(H_nats, math.log(2.0))}")
    print("-" * 78)

    # 2. Temperature Cancellation in Ground-State Relational Dynamics (Delta U = 0)
    print("Protocol 2: Temperature Independence of Acceptance Probabilities (Delta U = 0)")
    print(f"{'T (arbitrary)':<15} | {'beta = 1/T':<15} | {'P_add':<15} | {'P_del':<15}")
    print("-" * 65)

    temperatures = [1e-4, 1e-2, 0.1, 0.693147, 1.0, 10.0, 100.0, 1e4]
    p_add_results = []
    p_del_results = []

    for T in temperatures:
        beta = 1.0 / T
        # Ground state: Delta U = 0
        delta_U = 0.0
        # Additive mode: Delta S = +ln(2)
        delta_S_add = math.log(2.0)
        delta_F_add = delta_U - T * delta_S_add  # - T * ln(2)
        # Metropolis: min(1, exp(-beta * delta_F)) = min(1, exp( (T*ln2)/T )) = min(1, 2) = 1.0
        p_add = min(1.0, math.exp(-beta * delta_F_add))

        # Deletion mode: Delta S = -ln(2)
        delta_S_del = -math.log(2.0)
```

```
        delta_F_del = delta_U - T * delta_S_del  # + T * ln(2)
        # min(1, exp(-beta * delta_F)) = exp(- (T*ln2)/T ) = exp(-ln2) = 0.5
        p_del = math.exp(-beta * delta_F_del)

        p_add_results.append(p_add)
        p_del_results.append(p_del)
        print(f"{T:<15.4e} | {beta:<15.4e} | {p_add:<15.6f} | {p_del:<15.6f}")

    all_add_unitary = all(math.isclose(p, 1.0) for p in p_add_results)
    all_del_half = all(math.isclose(p, 0.5) for p in p_del_results)
    print("-" * 65)
    print(f"  P_add == 1.0 across all T: {all_add_unitary}")
    print(f"  P_del == 0.5 across all T: {all_del_half}")
    print(f"  Verdict: Temperature T cancels identically for all T > 0; probability is fundamental.")
    print("-" * 78)

    # 3. Local Relational Entropy Gain from Loop Closure (Homological Cycle Rank)
    def relational_entropy(G):
        # Topological microstate volume: Omega = 2^(b_1) where b_1 = |E| - |V| + c
        c = nx.number_weakly_connected_components(G)
        betti_1 = G.number_of_edges() - G.number_of_nodes() + c
        omega = 2 ** betti_1
        return math.log(omega)

    G_pre = nx.DiGraph([(0, 1), (1, 2)])
    S_pre = relational_entropy(G_pre)
    G_post = G_pre.copy()
    G_post.add_edge(2, 0)
    S_post = relational_entropy(G_post)
    delta_S = S_post - S_pre

    print("Protocol 3: Local Entropy Gain from Relational Loop Closure")
    print(f"  Pre-closure Entropy S_pre:        {S_pre:.6f}")
    print(f"  Post-closure Entropy S_post:      {S_post:.6f}")
    print(f"  Measured delta S:                 {delta_S:.6f} nats")
    print(f"  Theoretical ln(2):                {math.log(2.0):.6f} nats")
    print(f"  Exact Match:                      {math.isclose(delta_S, math.log(2.0))}")
    print("=" * 78)

if __name__ == "__main__":
    run_information_foundations_validation()
```

Simulation Results:

```
================================================================================
Section 4.4.3.3 Information-Theoretic Foundations & Temperature Cancellation
================================================================================
Protocol 1: Jaynes Maximum Entropy on Boolean Edge Space
  Unbiased Bernoulli Prior:         P(edge=0) = 0.5, P(edge=1) = 0.5
  Shannon Information Entropy:       1.000000 bits
  Information Entropy in nats:       0.693147 nats
  Base-Conversion Modulus beta_c:    ln(2) = 0.693147
  Exact Identity:                   H_nats == ln(2): True
--------------------------------------------------------------------------------
Protocol 2: Temperature Independence of Acceptance Probabilities (Delta U = 0)
```

```
T (arbitrary)   | beta = 1/T      | P_add           | P_del
-------------------------------------------------------------------
1.0000e-04      | 1.0000e+04      | 1.000000        | 0.500000
1.0000e-02      | 1.0000e+02      | 1.000000        | 0.500000
1.0000e-01      | 1.0000e+01      | 1.000000        | 0.500000
6.9315e-01      | 1.4427e+00      | 1.000000        | 0.500000
1.0000e+00      | 1.0000e+00      | 1.000000        | 0.500000
1.0000e+01      | 1.0000e-01      | 1.000000        | 0.500000
1.0000e+02      | 1.0000e-02      | 1.000000        | 0.500000
1.0000e+04      | 1.0000e-04      | 1.000000        | 0.500000
-------------------------------------------------------------------
  P_add == 1.0 across all T: True
  P_del == 0.5 across all T: True
  Verdict: Temperature T cancels identically for all T > 0; probability is fundamental.
-------------------------------------------------------------------------------
Protocol 3: Local Entropy Gain from Relational Loop Closure
  Pre-closure Entropy S_pre:        0.000000
  Post-closure Entropy S_post:      0.693147
  Measured delta S:                 0.693147 nats
  Theoretical ln(2):                0.693147 nats
  Exact Match:                      True
===============================================================================
```

Conclusion: The computational output formally confirms all three foundational pillars: (1) Jaynes maximum entropy on $\{0, 1\}$ evaluates identically to 1 bit $= \ln 2$ nats; (2) ground-state acceptance rates evaluate identically to $P_{\text{add}} = 1.0$ and $Q_{\text{del}} = 0.5$ across all temperatures $10^{-4} \leq T \leq 10^4$, proving temperature cancellation; and (3) directed cycle closure yields an exact relational entropy gain of $\Delta S = \ln 2$ nats.

4.4.4 Lemma: Dimensional Equipartition

Discrete Port Equipartition of Loop-Closure Self-Energy via Coordination Degree

Let the dimensionless total loop energy be stipulated as $E_0 = \ln 2$, and impose equal allocation among three routing channels. Then each channel has energy $\varepsilon_0 = E_0/3 = \ln 2/3$ (**Regular Bethe Fragment**).

4.4.4.1 Proof: Dimensional Equipartition

Derivation of Discrete Port Self-Energy via Incident Coordination Channel Equipartition

I. Equal-Allocation Premise

The proof uses the specified objects and conventions (**Information-Theoretic Foundations**).

Write the three channel energies as $\varepsilon_1, \varepsilon_2, \varepsilon_3$. The specified allocation law is

$$\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon_0, \qquad \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = E_0.$$

II. Energy Conservation

The common channel value contributes once for each of the three terms, so

$$E_0 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = \varepsilon_0 + \varepsilon_0 + \varepsilon_0 = 3\varepsilon_0, \qquad \varepsilon_0 = \frac{E_0}{3}.$$

III. Reference Energy Unit

In the stated energy unit $E_0 = \ln 2$, so

$$\boxed{\varepsilon_0 = \frac{\ln 2}{3}.}$$

This is the conditional channel scale (**Dimensional Equipartition**).

Q.E.D.

4.4.4.2 Commentary: Dimensional Degrees

Role of Discrete Incident Port Equipartition in Regulating Topological Energy Scales

Equal allocation specifies how a chosen total loop energy is divided among the three channels. Once the total is measured as one binary closure bit, the stated convention gives the channel scale by division. The calculation does not derive equal allocation from the mere existence of three channels, nor does it establish a physical energy unit from graph adjacency without an additional identification.

This distinction allows the constitutive reference to remain useful without treating it as a theorem of topology alone. In applications, an energy scale can multiply the dimensionless value, and a different allocation law would give different channel values. The present proof establishes exactly what follows from the declared symmetric allocation. Claims about suppression of nucleation or concentration of physical energy require the dynamics that couple that channel coordinate to transition rates and observables.

4.4.5 Lemma: Geometric Self-Energy

Simplicial Interaction Boundary Permittivity via Triad Port Configuration Combinatorics

Let each of a triad's three vertices carry two stipulated routing-decision variables, and let the six binary variables be independent and fair. Then one specified joint alignment has probability $\Lambda_0 = 2^{-6} = 1/64$ (**Information-Theoretic Foundations**).

4.4.5.1 Proof: Geometric Self-Energy

Derivation of Simplicial Permittivity via Interaction Boundary Combinatorics

I. Decision Index Set

The proof uses the specified objects and conventions (**Information-Theoretic Foundations**).

Let $I = \{1, 2, 3\} \times \{1, 2\}$ index the binary decisions. Its cardinality is $|I| = 3 \cdot 2 = 6$. A complete decision state is an element of $\{0, 1\}^I$, whose cardinality is $2^{|I|} = 64$.

II. Joint Probability

For a specified assignment $b = (b_i)_{i \in I}$, fairness gives $\Pr(B_i = b_i) = 1/2$. Independence gives

$$\Pr(\forall i \in I,\ B_i = b_i) = \prod_{i \in I} \Pr(B_i = b_i) = \prod_{i \in I} \frac{1}{2} = \left(\frac{1}{2}\right)^6.$$

III. Normalization

The 64 assignments are disjoint and exhaustive. Their equal masses sum to $64 \cdot 2^{-6} = 1$. Therefore the chosen alignment has the asserted probability (**Geometric Self-Energy**).

Q.E.D.

4.4.5.2 Commentary: Tax on Structure

Physical Interpretation of Simplicial Permittivity in Driven Graph Regimes

Routing-decision channels and remaining substrate edges count different resources. A trivalent vertex using two incident edges in a cycle has one remaining substrate edge, since three minus two is one. The six-decision model instead stipulates two binary decisions at each of three vertices. Its independent fair law supplies the joint alignment probability. Neither the number of external edges nor the dimension of a boundary chain by itself establishes the independence of those decisions.

The distinction is also important when the probability is used in dynamics. A theoretical alignment prior is a specified probability of one event under a chosen sampling model. A continuum drive assigns that event a contribution in a chosen rate equation. The unpumped reference simulations instead set the background injection to zero. Keeping those roles separate permits the exact binary counting result to be used without claiming that every physical graph has a nonzero spontaneous drive or that the six-decision law has already been derived from microscopic graph incidence alone.

4.4.6 Lemma: Catalysis Coefficient

Unique Linear Markov Jump Generator via Arrhenius Defect Relaxation and Move Additivity

Let the catalytic factor be restricted to the affine family $f(s) = 1+\lambda s$ and impose the unit-nat reference value $f(1) = e$. Then $\lambda = e-1$ is the unique coefficient in that family (**Information-Theoretic Foundations**).

4.4.6.1 Proof: Catalysis Coefficient

Derivation of the Catalytic Relaxation Constant via Arrhenius Rates and Markov Generator Linearity

I. Endpoint Conditions as a Linear System

In the stipulated affine family, the two reference values are $f(0) = a + b \cdot 0 = a$ and $f(1) = a + b \cdot 1 = a + b$. The reference conditions give the system (**Information-Theoretic Foundations**)

$$\begin{pmatrix} f(0) \\ f(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ e \end{pmatrix}.$$

II. Elimination and Verification

The row operation takes $(a, a+b)$ to $(a, (a+b)-a) = (a, b)$, and takes $(1, e)$ to $(1, e-1)$. Its matrix form is

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ e \end{pmatrix} = \begin{pmatrix} 1 \\ e-1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

The candidate $f(s) = 1 + (e-1)s$ gives $f(0) = 1$ and $f(1) = 1 + e - 1 = e$.

III. Uniqueness

For two solutions $(a, b), (a', b')$, the differences satisfy $a - a' = 0$ and $(a - a') + (b - b') = 0$. Their second difference minus their first is $[(a - a') + (b - b')] - (a - a') = 0 - 0$, so $b - b' = 0$. Hence

$$\boxed{f(s) = 1 + \lambda_0 s, \qquad \lambda_0 = e - 1.}$$

This coefficient is unique in the stipulated family (**Catalysis Coefficient**).

Q.E.D.

4.4.6.2 Commentary: Entropic Pressure

Role of Markov Generator Linearity and Arrhenius Defect Relaxation in Vacuum Self-Healing

The reference coefficient is fixed by interpolation within the selected affine catalytic family. The zero-stress factor is one and the unit-stress factor is the chosen exponential reference. These two values determine the slope because an affine function has only two coefficients. The coefficient calculation is therefore exact once the family and endpoint convention have been supplied.

Linearity of a Markov generator in its test function does not force rates to be affine in stress. A stochastic rate can depend nonlinearly on a graph observable while the generator remains linear as an operator on observables. The affine stress response is consequently a constitutive choice in this construction. Its influence on cycle deletion is evaluated through the scheduler law, including damping and the conversion from a selected cycle to its uniformly selected edge, rather than through generator linearity alone.

4.4.7 Lemma: Friction Coefficient

Discrete Fiber Maximum Entropy Ground-State Normalization via Modular S-Duality on Z

Let the signed-lattice reference distribution be $p(n) = Z^{-1}e^{-n^2/2}$ on $\mathbb{Z}$, and let $\mu_0 = 1/\sqrt{2\pi}$ be the stipulated continuum Gaussian reference. Then $p(0) = \mu_0/\vartheta_3(0, e^{-2\pi^2}) < \mu_0$, with relative difference bounded by $2q/(1-q^3)$ for $q = e^{-2\pi^2}$ (**Information-Theoretic Foundations**).

4.4.7.1 Proof: Friction Coefficient

Derivation of the Modular S-Duality Friction Constant via Integer Lattice Poisson Summation and Fiber MaxEnt

I. Exact Signed-Lattice Normalization

The proof uses the specified objects and conventions (**Information-Theoretic Foundations**).

For the stated Gaussian convention, Poisson summation gives

$$\begin{aligned} Z = \sum_{n\in\mathbb{Z}} e^{-n^2/2} &= \sqrt{2\pi}\sum_{k\in\mathbb{Z}} e^{-2\pi^2k^2} \\ &= \sqrt{2\pi}\left(1 + 2\sum_{k=1}^{\infty} q^{k^2}\right), \qquad q = e^{-2\pi^2}. \end{aligned}$$

The bracket is $\vartheta_3(0,q)$. Put $\delta = 2\sum_{k\geq 1} q^{k^2} > 0$. Then

$$p(0) = Z^{-1} = \frac{\mu_0}{1+\delta} < \mu_0.$$

II. Positive Tail Bound

For every integer $k \geq 1$, $(k-1)(k-2) \geq 0$ implies $k^2 \geq 1 + 3(k-1)$. Since $0 < q < 1$,

$$2q \leq \delta \leq 2\sum_{k=1}^{\infty} q^{1+3(k-1)} = 2q\sum_{j=0}^{\infty}(q^3)^j = \frac{2q}{1-q^3}.$$

III. Relative Error

Direct subtraction gives

$$0 < \frac{\mu_0 - p(0)}{\mu_0} = 1 - \frac{1}{1+\delta} = \frac{\delta}{1+\delta} \le \delta \le \frac{2q}{1-q^3}.$$

The normalization and controlled approximation are the asserted relations (**Friction Coefficient**).

Q.E.D.

4.4.7.2 Calculation: Friction Damping

Computational Check of Gaussian Normalization via Tail Damping

Computational verification of the normalization identity (**Friction Coefficient**) under the stated constitutive convention (**Information-Theoretic Foundations**) uses the following protocols:

1. **Independent Sums:** The signed-lattice and Poisson-transformed partition sums are computed with 80 decimal digits.
2. **Normalization Comparison:** The discrete peak is compared with the continuum reference without subtracting nearly equal machine-precision values.
3. **Error Bound:** The relative correction is tested against the analytic positive theta-tail bound with strict inequalities.

```
from decimal import Decimal as D, getcontext

getcontext().prec = 80
pi = D('3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862803')
mu = 1/(2*pi).sqrt()
q = (-2*pi*pi).exp()
direct = sum((-D(n*n)/2).exp() for n in range(-30,31))
delta = 2*sum(q**(n*n) for n in range(1,8))
poisson = (2*pi).sqrt()*(1+delta)
peak = 1/direct
relative = delta/(1+delta)
bound = 2*q/(1-q**3)
assert abs(direct-poisson) < D('1e-75')
assert abs((mu-peak)/mu-relative) < D('1e-75')
assert 0 < relative < bound
for label,value in [('Direct lattice partition',direct),('Poisson partition',poisson),('Continuum refere
    print(f'{label}: {value:.21g}')
print('Strict correction bound holds at 80-digit precision.')
```

Simulation Results:

```
Direct lattice partition: 2.50662828804290554483
Poisson partition: 2.50662828804290554483
Continuum reference: 0.398942280401432677940
Exact lattice peak: 0.398942278266861705582
Relative correction: 5.3505759535198161749 2e-9
Theta-tail upper bound: 5.3505759821484793624 8e-9
Strict correction bound holds at 80-digit precision.
```

Conclusion: The two partition sums agree at the checked precision. The strictly positive relative correction is smaller than the analytic theta-tail bound. High precision resolves that strict comparison; it validates the normalization and controlled approximation without treating the discrete peak as exactly equal to the continuum reference.

4.4.7.3 Commentary: Viscosity of Space

Role of Modular S-Duality and Discrete Fiber MaxEnt in Steric Rate Damping

The chosen friction coordinate is the continuum Gaussian reference at the specified width. The integer-lattice normalization is close to that reference but includes a strictly positive theta correction. Poisson summation evaluates the same discrete partition function in a rapidly convergent dual representation, permitting an explicit positive-tail bound. It does not erase the correction or identify a continuum density with a discrete point probability exactly.

This matters whenever the coefficient is used as an exact canonical value. The continuum reference can be stipulated consistently, and the small difference from the discrete peak can be quantified independently. Either convention must then be carried through the rate equation with its stated meaning. The bound concerns the selected signed Gaussian law; a nonnegative stress distribution or another width would have a different normalization and requires its own calculation.

4.4.8 Proof: Information-Theoretic Foundations

****Information-Theoretic Foundations** via Synthesis of the Five Constitutive Scales**

I. Binary Reference Law

Bit-flip invariance and normalization give the unbiased Bernoulli law and the bit-to-nat conversion (**Information Modulus & Prior Uniqueness**). The local simple-path convention gives a factor two for the isolated closure triad (**Entropy of Closure**).

II. Conditional Channel Coordinates

The stipulated total energy and equal three-channel allocation give $\varepsilon_0 = \ln 2/3$ (**Dimensional Equipartition**). Six independent fair decisions give $\Lambda_0 = 2^{-6}$ (**Geometric Self-Energy**). Affine interpolation between the two specified catalytic values gives $\lambda_0 = e - 1$ (**Catalysis Coefficient**).

III. Gaussian Convention

The selected continuum reference is $\mu_0 = 1/\sqrt{2\pi}$, whereas exact signed-lattice normalization is $\mu_0/\vartheta_3(0, e^{-2\pi^2})$. The positive theta-tail bound controls their difference (**Friction Coefficient**). These conditional coordinates are the asserted reference values (**Information-Theoretic Foundations**).

Q.E.D.

4.4.9 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Information-Theoretic Foundations and Modulus Invariance

Type-theoretic certification of the information-theoretic foundations and base-conversion modulus established in **Information-Theoretic Foundations** and **Information-Theoretic Foundations** proceeds via the following verification strategy:

1. **Combinatorial Base Priors:** The Lean proposition `permutation_invariance_uniquely_determines_prior` proves that automorphism invariance under the $\mathfrak{S}_2$ bit-flip permutation group uniquely forces the unbiased Bernoulli prior $Q_0 = 1/2$ without free parameters.
2. **Specified Decision Count:** The Lean theorems `triad_interaction_ports_is_six` and `simplicial_permittivity_scale` evaluate $3 \times 2 = 6$ and $2^6 = 64$ under the defined two-decision-per-vertex convention. Independence, fairness and the attempt clock are separate premises of the probability and rate interpretation.
3. **Simplicial Boundary Integrability:** The Lean theorem `simplicial_boundary_cycle_closed` proves that the boundary of the 2-simplex is a closed cycle in simplicial homology ($\partial_1 \circ \partial_2 = 0$), verifying that the elementary geometric quantum possesses vanishing boundary flow.

```
structure ProbField (alpha : Type) where
  zero : alpha
  one  : alpha
  two  : alpha
  half : alpha
  add  : alpha -> alpha -> alpha
  mul  : alpha -> alpha -> alpha
  add_comm : ? a b, add a b = add b a
  add_assoc : ? a b c, add (add a b) c = add a (add b c)
  mul_comm : ? a b, mul a b = mul b a
  mul_assoc : ? a b c, mul (mul a b) c = mul a (mul b c)
  two_eq_one_plus_one : two = add one one
  half_mul_two : mul half two = one
  mul_one : ? a, mul a one = a
  one_mul : ? a, mul one a = a
  add_mul_distrib : ? a b c, mul (add a b) c = add (mul a c) (mul b c)

variable {alpha : Type} (F : ProbField alpha)

structure BooleanDistribution (alpha : Type) (F : ProbField alpha) where
  p_false : alpha
  p_true  : alpha
  normalized : F.add p_false p_true = F.one

def IsBitFlipInvariant (d : BooleanDistribution alpha F) : Prop :=
  d.p_false = d.p_true

/--
THEOREM 9.7: Bit-Flip Permutation Invariance Uniquely Determines the Prior Q? = 1/2
-/
theorem permutation_invariance_uniquely_determines_prior
    (d : BooleanDistribution alpha F) (h_sym : IsBitFlipInvariant F d) :
    d.p_false = F.half ? d.p_true = F.half := by
  have h_norm := d.normalized
  dsimp [IsBitFlipInvariant] at h_sym
  have h_two_p_false : F.mul F.two d.p_false = F.one := by
    calc
      F.mul F.two d.p_false
        = F.mul (F.add F.one F.one) d.p_false := by rw [F.two_eq_one_plus_one]
      _ = F.add (F.mul F.one d.p_false) (F.mul F.one d.p_false) := by rw [F.add_mul_distrib]
      _ = F.add d.p_false d.p_false := by rw [F.one_mul]
      _ = F.add d.p_false d.p_true := by rw [h_sym]
      _ = F.one := h_norm
  have h_two_p_true : F.mul F.two d.p_true = F.one := by
    calc
      F.mul F.two d.p_true
        = F.mul (F.add F.one F.one) d.p_true := by rw [F.two_eq_one_plus_one]
      _ = F.add (F.mul F.one d.p_true) (F.mul F.one d.p_true) := by rw [F.add_mul_distrib]
      _ = F.add d.p_true d.p_true := by rw [F.one_mul]
      _ = F.add d.p_false d.p_true := by rw [<- h_sym]
      _ = F.one := h_norm
  constructor
  - calc
      d.p_false = F.mul F.one d.p_false := by rw [F.one_mul]
```

```
      _ = F.mul (F.mul F.half F.two) d.p_false := by rw [F.half_mul_two]
      _ = F.mul F.half (F.mul F.two d.p_false) := by rw [F.mul_assoc]
      _ = F.mul F.half F.one := by rw [h_two_p_false]
      _ = F.half := by rw [F.mul_one]
  - calc
      d.p_true = F.mul F.one d.p_true := by rw [F.one_mul]
      _ = F.mul (F.mul F.half F.two) d.p_true := by rw [F.half_mul_two]
      _ = F.mul F.half (F.mul F.two d.p_true) := by rw [F.mul_assoc]
      _ = F.mul F.half F.one := by rw [h_two_p_true]
      _ = F.half := by rw [F.mul_one]

def boundary_decision_channels_per_vertex : Nat := 2

def triad_interaction_ports (num_vertices : Nat) : Nat :=
  num_vertices * boundary_decision_channels_per_vertex

/--
THEOREM 9.4: Triad Interaction Boundary is Exactly Six Routing Ports
-/
theorem triad_interaction_ports_is_six :
    triad_interaction_ports 3 = 6 := by
  rfl

/--
THEOREM 9.5: Simplicial Permittivity Microstate Space
-/
theorem simplicial_permittivity_scale :
    2 ^ (triad_interaction_ports 3) = 64 := by
  rfl
```

Verification Summary:

The Lean block proves the bit-flip prior identity under its explicit algebraic field assumptions and evaluates the declared decision counts. Its decision-channel definition stipulates two variables per vertex; the arithmetic does not derive that convention from external edge incidence. The resulting 64 assignments have equal mass only when the independent fair law is supplied. Interpreting their probability as a continuum drive also needs the reference attempt clock. The block contains no admitted proof or custom axiom declaration, but its structures carry explicit premises. Lean kernel acceptance certifies the conditional prior identity (**Information Modulus & Prior Uniqueness**).

4.4.Z Implications and Synthesis

Information-Theoretic Foundations

Bit-flip symmetry and normalization determine the unbiased binary prior, while logarithmic base conversion supplies the bit-to-nat relation (**Information Modulus & Prior Uniqueness**). The local path-count convention gives the closure entropy ratio (**Entropy of Closure**). These information identities do not independently determine every physical rate.

Equal allocation of the stipulated total energy gives the three-channel scale (**Dimensional Equipartition**). Six independent fair decisions supply the alignment probability (**Geometric Self-Energy**). Affine interpolation at the prescribed endpoints gives the catalytic coefficient (**Catalysis Coefficient**).

The continuum Gaussian reference and exact lattice peak differ by the explicitly bounded theta correction (**Friction Coefficient**). Together these relations define the conditional reference coordinates

(**Information-Theoretic Foundations**). Using them in a graph scheduler or scalar closure retains their constitutive premises; the coordinate calculation alone is not a stability or phase-survival theorem.

4.5 Universal Constructor

We confront the operational necessity of designing a Universal Constructor that can execute topological rewrites while strictly respecting the axioms of causality. Defining an algorithm that mutates the graph requires transforming abstract entropic potential into a concrete mechanical sequence of edge additions and deletions. We must specify a deterministic update mechanism that evaluates the local configuration of the graph to generate a weighted distribution of valid successor states without violating the logical consistency of the timeline.

A constructor that acts randomly without filtering for paradoxes would immediately generate closed timelike curves and destroy the causal order of the universe. If we allowed every energetically favorable transition to occur the graph would quickly become riddled with logical contradictions that render the concept of a consistent history impossible. Furthermore a constructor that operates without thermodynamic modulation would fail to regulate the density of the graph and lead to a catastrophe where the universe collapses into a singularity of infinite connectivity. A mechanism that cannot balance the drive for creation with the necessity of consistency cannot produce a stable spacetime.

We solve this operational challenge by defining the Universal Constructor $\mathcal{R}$ as a multi-stage engine that scans for compliant sites and validates them against the Acyclic Effective Causality constraint and weights them according to their thermodynamic costs. By employing a scan-validate-weight cycle we ensure that every proposed change is both physically motivated and logically sound. This mechanism acts as a biased pump that draws structure from the vacuum and filters the raw potential of the graph through a sieve of thermodynamic and logical constraints to ensure that only robust geometries propagate forward.

4.5.1 Definition: Universal Constructor

Algorithmic Implementation of the Rewrite Rule $\mathcal{R}$ by Thermodynamic Modulation

The **Universal Constructor** $\mathcal{R}$ is defined as a stochastic map $\mathcal{R} : \mathbf{AnnCG} \to \mathcal{P}(\mathbf{CG})$ that transforms an annotated graph (G, σ) into a probability distribution over potential successor states. The constructor operates via a strictly defined sequence of **Scanning**, **Validation**, and **Weighting**, formally implemented by the following algorithm: **(Gillespie, 1977)**

```
def R(annotated_graph, T, mu, lambda_cat):
    r"""
    Takes an annotated graph T(G) = (G, \sigma) and returns a
    probability distribution over successor graphs $\mathbb{P}$(G_t+1).
    Constants T, mu, lambda_cat derived in the thermodynamic parameters section (Sec.4.4).
    """
    # --- 1. SCAN & FILTER (The "Brakes") ---
    # Find all PUC-compliant 2-paths (for Addition) and 3-cycles (for Deletion)
    compliant_2_paths = _find_compliant_sites(G)
    existing_3_cycles = _find_all_3_cycles(G)

    add_proposals = []
    del_proposals = []

    # --- 2. VALIDATE & CALCULATE PROBABILITIES (Engine + Friction) ---

    # A) Process all ADD proposals (Generative Drive)
```

```
    for (v, w, u) in compliant_2_paths:
        proposed_edge = (u, v)

        # A.1) The AEC Pre-Check (Axiom 3 "Brake")
        # Deterministically reject paradoxes before probability calculation
        if not pre_check_aec(G, proposed_edge):
            continue

        # A.2) The Thermodynamic "Engine"
        # Base probability is 1.0 (Barrierless Creation at Criticality)
        P_thermo_add = 1.0

        # A.3) The "Friction" (Modulation by Local Stress)
        stress = measure_local_stress(G, {v, w, u})
        f_friction = exp(-mu * stress)

        # The full probability for this single event
        P_acc = f_friction * P_thermo_add

        # Assign Monotonic Timestamp
        H_new = 1 + max([H[e] for e in G.in_edges(u)] or [0])
        add_proposals.append( (proposed_edge, H_new, P_acc) )

    # B) Process all DELETE proposals (Entropic Balance)
    for cycle in existing_3_cycles:
        # B.1) The Thermodynamic "Engine"
        # Base probability is 0.5 (Entropic Penalty of Erasure)
        P_del_thermo = 0.5

        # B.2) The "Catalysis" (Modulation by Tension)
        # Stress *excluding* this cycle's own contribution
        stress = measure_local_stress(G, cycle.nodes) - 1
        f_catalysis = (1 + lambda_cat * max(0, stress))

        # The full probability for this single event
        P_del = min(1.0, f_catalysis * P_del_thermo)
        del_proposals.append( (cycle, P_del) )

    # --- 3. RETURN THE PROBABILITY DISTRIBUTION ---
    # The output is the ensemble of weighted proposals.
    # The realization (sampling/collapse) occurs in the Evolution Operator U (Sec.4.6).
    return (add_proposals, del_proposals)
```

This implementation adheres to the Micro/Macro separation principle, operating exclusively on local variables with universal constants derived in **Thermodynamic Foundations** .

4.5.1.1 Commentary: Algorithmic Agency

Stochastic Partitioning of Proposal and Realization in Constructor Dynamics

Designing the Universal Constructor $\mathcal{R}$ to decouple proposal generation from stochastic collapse is essential for maintaining causal integrity across the pre-geometric graph. By separating the mechanical enumeration of candidate graph rewrites from their physical realization, the theory places the origin of thermodynamic irreversibility strictly within the state sampling step executed by the evolution operator $\mathcal{U}$.

Furthermore, the candidate proposal search space enforces strict local radius bounds of $O(1)$ centered around

active vertices, bounded by the Acyclicity Evaluation Check horizon cutoff $L_{\text{cut}} = \lfloor \log_2 N \rfloor + 3$ under parent uniqueness. This local restriction guarantees computational scalability while ensuring physical realism, microcausality, structural integrity, and strict spatial locality across the entire relational substrate.

This operational pipeline operates within a dual time architecture distinguishing global logical scheduling time $t_L \in \mathbb{N}_0$ from emergent physical proper time t_{phys}. When an execution tick produces no accepted additions and no accepted deletions ($\mathcal{A}_{t_L} = \emptyset \wedge \mathcal{D}_{t_L} = \emptyset$), the evolution operator acts as the identity map ($\mathcal{U}(G_{t_L}) = G_{t_L}$), and emergent physical time pauses ($\Delta t_{\text{phys}} \equiv 0$). In finite-lattice simulations, the operational homeostatic stopping time $\tau_{\text{homeo}} := \inf\{t_L \geq 1 \mid \mathcal{A}_{t_L} = \emptyset \wedge \mathcal{D}_{t_L} = \emptyset\}$ halts iteration at the physical stasis point, isolating the true Quasi-Stationary Distribution before open leaf boundary dissipation takes effect.

4.5.2 Definition: Catalytic Tension Factor

Syndrome-Response Function Modulating Base Probabilities via Catalytic Tension Factor

The **Catalytic Tension Factor**, denoted $\chi(\sigma_e)$, is defined as the scalar modulation function acting on the base transition probabilities. It is constructed as the product of two distinct terms:

$$\chi(\sigma_e) = \underbrace{\left(\prod_{s \in \mathcal{S}_{\text{sites},e}} (1 + \lambda_{\text{cat}} \cdot I[\Delta s(e) = +2]) \right)}_{\text{Catalysis Term}} \cdot \underbrace{\exp\left(-\mu \cdot \sum_{x \in \text{nbhd}(e)} I[\sigma_x = -1] \right)}_{\text{Friction Term}}$$

1. **Catalysis Term:** The product over the set of local sites where the proposed action resolves a syndrome excitation ($\Delta s = +2$). This term applies a linear scaling factor of $(1 + \lambda_{cat})$ for every resolved defect.
2. **Friction Term:** The exponential decay function of the total local stress, defined as the count of negative syndromes ($\sigma_x = -1$) within the immediate neighborhood nbhd(e). This term applies a damping factor with coefficient μ.

4.5.2.1 Commentary: Adaptive Feedback

Interpretation of Catalysis and Friction

The Catalytic Tension Factor serves as the critical interface between the Awareness Layer (diagnosis) and the Action Layer (dynamics). It transforms abstract diagnostic data (syndrome tuples) into concrete kinetic bias. The duality of this function, additive catalysis for relief and exponential friction for caution, embeds a sophisticated negative feedback loop directly into the micro-physics of the vacuum.

Consider the physical implications: High stress (indicated by negative syndromes) catalyzes deletions via the mode-specific application of λ_{cat}, effectively accelerating the decay of unstable structures. Simultaneously, friction curbs additions in these same dense regions, preventing the system from adding fuel to the fire. By explicitly separating these terms, the theory allows the universe to navigate the "Goldilocks zone" of density. It prevents both runaway crystallization (the Small World catastrophe where every point connects to every other) and total dissolution (where structure evaporates faster than it can form). This function is the thermostat of the cosmos.

4.5.3 Definition: Addition Mode

Constructive Operation Proposing Edge Additions via Addition Mode

The **Addition Mode** is defined as the constructive operation of the Action Layer, operating on a set of compliant **2-Path** structures. It generates a set of tuples `(proposed_edge, H_new, P_acc)`, where P_{acc} is the friction-damped probability derived from the **Catalytic Tension Factor** .

4.5.3.1 Commentary: Generative Drive

Bias Toward Complexity

Addition is the default drive of the system: the "inertial" tendency of the vacuum. Because the base probability is unity ($\mathbb{P} \to 1$) at the critical temperature, the vacuum naturally and aggressively seeks to close open paths. This "generative drive" is an intrinsic consequence of the bit-nat equivalence ($T = \ln 2$).

Crucially, the generative drive of edge additions is strictly audited by the Acyclic Effective Causality (AEC) pre-check, which acts as the absolute guardian of the timeline. The pre-check deterministically rejects any proposed edge that would close a causal loop, elevating the arrow of time to a hard, logical constraint rather than a statistical average. This gatekeeping mechanism introduces a fundamental physical asymmetry: while edge additions must undergo verification to prevent retroactive paradoxes, edge deletions require no such check. Removing connections can never introduce cycles or closed loops. The asymmetry between constructing and dismantling structure establishes a non-trivial directionality in the evolution of spacetime, demonstrating that thermodynamic irreversibility is deeply intertwined with logical consistency.

4.5.4 Definition: Deletion Mode

Destructive Operation Proposing Edge Removals via Deletion Mode

The **Deletion Mode** is defined as the destructive operation of the Action Layer, acting on directed 3-cycles governed by the **Geometric Quantum** . It generates a set of tuples `(target_edge, P_del)`, where P_{del} is the catalysis-boosted probability derived from the **Catalytic Tension Factor** .

4.5.4.1 Commentary: Pruning and Balance

Prevention of the Small World Catastrophe

Without the counterbalancing process of structural edge deletion, the uninhibited generative drive would relentlessly saturate the graph with adjacencies until it collapsed into a complete graph (K_N). This runaway crystallization would destroy all spatial locality, topological metrics, and dimensional distinctions across the substrate. Deletion mode provides the essential pruning mechanism required to maintain network sparsity, metric stability, and low-complexity vacuum baselines.

Crucially, the deletion operator targets existing geometric **3-cycles** rather than removing arbitrary spatial edges. This geometric specificity ensures that the substrate dissolves area quanta back into the vacuum without severing vital causal pathways or leaving orphaned vertices. Targeted simplicial dissolution maintains the topological integrity of the emergent manifold while regulating geometric density, analogous to cellular apoptosis preserving biological form in living tissues.

4.5.5 Theorem: Universal Constructor

Thermodynamic Transition Probabilities by Feedback Modulation of the Rewrite Map

Let $\mathcal{R}$ denote the Universal Constructor stochastically mapping annotated graphs. Then the base thermodynamic acceptance probability is $\mathbb{P}_{\text{acc,thermo}} = 1$ for edge addition and $\mathbb{P}_{\text{del,thermo}} = 1/2$ for edge deletion; moreover, the local rewrite rates are modulated by the Catalytic Tension Factor.

4.5.5.1 Commentary: Argument Outline

Structure of the Universal Constructor Argument via Addition Probability and Deletion Probability

The proof proceeds via Direct Construction, demonstrating that the base transition probabilities are established by entropic dominance and local stress feedback.

```
o 4.5.5 Theorem Universal Constructor  [by construction]
|
+-- 4.5.6 Lemma: Addition Probability
|   +-- 4.5.6.1 Proof: Addition Probability
|   +-- 4.5.6.2 Commentary: Generative Drive
|
+-- 4.5.7 Lemma: Deletion Probability
|   +-- 4.5.7.1 Proof: Deletion Probability
|   +-- 4.5.7.2 Commentary: Detailed Balance
|
+-- 4.5.8 Proof: Universal Constructor
|
+-- 4.5.9 Validation: Lean 4 Core
```

4.5.6 Lemma: Addition Probability

Unitary Thermodynamic Acceptance Probability via Edge Creation

Let $\mathbb{P}_{\text{acc,thermo}}$ denote the base thermodynamic acceptance probability for edge creation in the critical vacuum regime under the barrierless free energy condition of **Information Modulus & Prior Uniqueness** . Then $\mathbb{P}_{\text{acc,thermo}}$ is identically equal to 1.

4.5.6.1 Proof: Addition Probability

Derivation of Barrierless Addition from Free Energy Minimization

I. Thermodynamic Factor and Barrierless Condition

Let $T > 0$ and let the thermodynamic acceptance factor be

$$P_{thermo} = \min\{1, e^{-\Delta F/T}\}, \qquad \Delta F = \Delta E - T\Delta S$$

in the supplied kernel (**Addition Mode**). For the bit-creation baseline, $\Delta S = \ln 2$ (**Information Modulus & Prior Uniqueness**). Therefore

$$-\frac{\Delta F}{T} = \ln 2 - \frac{\Delta E}{T}, \qquad P_{thermo} = \min\{1, 2e^{-\Delta E/T}\}.$$

II. Exact Saturation Range

The barrierless condition is $\Delta F \le 0$. Since $T > 0$, it is equivalent to

$$\Delta E - T\ln 2 \le 0 \iff \frac{\Delta E}{T} \le \ln 2.$$

In that range,

$$\ln 2 - \frac{\Delta E}{T} \ge 0 \implies e^{-\Delta F/T} \ge 1 \implies P_{thermo} = 1.$$

In particular, the declared $\Delta E = 0$ baseline gives $e^{-\Delta F/T} = e^{\ln 2} = 2$ and hence $P_{thermo} = 1$ exactly, without a large-volume limit.

III. The Stated Channel-Energy Examples

For the supplied microscopic channel value $\varepsilon_{geo} = \ln 2/3$ and $T_c = \ln 2$ (**Dimensional Equipartition**),

$$\frac{\varepsilon_{geo}}{T_c} = \frac{1}{3}.$$

The logarithm satisfies

$$\ln 2 = \int_1^2 \frac{\mathrm{d}x}{x} > \int_1^2 \frac{1}{2}\,\mathrm{d}x = \frac{1}{2} > \frac{1}{3}.$$

Therefore $\Delta F/T_c = 1/3 - \ln 2 < 0$ and

$$P_{thermo} = \min\{1, e^{\ln 2 - 1/3}\} = 1.$$

For the separately stated quarter-unit example, $\Delta E/T_c = 1/4 < 1/2 < \ln 2$. Therefore

$$-\Delta F/T_c = \ln 2 - \tfrac{1}{4} > 0, \qquad P_{thermo} = \min\{1, e^{\ln 2 - 1/4}\} = 1.$$

Both conclusions are conditional on their supplied energy normalizations.

IV. Scope of the Factor

Writing the full admissible-proposal acceptance as $P_{acc} = \chi(\sigma) P_{thermo}$ gives, within the barrierless range,

$$P_{acc} = \chi(\sigma) \cdot 1 = \chi(\sigma).$$

The exact claim about the thermodynamic factor is consequently (**Addition Probability**)

$$\boxed{\Delta F \le 0, \quad T > 0 \implies P_{thermo} = 1.}$$

Q.E.D.

4.5.6.2 Commentary: Generative Drive

Interpretation of Unitary Creation Probability in Graph Expansion

We show that the base probability for edge creation is unity at the critical temperature. This generative drive represents the fundamental bias of the vacuum toward establishing new relations, establishing an arrow of structural growth. The cost of instantiating a new relation is exactly balanced by the entropic gain of the new configuration.

Crucially, this drive is audited by the Acyclic Effective Causality (AEC) pre-check, which serves as the absolute guardian of the timeline. The pre-check rejects any proposed edge that would close a causal loop, elevating the arrow of time to a hard, logical constraint rather than a statistical average. This gatekeeping mechanism introduces a fundamental physical asymmetry: while edge additions must undergo verification to prevent retroactive paradoxes, edge deletions require no such check, creating a non-trivial directionality in the evolution of spacetime.

Unity here is the saturated thermodynamic factor under the stated barrierless condition. It is not a statement that every candidate is legal, that every site is sampled or that the stress-modulated probability is one. The comparison with the channel-energy examples consumes their proposed normalizations; it does not derive the energy scales or extend saturation to arbitrary energy barriers. This distinction keeps the arithmetic acceptance result separate from the physical arguments determining the available proposals.

4.5.7 Lemma: Deletion Probability

Half-unit thermodynamic deletion probability via Deletion Probability

Let $\mathbb{P}_{\text{del,thermo}}$ denote the base thermodynamic deletion probability for geometric quanta in the critical vacuum regime. Then $\mathbb{P}_{\text{del,thermo}}$ is identically equal to $1/2$ (**Entropy of Closure**).

4.5.7.1 Proof: Deletion Probability

Limit Evaluation via Entropic Dominance

I. Entropic Baseline

The base deletion convention uses $T > 0$, $\Delta E = 0$ and the loss of one bit, $\Delta S = -\ln 2$ (**Information Modulus & Prior Uniqueness**), with the bit-loss input supplied by the closure-entropy model (**Entropy of Closure**). The thermodynamic rule gives

$$\begin{aligned} \Delta F_{base} &= 0 - T(-\ln 2) = T \ln 2, \\ -\Delta F_{base}/T &= -\ln 2, \\ Q_{base} &= \min\{1, e^{-\ln 2}\} = \min\{1, \tfrac{1}{2}\} = \tfrac{1}{2}. \end{aligned}$$

II. Separation from a Nonzero Released Energy

For a released internal energy $\varepsilon \geq 0$, the energy change is instead $\Delta E = -\varepsilon$. Keeping the same entropy change gives

$$\begin{aligned} \Delta F(\varepsilon) &= -\varepsilon + T \ln 2, \\ -\Delta F(\varepsilon)/T &= \frac{\varepsilon}{T} - \ln 2, \\ Q(\varepsilon) &= \min\{1, \tfrac{1}{2} e^{\varepsilon/T}\}. \end{aligned}$$

The cases are therefore

$$Q(\varepsilon) = \begin{cases} \frac{1}{2}, & \varepsilon = 0, \\ \frac{1}{2} e^{\varepsilon/T} \in (\frac{1}{2}, 1), & 0 < \varepsilon < T \ln 2, \\ 1, & \varepsilon \geq T \ln 2. \end{cases}$$

For the supplied channel energy at $T_c = \ln 2$, the ratio is $(\ln 2/3)/\ln 2 = 1/3 < \ln 2$. The unclipped value is therefore $Q = \frac{1}{2} e^{1/3}$. The first five terms of the exponential series give

$$\sum_{j=0}^{4} \frac{(1/3)^j}{j!} = 1 + \frac{1}{3} + \frac{1}{18} + \frac{1}{162} + \frac{1}{1944} = \frac{2713}{1944}.$$

The first omitted term is $1/29160$. Each following term is at most $1/18$ times the preceding term, so

$$\frac{2713}{3888} < Q \leq \frac{2713}{3888} + \frac{1}{2 \cdot 29160} \frac{1}{1 - 1/18}.$$

These endpoints lie between 0.69775 and 0.69785, so $Q \approx 0.6978$ to four decimal places.

This is the nonzero-energy correction, rather than the base factor.

III. Limit and Baseline Identity

If a family of energy normalizations satisfies $\varepsilon/T \to 0$, continuity of the exponential gives

$$\lim_{\varepsilon/T \to 0} Q(\varepsilon) = \min\{1, \tfrac{1}{2}e^0\} = \tfrac{1}{2}.$$

The base value itself requires no limiting argument. Together with the base addition value, it satisfies

$$P_{base} = 1, \qquad Q_{base} = \tfrac{1}{2}, \qquad \frac{Q_{base}}{P_{base}} = \frac{1}{2}.$$

Thus the specified thermodynamic deletion baseline is exactly (**Deletion Probability**)

$$\boxed{Q^{base}_{\text{del,thermo}} = \tfrac{1}{2} \qquad (T > 0,\ \Delta E = 0,\ \Delta S = -\ln 2).}$$

Q.E.D.

4.5.7.2 Commentary: Detailed Balance

Engine of Growth

The fundamental asymmetry between Addition (1.0) and Deletion (0.5) constitutes the thermodynamic engine of the universe. It creates a net flow towards structure, a "pressure" to evolve. The universe builds twice as fast as it decays provided the local stress is low.

Equilibrium is only reached when the friction from rising density (μ) suppresses the addition rate enough to match the deletions or when catalysis (λ_{cat}) boosts the deletion rate to match the additions. This dynamic balance defines the emergent geometry. The "shape" of space is effectively the surface where these two opposing forces, the drive to connect and the drive to simplify, reach a standoff. This is why the universe is not a static crystal but a dynamic foam, constantly seething with creation and destruction even at equilibrium.

The factor one half is the declared purely entropic baseline. A nonzero released energy changes the thermodynamic acceptance even at fixed critical temperature, as the explicit expression shows. Increasing graph size does not by itself send that local energy to zero; a system-size limit needs a specified energy normalization. Finally, a ratio of two acceptance factors alone is not detailed balance for a full graph transition kernel unless the forward and reverse proposal probabilities and state weights have also been accounted for.

4.5.8 Proof: Universal Constructor

Synthesis of Transition Probabilities via Feedback Loops in Constructor Dynamics

I. Stochastic Update Map

Let the annotated graph (G, σ) evolve stochastically under the constructor map $\mathcal{R}$. The transition probabilities decompose into a base thermodynamic factor and a local syndrome-response factor.

II. Base Probability Calibration

The base thermodynamic probabilities are calibrated at the critical vacuum temperature. Edge additions occur barrierless with unitary probability $\mathbb{P}_{\text{acc,thermo}} = 1$ according to **Addition Probability** . Edge deletions face an entropic barrier, yielding a half-unit probability $\mathbb{P}_{\text{del,thermo}} = 1/2$ according to **Deletion Probability** .

Writing p_{bare} and q_{bare} for the two thermodynamic baselines, the zero addition barrier and one-bit deletion cost give

$$p_{\text{bare}} = e^0 = 1,$$
$$q_{\text{bare}} = e^{-\log 2} = \frac{1}{e^{\log 2}} = \frac{1}{2}.$$

III. Dynamic Modulation

The base probabilities are modulated by the Catalytic Tension Factor defined in **Catalytic Tension Factor** . Adding edges is damped exponentially by local stress, whereas deleting edges is catalyzed linearly by syndrome resolution.

Let s_a be the stress of an eligible addition and s_c the stress of an old triangle deletion trial. Multiplying the baselines by the stress factors and clipping the deletion trial probability at one gives

$$\begin{aligned} p_a &= p_{\text{bare}} e^{-\mu s_a} = e^{-\mu s_a}, \\ \tilde{q}_c &= q_{\text{bare}}(1+\lambda s_c)e^{-\mu s_c} = \tfrac{1}{2}(1+\lambda s_c)e^{-\mu s_c}, \\ q_c &= \min\{1, \tilde{q}_c\}. \end{aligned}$$

For nonnegative stress and parameters, $-\mu s_a \le 0$ and $1+\lambda s_c \ge 1$. Therefore

$$0 < p_a \le 1, \qquad 0 \le \tilde{q}_c, \qquad 0 \le q_c \le 1.$$

The addition outcome probabilities sum to $(1-p_a)+p_a = 1$. Under the uniform edge choice in **Evolution Operator** , the old triangle trial assigns probability $1-q_c$ to no trigger and probability $q_c/3$ to each edge choice. Explicitly,

$$\Pr(\text{choose } e_j) = \Pr(\text{trigger})\Pr(\text{choose } e_j \mid \text{trigger}) = q_c \cdot \frac{1}{3} = \frac{q_c}{3}, \qquad j \in \{1,2,3\},$$
$$(1-q_c) + \sum_{j=1}^{3} \frac{q_c}{3} = (1-q_c) + q_c = 1.$$

At zero stress these factors recover the baselines,

$$p_a\big|_{s_a=0} = 1 \cdot e^0 = 1, \qquad q_c\big|_{s_c=0} = \min\{1, \tfrac{1}{2}(1+\lambda\cdot 0)e^0\} = \tfrac{1}{2}.$$

IV. Microscopic Rate-Inversion Threshold

For a comparison at common stress $s = s_a = s_c \ge 0$ and positive μ, λ, the addition probability is positive. Dividing the clipped deletion probability by it gives

$$\frac{q_c(s)}{p_a(s)} = \frac{\min\{1, \frac{1}{2}(1+\lambda s)e^{-\mu s}\}}{e^{-\mu s}} = \min\left\{e^{\mu s}, \frac{1+\lambda s}{2}\right\}.$$

The second term equals one exactly when $1+\lambda s = 2$, or $s = s_* = 1/\lambda$. The first term is at least one for $s \ge 0$ and strictly greater than one for $s > 0$. Therefore

stress	$(1+\lambda s)/2$	channel comparison
$0 \le s < s_*$	< 1	$q_c(s) < p_a(s)$
$s = s_*$	$= 1$	$q_c(s) = p_a(s)$
$s > s_*$	> 1	$q_c(s) > p_a(s)$

In the last row both arguments of the minimum exceed one, so the strict inequality also holds when the deletion probability is clipped. At the canonical $\lambda_0 = e - 1$ from **Catalysis Coefficient** , the series

$e = 1 + 1 + \sum_{j\geq 2} 1/j! > 2$ gives $0 < s_* < 1$. In particular the equal-stress comparison at the isolated triad's stress two is

$$\frac{q_c(2)}{p_a(2)} = \min\{e^{2\mu_0}, \tfrac{1}{2} + \lambda_0\} = \min\{e^{2/\sqrt{2\pi}}, e - \tfrac{1}{2}\} > 1.$$

Q.E.D.

4.5.9 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Isolated Triad Self-Stress and Deletion Probability

Type-theoretic certification of the constitutive deletion self-stress functional established in **Universal Constructor** and **Deletion Probability** proceeds via the following verification strategy:

1. **Canonical Isolated Triad Topology:** The structure `TriadGraph` and definition `canonical_isolated_3cycle` encode the directed 3-cycle topology $0 \to 1 \to 2 \to 0$ over a 3-vertex carrier.
2. **Topological Vertex Participation:** The function `vertex_cycle_participation` evaluates the participation of each constituent vertex in closed directed cycles, and theorem `canonical_triad_vertex_participation` proves that every vertex in an isolated 3-cycle participates in exactly one cycle.
3. **Self-Stress Derivation:** The function `topological_deletion_stress` implements the constitutive functional $s_{\text{del}} = (\sum_x |\mathcal{C}_3(x)|) - 1$, and theorem `isolated_3cycle_self_stress_eq_two` certifies by definitional reduction (`rfl`) that an isolated 3-cycle yields exactly $s_{\text{del}} = 2$, evaluating the deletion probability to $Q_{\text{del}}(2) \approx 0.999$.

```
def TriadGraph := Fin 3 -> Fin 3 -> Bool

def canonical_isolated_3cycle : TriadGraph :=
  fun u v =>
    match u.1, v.1 with
    | 0, 1 => true
    | 1, 2 => true
    | 2, 0 => true
    | _, _ => false

def vertex_cycle_participation (G : TriadGraph) (_v : Fin 3) : Nat :=
  if G <0, by omega> <1, by omega> && G <1, by omega> <2, by omega> && G <2, by omega> <0, by omega> th

theorem canonical_triad_vertex_participation (v : Fin 3) :
    vertex_cycle_participation canonical_isolated_3cycle v = 1 := by
  rfl

def topological_deletion_stress (G : TriadGraph) : Nat :=
  (vertex_cycle_participation G <0, by omega> +
   vertex_cycle_participation G <1, by omega> +
   vertex_cycle_participation G <2, by omega>) - 1

/--
THEOREM 8.2: Topological Derivation of Isolated Cycle Self-Stress
Formally derives that for any canonical isolated 3-cycle graph, the constitutive
deletion self-stress functional s_del evaluates to exactly 2 from graph topology.
-/
theorem isolated_3cycle_self_stress_eq_two :
```

```
  topological_deletion_stress canonical_isolated_3cycle = 2 := by
rfl
```

Verification Summary: The formal proofs certified in Lean 4 demonstrate that the constitutive deletion self-stress functional evaluates strictly to $s_{\mathrm{del}} = (1+1+1)-1 = 2$ from raw graph topology with zero empirical parameters (`isolated_3cycle_self_stress_eq_two`). Under the canonical reference priors $(\lambda_0, \mu_0) = (e-1, 1/\sqrt{2\pi})$, this guarantees that isolated single-cycle excitations face a deletion probability $Q_{\mathrm{del}}(2) \approx 0.99885$, ensuring that unassisted topological fluctuations rapidly quench into the absorbing vacuum state.

4.5.Z Implications and Synthesis

Action Layer

The operationalization of the thermodynamic mandates into a concrete algorithm is achieved via the **Universal Constructor** . The action layer functions as a biased, self-regulating pump that draws compliant paths from the vacuum and crystallizes them into geometry with a base probability of unity, proposing additions via the **Addition Mode** , while dissolving existing structures via the **Deletion Mode** with a probability of one-half. This fundamental asymmetry drives the arrow of complexity, while the Catalytic Tension Factor provides the necessary brakes and accelerators to navigate the phase transition without collapsing into chaos.

This mechanism produces a distribution of potential futures, separating the proposal of change (governed by the stochastic constructor $\mathcal{R}$) from its realization (governed by the evolution operator $\mathcal{U}$). This separation is crucial, as it locates the source of physical irreversibility in the eventual collapse of this distribution rather than in the mechanical generation of options. Furthermore, the search space for proposals enforces strict locality, focusing modifications on neighborhoods of radius $O(1)$ centered around active vertices to maintain computational scalability and physical realism. By filtering this localized raw potential through a sieve of logical and thermodynamic constraints, the constructor ensures that only robust geometries propagate forward. The crossover at common stress gives the microscopic rate comparison. The graph-level balance uses the actual candidate opportunities and triangle losses (**Conditional Triangle Drift**). The scalar equilibrium follows under its own closure (**Global Stability**).

The operationalization of the rewrite rule as a stochastic process governed by local stress completes the microscopic definition of the dynamics. It establishes the universe as a computational engine that actively seeks to maximize its internal complexity while minimizing logical contradictions. This biased random walk through the configuration space of graphs is the microscopic origin of the macroscopic laws of evolution, driving the system inevitably toward the geometric phase where matter and spacetime can emerge.

4.6 Single Tick of Logical Time

We face the final dynamical problem of defining the tick of logical time as an irreversible physical event that locks the present into the past. We must integrate the distinct processes of awareness, proposal, merge, and deletion into a unified operator that advances the state of the universe by one discrete step to ensure the continuity of existence. We are forced to describe the process that selects a realized trajectory from the branching space of potential successor configurations without relying on an external observer.

A universe that preserves all possible branching trajectories without selection fails to manifest a concrete causal history and leaves the specific evolution of the cosmos undefined. If the branching tree of candidate graph updates is never sampled, history remains an abstract ensemble of unrealized alternatives and the macroscopic thermodynamic arrow of time cannot emerge from discrete graph transitions. Treating time as a continuous flow obscures the discrete computational nature of the underlying process and fails to account for the generation of entropy associated with the reduction of possibilities. Without a mechanism to irrevocably commit to a specific path, the universe would lack a definite past and a determinate future.

We resolve this by defining the evolution operator $\mathcal{U}$ as the four-step composition of awareness diagnostic mapping, parallel stochastic proposal, symmetric reciprocal addition merge, and intermediate deletion excision. This operator enforces the constitutive rules of the substrate by evaluating local stress diagnostics and stochastically sampling candidate topological rewrites. This cycle generates the thermodynamic arrow of time through the information loss inherent in stochastic selection collapse and monotonic historical accumulation (**Hist**), ensuring that the universe evolves as a distinct, race-free, and irreversible sequence of events.

4.6.1 Definition: Evolution Operator

Composition of Awareness, Proposal, Addition Merge, and Deletion Excision as the Logical Tick

Let the **Evolution Operator** be the Markov kernel induced by a deterministic four-stage realization map $\mathcal{U}(G, \xi)$ on the complete active graph state, including its current heights. All proposals and their probabilities are evaluated against the same pre-tick snapshot.

1. **Awareness:** Enumerate legal addition sites $\mathcal{A}(G)$, directed triangles $\mathcal{C}_3(G)$, and vertex cycle incidences $s_G(v)$. The addition stress is the sum over its three vertices; a triangle's deletion stress is $\max(0, \sum_{v \in c} s_G(v) - 1)$.
2. **Independent selection:** For each addition site choose $X_a \in \{0, 1\}$ with probability $p_a = e^{-\mu s_a}$. For each triangle choose $J_c \in \{\perp\} \cup E(c)$ with probabilities $\Pr(J_c = \perp) = 1 - q_c$ and $\Pr(J_c = e) = q_c/3$, where $q_c = \min(1, \frac{1}{2}(1 + \lambda s_c)e^{-\mu s_c})$.
3. **Symmetric merge:** Let $B = \{e_a : X_a = 1\}$ and retain $B_{\text{filt}} = \{(u, v) \in B : u \neq v,\ (v, u) \notin B\}$. Insert these edges with source-side heights from the pre-tick snapshot and record their birth occurrences.
4. **Deletion purge:** Let $D = \{J_c : J_c \neq \perp\}$. Remove D from the intermediate active edge set. Multiple selections of the same edge merge by set union; they do not create repeated deletions or erase historical records.

The complete realization is $\xi = (X_{\mathcal{A}}, J_{\mathcal{C}})$. The graph kernel is the pushforward of its normalized product measure (**Euclidean Transition Measure**). The raw reciprocal filter rejects both accepted directions, rather than choosing an order-dependent winner.

4.6.1.1 Commentary: Anatomy of the Tick

Decomposition separating the logical stages of time evolution into distinct physical roles

The tick of logical time represents a structured, non-unitary physical cycle composed of four distinct operational stages. Each stage fulfills a necessary, specialized role in advancing the relational state of the universe while preserving background independence, causal consistency, and thermodynamic stability across the network.

- **Awareness (Diagnostic Sensing):** This step transforms the static topology into a self-referential state. By computing $\mathcal{C}_3(G_t)$ and mapping local stress counts stress_map(x), the rewrite engine guarantees that subsequent dynamics are driven entirely by the graph's internal diagnostics rather than arbitrary external coordinates.
- **Proposal (Stochastic Sampling):** This step explores candidate updates via independent Bernoulli trials. By evaluating addition probabilities P_{acc} and catalytic deletion probabilities Q_{del} across disjoint topological sites, the engine samples from the physical state space without violating local move independence.
- **Merge (Idempotent Addition):** This step applies a symmetric reciprocal filter A_{filt} to eliminate bidirectional edge hazards and merges accepted directed chords into the intermediate graph G'. Because $A_{\text{edges}} \cap E(G_t) = \emptyset$, additions introduce strictly new edges without mutating pre-existing edges.
- **Deletion (Deterministic Excision):** This step excises the accepted deletion set $D \subseteq E(G_t)$ from the intermediate graph G'. Because additions merge in Step 3 before deletions act in Step 4, and

because $A_{\text{edges}} \cap D = \emptyset$, no edge added in tick t can be removed in tick t, guaranteeing race-free parallel execution.

4.6.2 Theorem: Emergent Dynamics

Emergence of Classical Transition Probabilities and Entropic Arrow from the Evolution Operator

Let a finite-horizon graph process use the four-stage realization map and its product sampling law (**Evolution Operator**). Then its transition kernel is the pushforward of that law, its full-trajectory Shannon entropy is nondecreasing, and KL divergence contracts when two state distributions evolve under the same kernel.

4.6.2.1 Commentary: Argument Outline

Structure of the Emergent Dynamics Argument via Classical Product-Rule Weights and the Thermodynamic Arrow of Time

The proof proceeds via Direct Construction, synthesizing the local independence of rewrite events with the information-theoretic irreversibility of sampling collapse and cumulative historical inclusion.

```
o 4.6.2 Theorem Emergent Dynamics  [by construction]
|
+-- 4.6.3 Lemma: Euclidean Transition Measure
|   +-- 4.6.3.1 Proof: Euclidean Transition Measure
|   +-- 4.6.3.2 Calculation: Euclidean Action Integration
|   +-- 4.6.3.3 Commentary: Thermodynamic Origin of the Modulus
|
+-- 4.6.4 Lemma: Thermodynamic Arrow
|   +-- 4.6.4.1 Proof: Thermodynamic Arrow
|   +-- 4.6.4.2 Commentary: Macroscopic Irreversibility
|   +-- 4.6.4.3 Calculation: Irreversibility Check
|   +-- 4.6.4.4 Diagram: Thermodynamic Arrow
|
+-- 4.6.5 Lemma: Conditional Triangle Drift
|   +-- 4.6.5.1 Proof: Conditional Triangle Drift
|   +-- 4.6.5.2 Calculation: Degree-Three Drift Calculation
|   +-- 4.6.5.3 Commentary: Foundation for the Continuum Limit
|
+-- 4.6.6 Lemma: Exact Unpumped Quiescence
|   +-- 4.6.6.1 Proof: Exact Unpumped Quiescence
|   +-- 4.6.6.2 Calculation: Reciprocal-Aware Quiescence Fixtures
|   +-- 4.6.6.3 Commentary: Quiescence and Candidate Cancellation
|
+-- 4.6.7 Lemma: Finite-Horizon Height Bound
|   +-- 4.6.7.1 Proof: Finite-Horizon Height Bound
|   +-- 4.6.7.2 Commentary: State Variables on a Finite Horizon
|
+-- 4.6.8 Lemma: Common-Height Shift Equivariance
|   +-- 4.6.8.1 Proof: Common-Height Shift Equivariance
|   +-- 4.6.8.2 Commentary: Relative Heights and Finite Reduction
|
+-- 4.6.9 Proof: Emergent Dynamics
|
+-- 4.6.10 Validation: Lean 4 Core
```

4.6.3 Lemma: Euclidean Transition Measure

Emergence of Path Integral Weighting from Markovian Transition Probabilities

Let ξ comprise the independent addition and cycle-deletion outcomes (**Evolution Operator**). Then $K(G, G') = \sum_{\xi:\mathcal{U}(G,\xi)=G'} \Pr(\xi \mid G)$ is normalized, and every positive realization probability has the additive negative-log action $-\log \Pr(\xi \mid G)$.

4.6.3.1 Proof: Euclidean Transition Measure

Derivation of the Exponential Action Functional from Local Probabilities

I. Local Outcome Normalization

The proof uses the specified objects and conventions (**Evolution Operator**).

For each addition, $\Pr(X_a = 1) = p_a$ and $\Pr(X_a = 0) = 1-p_a$, with sum one. Each triangle has three distinct edges and

$$\Pr(J_c = \perp) + \sum_{e \in E(c)} \Pr(J_c = e) = (1 - q_c) + 3(q_c/3) = 1.$$

II. Product Realization Measure

For a specified realization, let $A = \{a : X_a = 1\}$ and $C_D = \{c : J_c \neq \perp\}$. Independence gives

$$\Pr(\xi \mid G) = \prod_{a \in A} p_a \prod_{a \notin A} (1 - p_a) \prod_{c \in C_D} \frac{q_c}{3} \prod_{c \notin C_D} (1 - q_c).$$

The indices run through the fixed candidate sets of the input snapshot. Distributivity of the finite sums yields

$$\sum_{\xi} \Pr(\xi \mid G) = \prod_{a} [p_a + (1 - p_a)] \prod_{c} [(1 - q_c) + 3q_c/3] = 1.$$

III. Pushforward and Collision Preimages

The deterministic map assigns each realization exactly one successor. The preimage sets $F_{G'} = \{\xi : \mathcal{U}(G, \xi) = G'\}$ are disjoint and cover the realization space. Hence

$$\sum_{G'} K(G, G') = \sum_{G'} \sum_{\xi \in F_{G'}} \Pr(\xi \mid G) = \sum_{\xi} \Pr(\xi \mid G) = 1.$$

Shared deletion edges and reciprocal filtering can make $|F_{G'}| > 1$; the summation therefore cannot generally be replaced by one product.

IV. Realization Action

For a positive-probability realization, logarithmic additivity gives

$$\begin{aligned} \mathcal{S}(\xi \mid G) &= -\log \Pr(\xi \mid G) \\ &= -\sum_{a \in A} \log p_a - \sum_{a \notin A} \log(1 - p_a) \\ &\quad - \sum_{c \in C_D} \log(q_c/3) - \sum_{c \notin C_D} \log(1 - q_c). \end{aligned}$$

Thus $\Pr(\xi \mid G) = e^{-\mathcal{S}(\xi|G)}$ and $K(G, G') = \sum_{\xi \in F_{G'}} e^{-\mathcal{S}(\xi|G)}$. The accepted-addition contribution alone is $-\sum_{a \in A} \log p_a = \mu \sum_{a \in A} s_a$ (**Euclidean Transition Measure**).
Q.E.D.

4.6.3.2 Calculation: Euclidean Action Integration

Computational Verification of the Exponential Action Scaling Relation through Euclidean Action Integration

Computational verification of the complete realization law established in (**Euclidean Transition Measure**) and the fixed-snapshot selection rule (**Evolution Operator**) is based on the following protocols:

1. **Exact Outcome Weights:** Rational probabilities enumerate accepted and rejected addition outcomes and four outcomes per triangle.
2. **Pushforward Aggregation:** Deterministic reciprocal filtering and deletion merging aggregate every realization by its successor edge set.
3. **Normalization and Collision Checks:** The audit compares exact total mass, isolated-edge probability and shared-edge deletion preimages.

```
from fractions import Fraction as F
from itertools import product

def graph_kernel(edges, additions, triangles):
    # Rate parameters are supplied to audit the finite sampling law.
    factors = []
    for edge, p in additions:
        factors.append([(None, 1-p), (edge, p)])
    for cycle, q in triangles:
        factors.append([(None, 1-q)] + [(edge, q/3) for edge in cycle])
    law = {}
    count = {}
    for outcome in product(*factors):
        weight = F(1)
        for _, probability in outcome:
            weight *= probability
        if weight == 0:
            continue
        accepted = {edge for edge, _ in outcome[:len(additions)] if edge is not None}
        filtered = {edge for edge in accepted if edge[::-1] not in accepted and edge[0] != edge[1]}
        deleted = {edge for edge, _ in outcome[len(additions):] if edge is not None}
        successor = frozenset((set(edges) | filtered) - deleted)
        law[successor] = law.get(successor, F(0)) + weight
        count[successor] = count.get(successor, 0) + 1
    assert sum(law.values()) == 1
    return law, count

triangle = ((0,1),(1,2),(2,0))
law, _ = graph_kernel(triangle, [], [(triangle,F(3,5))])
specific = frozenset(set(triangle)-{(0,1)})
assert law[specific] == F(1,5)
print('Isolated triangle: total mass = 1; specified-edge deletion =', law[specific])

other = ((0,1),(1,3),(3,0))
edges = set(triangle) | set(other)
law, preimages = graph_kernel(edges, [], [(triangle,F(3,5)),(other,F(1,2))])
```

```
shared = frozenset(edges-{(0,1)})
assert preimages[shared] == 3
print('Shared triangles: total mass = 1; shared-edge-only preimages =', preimages[shared])

# Abstract reciprocal merge fixture: this checks merging, not proposal admission.
law, preimages = graph_kernel([], [((0,1),F(1,3)),((1,0),F(2,5))], [])
assert preimages[frozenset()] == 2
print('Reciprocal merge fixture: total mass = 1; unchanged-state preimages =', preimages[frozenset()])
```

Simulation Results:

```
Isolated triangle: total mass = 1; specified-edge deletion = 1/5
Shared triangles: total mass = 1; shared-edge-only preimages = 3
Reciprocal merge fixture: total mass = 1; unchanged-state preimages = 2
```

Conclusion: All three exact finite laws have total probability one. The isolated triangle gives probability one fifth for a specified deletion edge, including the uniform choice factor. Shared-edge and reciprocal fixtures expose three and two realization preimages respectively. These values validate the normalization and pushforward identities; proposal admission is outside the abstract reciprocal fixture.

4.6.3.3 Commentary: Thermodynamic Origin of the Modulus

Distinguishing Classical Transition Measures from Quantum Interference

The action here is the negative logarithm of a complete classical realization probability. Rejected trials and the uniform edge choice are part of that probability. A successor graph can have multiple realization preimages, so its effective negative-log weight is a logarithm of a sum rather than the action of one arbitrarily selected update set. This positive Markov measure is distinct from a unitary amplitude or a Born-rule derivation.

Candidate-relative PUC implies that distinct legal addition sites have distinct directed target chords: equal targets fix the two endpoints, and the uniqueness clause forces equality of the intermediate vertices. This eliminates duplicate directed-addition preimages at the candidate stage. It does not eliminate reciprocal-filter or shared-deletion preimages at the full scheduler stage. The full kernel remains normalized even when these collision preimages are numerous, because they partition the already normalized realization space.

4.6.4 Lemma: Thermodynamic Arrow

Irreversibility from entropy production in the evolution operator

Let $G_0, \dots, G_{t+1}$ be a Markov trajectory with finite entropies and kernel K, and let p, q be two distributions on the finite-horizon state class propagated by that same kernel. Then $H(G_0, \dots, G_{t+1}) - H(G_0, \dots, G_t) = H(G_{t+1} \mid G_t) \geq 0$ and $D_{\mathrm{KL}}(pK \parallel qK) \leq D_{\mathrm{KL}}(p \parallel q)$ (**Evolution Operator**).

4.6.4.1 Proof: Thermodynamic Arrow

Decomposition into Non-invertible Components via Thermodynamic Arrow

I. Chain Rule on Full Trajectories

The proof uses the specified objects and conventions (**Evolution Operator**).

For fixed vertex count and horizon T, every active edge is absent or has a height in $\{0, \dots, T\}$, giving at most $(T+2)^{N(N-1)}$ labeled configurations. The distributions below are on this finite class. Write $\Gamma_t = (G_0, \dots, G_t)$. For each trajectory and successor of positive probability,

$$\Pr(\Gamma_t, g') = \Pr(\Gamma_t) K(G_t, g'), \qquad \log \Pr(\Gamma_t, g') = \log \Pr(\Gamma_t) + \log K(G_t, g').$$

Multiply by the joint probability, sum, and use $\sum_{g'} K(G_t, g') = 1$. The result is

$$H(\Gamma_{t+1}) = H(\Gamma_t) - \sum_g p_t(g) \sum_{g'} K(g, g') \log K(g, g').$$

The final sum with its minus sign is $H(G_{t+1} \mid G_t)$. Every summand $-K \log K$ is nonnegative for $0 \le K \le 1$, using $0 \log 0 = 0$. It is positive exactly when a positive-mass current state has a nondegenerate successor distribution.

II. Log-Sum Inequality

For nonnegative finite families a_i, b_i with positive sums A, B, put $\alpha_i = a_i/A$ and $\beta_i = b_i/B$. If some $a_i > 0 = b_i$, the divergence is infinite and the inequality is immediate. Otherwise, $-\log z \ge 1 - z$ gives

$$\begin{aligned} \sum_i a_i \log \frac{a_i}{b_i} - A \log \frac{A}{B} &= A \sum_{i:\alpha_i>0} \alpha_i \log \frac{\alpha_i}{\beta_i} \\ &\ge A \sum_{i:\alpha_i>0} (\alpha_i - \beta_i) \\ &= A \left(1 - \sum_{i:\alpha_i>0} \beta_i \right) \ge 0. \end{aligned}$$

III. Common-Kernel Contraction

At each output y, apply the inequality to $a_x = p(x)K(x, y)$ and $b_x = q(x)K(x, y)$. Summing over y gives

$$\begin{aligned} D_{\mathrm{KL}}(pK \parallel qK) &\le \sum_{x,y} p(x)K(x, y) \log \frac{p(x)K(x, y)}{q(x)K(x, y)} \\ &= \sum_x p(x) \log \frac{p(x)}{q(x)} \sum_y K(x, y) \\ &= D_{\mathrm{KL}}(p \parallel q). \end{aligned}$$

Zero-probability terms have their usual limiting values. If the input divergence is infinite, contraction is understood as an extended-value inequality. This proves both asserted information laws (**Thermodynamic Arrow**).

Q.E.D.

4.6.4.2 Commentary: Macroscopic Irreversibility

Thermodynamic Arrow as the Result of Stochastic Selection and Historical Accumulation

Trajectory entropy measures uncertainty in a sequence of graph states rather than uncertainty in its final member alone. Appending the next state adds a nonnegative conditional entropy, while concentration toward an absorbing class can reduce the instantaneous snapshot entropy. The two statements are compatible. Common-kernel KL contraction likewise compares two propagated ensembles and can hold with equality. Its physical interpretation depends on what the comparison ensemble represents, not just on the existence of a stochastic update.

The creation record and full trajectory have distinct information content unless a reconstruction map is supplied. A trajectory can encode rejected trials only through their visible effects, and different realizations can have the same successor graph. Discarding those random details does not by itself identify two predecessor graphs or establish thermodynamic dissipation. A physical entropy-production law needs its forward and

reverse processes and its environment accounting. The proved path and relative-entropy identities provide exact information-theoretic statements while keeping that additional thermodynamic identification explicit.

4.6.4.3 Calculation: Irreversibility Check

Computational Verification of Shannon Entropy Loss via Stochastic Selection

Computational verification of the trajectory and common-kernel information inequalities established in (**Thermodynamic Arrow**) uses the normalized Markov law (**Euclidean Transition Measure**) through the following protocols:

1. **Distribution Construction:** A binary stochastic kernel propagates two strictly positive input distributions.
2. **Entropy Accounting:** The audit evaluates initial entropy, final entropy and the conditional contribution to joint trajectory entropy separately.
3. **Relative Entropy Comparison:** The propagated divergence is compared with the input divergence, while an explicit concentration example tests decreasing snapshot entropy.

```
import math

def entropy(p):
    return -sum(x*math.log(x) for x in p if x > 0)

def push(p, K):
    return [sum(p[i]*K[i][j] for i in range(len(p))) for j in range(len(K[0]))]

def kl(p,q):
    return sum(a*math.log(a/b) for a,b in zip(p,q) if a>0)

K = [[0.9,0.1],[0.9,0.1]]
p = [0.5,0.5]
q = [0.8,0.2]
after = push(p,K)
conditional = sum(p[i]*entropy(K[i]) for i in range(2))
joint = entropy([p[i]*K[i][j] for i in range(2) for j in range(2)])
assert abs(joint-entropy(p)-conditional)<1e-12
assert entropy(after)<entropy(p)
assert joint>=entropy(p)
assert kl(push(p,K),push(q,K))<=kl(p,q)+1e-12
print(f'Snapshot entropy change: {entropy(after)-entropy(p):.9f} nats')
print(f'Full trajectory entropy increment: {joint-entropy(p):.9f} nats')
print(f'Conditional entropy: {conditional:.9f} nats')
print(f'KL input / output: {kl(p,q):.9f} / {kl(push(p,K),push(q,K)):.9f}')
```

Simulation Results:

```
Snapshot entropy change: -0.368064207 nats
Full trajectory entropy increment: 0.325082973 nats
Conditional entropy: 0.325082973 nats
KL input / output: 0.223143551 / -0.000000000
```

Conclusion: The concentration example decreases snapshot entropy by approximately 0.368 nats while increasing full trajectory entropy by approximately 0.325 nats. The latter equals the conditional entropy to numerical precision. KL divergence contracts from approximately 0.223 nats to zero. These values validate the separated information identities rather than an inequality for snapshot entropy production.

4.6.4.4 Diagram: Thermodynamic Arrow

Visualization of Irreversibility via Information Loss in Stochastic Selection and History Accumulation

```
Why the evolution cycle cannot be inverted
------------------------------------------

FORWARD (t -> t+1):
Stochastic Bernoulli sampling and idempotent chord merging
map many candidate branch trajectories to the same successor graph.

   Branch_A (2^M proposals) --\
                               \
   Branch_B ------------------> Realized State G_{t+1}
                               /
   Branch_C ------------------/

REVERSE (t+1 -> t):
Given only active routing G_{t+1}, which predecessor state and unselected
alternative branches produced it?

   G_{t+1}  ---->  ??? (Cannot reconstruct G_t without external record)

RESULT: Unselected alternatives are permanently discarded;
        cumulative history Hist grows monotonically.
        Entropy increases (DeltaS_tick >= 0). Time is directed.
```

4.6.5 Lemma: Conditional Triangle Drift

Exact Conditional Triangle Drift via Frozen Trial Probabilities

Let G be a fixed-vertex snapshot of the frozen unpumped tick, with old triangles $\mathcal{C}_0$, eligible additions $\mathcal{A}$, and triangles $\mathcal{T}$ in the full candidate supergraph, where p_e is the selection probability of e with value zero outside $\mathcal{A}$, q_C is the deletion trigger probability of $C \in \mathcal{C}_0$, ν_e counts members of $\mathcal{T}$ containing e, and $U(G)$ counts eligible chords of selection probability one with ineligible reverses and both supporting edges outside every old triangle. Then the conditional triangle drift satisfies the exact identity and bounds

$$a_T = \prod_{e \in E(T) \setminus E(G)} p_e(1 - p_{\bar{e}}), \qquad d_T = \prod_{C \in \mathcal{C}_0} \left(1 - \frac{q_C}{3} |E(C) \cap E(T)|\right),$$

$$\mathbb{E}[\Delta N_3 \mid G] = \sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} a_T d_T - \sum_{T \in \mathcal{C}_0} (1 - d_T),$$

$$U(G) - |\mathcal{C}_0| \leq \mathbb{E}[\Delta N_3 \mid G] \leq \sum_{e \in \mathcal{A}} p_e \nu_e - \sum_{C \in \mathcal{C}_0} q_C,$$

where $\bar{e}$ is the reverse of e.

4.6.5.1 Proof: Conditional Triangle Drift

Triangle Presence via Reciprocal Filtering and Categorical Deletion Products

I. New and Lost Triangles

The tick uses frozen candidate selection and old-triangle excision (**Conditional Triangle Drift**) (**Evolution Operator**).

Let $A \subseteq \mathcal{A}$ be the retained batch and $D \subseteq E(G)$ the excisions. The final edge set is $(E(G) \cup A) \setminus D$. If $\mathcal{C}_3(G)$ and $\mathcal{C}_3(F)$ are the old and final triangle sets, respectively, then

$$N_3(F) - N_3(G) = |\mathcal{C}_3(F) \setminus \mathcal{C}_3(G)| - |\mathcal{C}_3(G) \setminus \mathcal{C}_3(F)|.$$

II. Probability of Triangle Presence

For $e \notin E(G)$, let X_e be the indicator that e was selected, with $X_e = 0$ if $e \notin \mathcal{A}$. For any $T \in \mathcal{T}$, each missing edge must survive reciprocal filtering. The sets $\{e, \bar{e}\}$ for different edges of a directed triangle are pairwise disjoint. The conditional selection law (**Euclidean Transition Measure**) gives

$$\begin{aligned}\Pr(E(T) \setminus E(G) \subseteq A \mid G) &= \Pr\left(\bigcap_{e \in E(T) \setminus E(G)} \{X_e = 1,\ X_{\bar{e}} = 0\} \,\middle|\, G\right) \\ &= \prod_{e \in E(T) \setminus E(G)} p_e(1 - p_{\bar{e}}) = a_T.\end{aligned}$$

Let J_C be the outcome of the deletion trial for C, with $J_C = \emptyset$ for no trigger. With $m_{C,T} = |E(C) \cap E(T)|$, the trial spares T with probability

$$\Pr(J_C \notin E(T) \mid G) = (1 - q_C) + (3 - m_{C,T})\frac{q_C}{3} = 1 - \frac{q_C m_{C,T}}{3}.$$

The deletion union meets $E(T)$ exactly when at least one trial chooses an edge of T. Independence across deletion trials and between the selection and deletion families gives

$$\begin{aligned}\Pr(D \cap E(T) = \emptyset \mid G) &= \prod_{C \in \mathcal{C}_0} \Pr(J_C \notin E(T) \mid G) = d_T, \\ \Pr(T \in \mathcal{C}_3(F) \mid G) &= \Pr(E(T) \setminus E(G) \subseteq A,\ D \cap E(T) = \emptyset \mid G) \\ &= a_T d_T.\end{aligned}$$

III. Exact Drift and Sufficient Sign Conditions

For $T \in \mathcal{C}_0$ the empty product gives $a_T = 1$. Taking expectations of the triangle counts in Part I gives

$$\begin{aligned}\mathbb{E}[\Delta N_3 \mid G] &= \sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} \Pr(T \in \mathcal{C}_3(F) \mid G) - \sum_{T \in \mathcal{C}_0} \Pr(T \notin \mathcal{C}_3(F) \mid G) \\ &= \sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} a_T d_T - \sum_{T \in \mathcal{C}_0} (1 - d_T).\end{aligned}$$

A chord counted by $U(G)$ specifies a triangle with exactly one missing edge. Distinct such chords specify distinct triangles because the missing edge identifies the chord. Its addition factor is $1(1 - 0) = 1$. Its old support edges lie outside every old triangle, so $m_{C,T} = 0$ for all C and $d_T = 1$. Thus

$$\sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} a_T d_T \geq U(G), \qquad \sum_{T \in \mathcal{C}_0} (1 - d_T) \leq |\mathcal{C}_0|.$$

For each new T, every missing edge e gives $a_T d_T \leq p_e$. Summing over all missing edges and interchanging the sums yields

$$\sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} a_T d_T \leq \sum_{T \in \mathcal{T} \setminus \mathcal{C}_0} \sum_{e \in E(T) \setminus E(G)} p_e = \sum_{e \in \mathcal{A}} p_e \nu_e.$$

For each old T the factor with $C = T$ is $1-q_T$. All other factors belong to $[0,1]$, so $d_T \le 1-q_T$. Substituting gives

$$\begin{aligned} U(G) - |\mathcal{C}_0| \le \mathbb{E}[\Delta N_3 \mid G] &\le \sum_{e\in\mathcal{A}} p_e\nu_e - \sum_{C\in\mathcal{C}_0} q_C, \\ U(G) > |\mathcal{C}_0| &\implies \mathbb{E}[\Delta N_3 \mid G] > 0, \\ \sum_{e\in\mathcal{A}} p_e\nu_e < \sum_{C\in\mathcal{C}_0} q_C &\implies \mathbb{E}[\Delta N_3 \mid G] < 0. \end{aligned}$$

For fixed $N > 0$, the conditional density drift is

$$\mathbb{E}[\Delta(N_3/N) \mid G] = N^{-1}\mathbb{E}[\Delta N_3 \mid G].$$

IV. Degree and Stress Envelope

If the old undirected degree is at most Δ, each new neighbor is at old distance two. For each of at most Δ first neighbors, at most $\Delta - 1$ second neighbors differ from the starting vertex. The merged undirected degree is bounded by

$$\deg_{\text{merged}}(v) \le \Delta + \Delta(\Delta - 1) = \Delta^2.$$

If $|\mathcal{A}| \le M$, $s_e \ge s_*$, $\nu_e \le \nu_*$, and $q_C \ge q_*$, then for $\mu \ge 0$,

$$\begin{aligned} \mathbb{E}[\Delta N_3 \mid G] &\le \sum_{e\in\mathcal{A}} \nu_e e^{-\mu s_e} - \sum_{C\in\mathcal{C}_0} q_C \\ &\le |\mathcal{A}|\nu_* e^{-\mu s_*} - |\mathcal{C}_0| q_* \\ &\le M\nu_* e^{-\mu s_*} - |\mathcal{C}_0| q_*. \end{aligned}$$

V. Exact Drift in the Degree-Three Fixture

The old edge heights are $H(a,b) = H(b,c) = H(c,d) = 0$ and $H(c,a) = 1$. The complete two-path and return-edge lists are

support	return edge	old membership
(a,b,c)	(c,a)	present
(b,c,a)	(a,b)	present
(c,a,b)	(b,c)	present
(b,c,d)	(d,b)	absent

The candidate (d,b) has height $1+H(c,d) = 1$. The simple reverse route $b \to c \to d$ has height pair $(0,0)$ and fails $0 < 0$. The route through $c \to a$ can only continue to b, repeating the start. Thus the reverse-witness test passes. The old triangle counts at a, b, c, d are $1, 1, 1, 0$, giving

$$s_{\text{add}}(d,b) = 1 + 1 + 0 = 2, \qquad s_{\text{del}}(a,b,c) = 1 + 1 + 1 - 1 = 2.$$

The respective probabilities follow from (**Addition Probability**) (**Deletion Probability**)

$$p = e^{-2\mu_0}, \qquad q = \min\{1, \tfrac{1}{2}(1 + 2\lambda_0)e^{-2\mu_0}\}.$$

The old triangle is lost with probability q. The new triangle survives when its candidate is selected and edge (b,c) is kept, which has probability $p(1 - q/3)$ by independence and the uniform deletion choice. The exact triangle drift is therefore

$$\mathbb{E}[\Delta N_3 \mid G] = p(1-q/3) - q \le p - q < 0$$

at the canonical parameters, since $\frac{1}{2}(1+2\lambda_0) > 1$ and $p < 1$.

VI. Exact Drift at Breadth-First Ignition

Label the vacuum tree in breadth-first order from root 0, with three children at the root and two per subsequent parent up to vertex $N-1$ (**Regular Bethe Fragment**). All tree edges have height zero. Let G_1 be the tree after adding $(4,0)$ of height one, so $\mathcal{C}_0 = \{C_0\}$ with $C_0 = (0,1,4)$. The path $(0,1,4)$ already has its closing edge. The two-paths $(1,4,0)$ and $(4,0,1)$ also have their closing edges. For $N \ge 11$ the remaining candidate supports are the other outward tree two-paths and the two paths $(4,0,2)$ and $(4,0,3)$.

An outward tree two-path ending at u has a unique tree parent of u as intermediate vertex. The added edge $(4,0)$ cannot supply a second intermediate vertex on such a path. For each path $(4,0,j)$ with $j \in \{2,3\}$, the only edge entering j is $(0,j)$, so the support is unique. Each missing closing edge has a source of maximum incoming height zero, giving proposed height one. A strictly increasing reverse path ending below height one would consist of a single height-zero edge. The reverse edge is absent for each of these proposals. Thus, writing $\ell(v)$ for the original tree depth,

$$\begin{aligned} \#\{\text{outward tree two-paths}\} &= \#\{u : \ell(u) \ge 2\} = N-4, \\ \#\{\text{eligible outward tree closures}\} &= (N-4) - 1 = N-5, \\ |\mathcal{A}| &= (N-5) + 2 = N-3. \end{aligned}$$

Tree edges and the two added edges into 4 have depth increment $+1$. The ignition edge and all other proposals have increment -2. In a triangle with k edges of increment -2, the sum of increments is $(3-k)-2k = 3-3k$, giving $k = 1$. If a triangle uses a cross-branch addition $(j,4)$, its other advancing edge either enters j from root 0 or leaves 4 for one of its children. In the first case the retreat is the old edge $(4,0)$. In the second case the retreat from the child of 4 goes to its grandparent 1, which differs from j. A triangle without a cross-branch addition has two old tree edges and its single retreat. Thus each new triangle uses exactly one proposal. The reverse increments -1 and $+2$ are absent, so each reciprocal factor equals one.

The triangle vertices $0, 1, 4$ each have one old incidence. The table groups all proposals of positive stress once their endpoint labels are present.

s_e	supporting paths	$\lvert E(T) \cap E(C_0)\rvert$
1	$(0,2,6),\ (0,2,7),\ (0,3,8),\ (0,3,9)$	0
1	$(1,5,12),\ (1,5,13)$	0
1	$(4,10,22),\ (4,10,23),\ (4,11,24),\ (4,11,25)$	0
2	$(0,1,5),\ (1,4,10),\ (1,4,11)$	1
2	$(4,0,2),\ (4,0,3)$	1

For $N \ge 26$ the table gives ten stress-one and five stress-two proposals. The other $(N-3)-10-5 = N-18$ proposals have stress zero. With $a = e^{-\mu_0}$, the old trigger probability and the drift are

$$\begin{aligned} s_{\mathrm{del}}(C_0) &= 1+1+1-1 = 2, \\ q &= \min\{1, \tfrac{1}{2}(1+2\lambda_0)a^2\}, \\ \mathbb{E}[\Delta N_3 \mid G_1] &= (N-18)\cdot 1 + 10a\cdot 1 + 5a^2(1-q/3) - q \\ &= N - 10.786755982\ldots. \end{aligned}$$

For $11 \le N < 26$, the four paths in the first row are present and each contributes a expected births. The old loss is $q \le 1$. Since $\mu_0 = 1/\sqrt{2\pi} < 1/2$,

$$a = e^{-\mu_0} \geq 1 - \mu_0 > \frac{1}{2}, \qquad \mathbb{E}[\Delta N_3 \mid G_1] \geq 4a - q > 1 > 0.$$

Q.E.D.

4.6.5.2 Calculation: Degree-Three Drift Calculation

Foster-Lyapunov Drift Verification via Exact Event Enumeration of the Degree-Three Drift Fixture

Verification of the degree-three triangle drift fixture (**Conditional Triangle Drift**) (**Evolution Operator**) is based on the following protocols:

1. **Fixture:** Use the directed triangle with one outgoing tail and its single closing proposal.
2. **Outcome Law:** Evaluate the canonical addition and old-triangle deletion probabilities.
3. **Drift:** Sum the changes over the two addition outcomes and four deletion outcomes.

```
from itertools import product
from math import e, exp, isclose, pi, sqrt

mu, lam = 1 / sqrt(2 * pi), e - 1
p = exp(-2 * mu)
q = min(1, (1 + 2 * lam) * exp(-2 * mu) / 2)
old_triangle = {("a", "b"), ("b", "c"), ("c", "a")}
new_triangle = {("b", "c"), ("c", "d"), ("d", "b")}
old = old_triangle | {("c", "d")}
delete_choices = [(None, 1 - q)] + [(edge, q / 3) for edge in sorted(old_triangle)]
drift = 0.0
for (add, add_weight), (edge, del_weight) in product(
    [(False, 1 - p), (True, p)], delete_choices
):
    final = (old | ({("d", "b")} if add else set())) - {edge}
    change = int(old_triangle <= final) + int(new_triangle <= final) - 1
    drift += add_weight * del_weight * change

assert isclose(drift, p * (1 - q / 3) - q)
assert drift <= p - q < 0
print(f"Enumerated one-tick triangle drift: {drift:.8f}")
print(f"Site-summed upper bound: {p - q:.8f}")
```

Simulation Results:

```
Enumerated one-tick triangle drift: -0.69848930
Site-summed upper bound: -0.54856855
```

Conclusion:

The enumerated drift is negative and lies below the site-summed bound. The fixture confirms the analytic one-tick value and inequality on this snapshot.

4.6.5.3 Commentary: Foundation for the Continuum Limit

One-Tick Drift and Long-Time State Laws

The exact product separates candidate supply from survival of the supports. An eligible chord with no stress or reciprocal competitor contributes a certain birth when its supporting edges are protected. When several deletion trials target the same edge, the categorical product gives its combined effect on each triangle. The

site sum still provides a useful upper bound because it counts every possible new triangle containing an eligible chord.

The ignition tree supplies a positive-drift example with protected supports. The degree-three fixture gives the opposite sign when an old triangle contributes a larger loss than the available birth. Extending either sign to a region of reachable states requires uniform control of those opportunities and losses. A driven scalar equilibrium describes the stipulated mean-field closure, and identifying it with a microscopic stationary law requires an estimate for the driven kernel.

4.6.6 Lemma: Exact Unpumped Quiescence

Exact Unpumped Quiescence via Fixed Points of the Finite Unpumped Snapshot Kernel

Let G be a finite active snapshot for the canonical unpumped tick. Then $K(G, G) = 1$ exactly when there are no old triangles and every eligible addition candidate has an eligible reciprocal partner.

4.6.6.1 Proof: Exact Unpumped Quiescence

Exact Unpumped Quiescence via Positive Deletion Trials and Reciprocal Filtering

I. One-Tick Outcomes

In the canonical unpumped tick, every retained addition belongs to an absent ordered pair of G, and the deletion union contains only old edges selected by old triangles (**Evolution Operator**). Write $\mathcal{C}$ for the old triangles and A for the eligible additions. If D is the deletion union and A_{ret} the retained additions, the final edge set is

$$E' = (E \cup A_{\mathrm{ret}}) \setminus D, \qquad A_{\mathrm{ret}} \cap E = \emptyset, \quad D \subseteq E.$$

Thus $E' = E$ holds exactly when both A_{ret} and D are empty, in which case all active heights are also unchanged.

II. Positive Deletion Event

If $c \in \mathcal{C}$, its canonical trigger weight is

$$q_c = \min\{1, \tfrac{1}{2}(1 + \lambda_0 s_c)e^{-\mu_0 s_c}\} > 0,$$

because the stress is finite and nonnegative and $\lambda_0, \mu_0 > 0$. If this trial triggers, one of its old edges belongs to D, regardless of the outcomes of other trials. Consequently

$$K(G, G) = \Pr(E' = E \mid G) \leq 1 - q_c < 1.$$

III. Triangle-Free Addition Outcomes

If $\mathcal{C} = \emptyset$, then $D = \emptyset$ and each candidate has zero stress. Its selection probability is $e^0 = 1$, so all of A is selected with probability one. The retained set is then deterministic,

$$A_{\mathrm{ret}} = \{(u, v) \in A : (v, u) \notin A\}.$$

Thus the output equals the input exactly when this set is empty. Together with Section II this gives the stated equivalence (**Exact Unpumped Quiescence**).

Q.E.D.

4.6.6.2 Calculation: Reciprocal-Aware Quiescence Fixtures

Reciprocal-Aware Quiescence Fixtures via Finite Quiescence Checks on Three Timestamped Snapshots

Verification of quiescence on the three explicit inputs (**Exact Unpumped Quiescence**) (**Evolution Operator**) is based on the following protocols:

1. **Fixtures:** Construct a four-cycle, a single two-path, and an old triangle.
2. **Admission:** Check candidate uniqueness and the bounded increasing reverse path.
3. **Kernel:** Evaluate reciprocal cancellation in the triangle-free fixtures and test the triangle deletion event.

```
from math import e, exp, log2, pi, sqrt

import networkx as nx

def candidates(graph):
    result = set()
    cutoff = int(log2(len(graph))) + 3
    for v in graph:
        for w in graph[v]:
            for u in graph[w]:
                if len({v, w, u}) < 3 or graph.has_edge(v, u) or graph.has_edge(u, v):
                    continue
                if any(z != w and graph.has_edge(z, u) for z in graph[v]):
                    continue
                height = 1 + max(data["height"] for _, _, data in graph.in_edges(u, data=True))
                blocked = False
                for path in nx.all_simple_paths(graph, v, u, cutoff=cutoff):
                    heights = [graph[a][b]["height"] for a, b in zip(path, path[1:])]
                    if (len(heights) >= 2 and heights[-1] < height
                            and all(a < b for a, b in zip(heights, heights[1:]))):
                        blocked = True
                        break
                if not blocked:
                    result.add((u, v))
    return result

for name, edges, count, expected in [
    ("four-cycle", [(0, 1, 0), (1, 2, 0), (2, 3, 0), (3, 0, 0)], 4, True),
    ("two-path", [(0, 1, 0), (1, 2, 0)], 1, False),
    ("triangle", [(0, 1, 0), (1, 2, 0), (2, 0, 1)], 0, False),
]:
    graph = nx.DiGraph()
    graph.add_weighted_edges_from(edges, weight="height")
    eligible = candidates(graph)
    assert len(eligible) == count
    retained = {(u, v) for u, v in eligible if (v, u) not in eligible}
    triangles = [cycle for cycle in nx.simple_cycles(graph) if len(cycle) == 3]
    positive_deletion = bool(triangles)
    if positive_deletion:
        q = min(1, (1 + 2 * (e - 1)) * exp(-2 / sqrt(2 * pi)) / 2)
        assert q > 0
    quiescent = not positive_deletion and not retained
    assert quiescent == expected
```

```
    print(f"{name}: {len(eligible)} eligible, quiescent={quiescent}")
```

Simulation Results:

```
four-cycle: 4 eligible, quiescent=True
two-path: 1 eligible, quiescent=False
triangle: 0 eligible, quiescent=False
```

Conclusion:

The four-cycle is quiescent, the path has a surviving addition, and the triangle has a positive deletion event. These outputs confirm the quiescence classification for all three fixtures.

4.6.6.3 Commentary: Quiescence and Candidate Cancellation

Quiescence and Candidate Cancellation

A cycle-free snapshot can still have eligible closing proposals, because its open two-paths can support chords. Zero stress makes those candidates certain to be selected in the unpumped rule. Whether their selection changes the graph depends on the reciprocal filter. In the four-cycle fixture every selected chord is canceled by its reciprocal partner.

An old triangle gives a separate way to leave a state. Its positive deletion chance can remove an edge that was already present, whereas the addition channel targets pairs absent from the frozen input. The criterion concerns the one-step transition kernel at a specified active snapshot. Reaching the set of quiescent states from an initial ensemble remains a question about the global dynamics.

4.6.7 Lemma: Finite-Horizon Height Bound

Finite Active State Space through a Fixed Horizon

Let the vertex set be finite and fixed, and the active heights be nonnegative integers. After t unpumped ticks their maximum is bounded by $H_{\max}(0)+t$, with the maximum of an empty graph taken as zero, so the reachable active snapshots through any fixed horizon form a finite set.

4.6.7.1 Proof: Finite-Horizon Height Bound

Finite-Horizon Height Bound via Induction on the Maximum Active Height

I. Domain and Objects

An admitted addition uses its incoming source maximum (**Finite-Horizon Height Bound**) (**Creation Timestamp**).

II. Single-Tick Maximum

Each retained chord has height one plus an incoming source maximum in the frozen snapshot, at most $H_{\max}(t)+1$. A retained old edge keeps its height and deletion cannot increase the maximum. Thus $H_{\max}(t+1) \le H_{\max}(t)+1$.

III. Finite Horizon

At $t=0$ the bound is equality. If $H_{\max}(t) \le H_{\max}(0)+t$, then

$$H_{\max}(t+1) \le H_{\max}(t)+1 \le H_{\max}(0)+t+1.$$

Hence the bound holds through the horizon T. For each of the P directed pairs, the possibilities are one absent state and $H_{\max}(0)+T+1$ present heights, including zero. Taking the product over pairs bounds the number of snapshots by

$$\left(H_{\max}(0) + T + 2\right)^P$$

active snapshots in that horizon.

Q.E.D.

4.6.7.2 Commentary: State Variables on a Finite Horizon

State Variables on a Finite Horizon

Equal adjacency can conceal different incoming height comparisons. For a fixed two-path, assigning equal heights to its edges prevents it from serving as a strict reverse witness, whereas assigning increasing heights can make it one. The active state therefore carries the heights as well as the present directed edges.

A bound on the highest active height bounds the number of assignments for each adjacency through a chosen horizon. The counting argument applies even when the same pair is deleted and recreated many times, since every new assignment grows at most one beyond the old maximum. The cumulative ledger is a different object. The bound concerns the active snapshot used by the transition kernel. The observation horizon can be fixed independently of the spatial vertex count.

4.6.8 Lemma: Common-Height Shift Equivariance

Common-Height Shift Equivariance via Common-Shift Quotient of the Active Unpumped Kernel

Let two unpumped snapshots agree in adjacency and differ by a common integer shift on active heights, with both assignments nonnegative. Their transition probabilities agree on shift classes, and the quotient is finite on any invariant class with fixed vertices and a uniform finite height span.

4.6.8.1 Proof: Common-Height Shift Equivariance

Coupling by Shared Tick Outcomes

I. Translation of the Eligibility Tests

Let $H' = H + c$ on the common active edge set E. For an eligible return chord $e = (u, v)$ the old incoming support at u is nonempty (**Common-Height Shift Equivariance**) (**Evolution Operator**). Writing h_e and h'_e for the two candidate heights gives

$$h'_e = 1 + \max_{(x,u)\in E}\big(H(x,u) + c\big) = 1 + \max_{(x,u)\in E} H(x,u) + c = h_e + c.$$

For an old witness with edge list $e_1, \dots, e_\ell$, the corresponding differences are

$$H'(e_{j+1}) - H'(e_j) = H(e_{j+1}) - H(e_j), \qquad h'_e - H'(e_\ell) = h_e - H(e_\ell).$$

Every strict comparison therefore has the same truth value. The two-path counts and triangle stresses are functions of E, so both inputs also have the same uniqueness tests and trial probabilities.

II. Equality on Output Classes

Use the same vector ω of trial outcomes for the two inputs. It has the same law by equality of the snapshot-conditioned probabilities. The reciprocal filter and deletion rule applied to this vector produce the same set E_ω in both cases. For a surviving old edge the heights differ by c by hypothesis. For a retained new edge they differ by c by Section I. Thus the outputs $F(E, H; \omega)$ and $F(E, H'; \omega)$ lie in the same shift class for every ω. For any shift class $\mathcal{C}$, this gives

$$\Pr(F(E,H;\omega)\in\mathcal{C}) = \mathbb{E}[\mathbf{1}_{\mathcal{C}}(F(E,H;\omega))] = \mathbb{E}[\mathbf{1}_{\mathcal{C}}(F(E,H';\omega))] = \Pr(F(E,H';\omega)\in\mathcal{C}).$$

III. Counting Representatives under a Span Bound

For nonempty E with span at most B, subtract $\min_{e\in E} H(e)$ to get a representative whose active heights lie in $\{0,\dots,B\}$. If m is the number of possible directed edges on the fixed vertex set, the number of quotient states is at most

$$1+\sum_{j=1}^{m}\binom{m}{j}(B+1)^j = (B+2)^m.$$

The term one counts the empty adjacency. Hence the invariant quotient is finite.

Q.E.D.

4.6.8.2 Commentary: Relative Heights and Finite Reduction

Relative Heights and Finite Reduction

A common shift preserves comparisons among active heights and the comparison with the height assigned to a new supported chord. The candidate source has an old incoming edge, so adding the same constant shifts the maximum used for the assignment. Using the same trial outcomes therefore couples the two updated snapshots modulo that shift.

Normalizing a representative by subtracting its smallest active height removes the absolute offset. Relative gaps still remain and can range over arbitrarily many integers without a preserved span bound. A proposed compression of those gaps must preserve transition probabilities on the proposed reduced states. The common-shift proof only identifies the equivalence classes for which the given coupling applies. An invariant bound would have to control the span after every possible retained batch.

4.6.9 Proof: Emergent Dynamics

Synthesis of Transition Probabilities via Entropy Production in the Evolution Cycle

I. Classical Transition Law

The independent finite outcome law is normalized, and its deterministic pushforward gives the normalized graph kernel (**Euclidean Transition Measure**). Positive realization weights have the additive negative-log action displayed there.

II. Information Monotonicity

The trajectory chain rule gives a nonnegative conditional-entropy increment, while the log-sum inequality gives common-kernel KL contraction (**Thermodynamic Arrow**). These are the stated information-theoretic conclusions (**Emergent Dynamics**).

III. Scope of the Dynamical Bound

The finite-horizon bound supplies a finite state domain (**Finite-Horizon Height Bound**). Common shifts preserve the unpumped transition on height classes (**Common-Height Shift Equivariance**).

Quiescent states are identified by reciprocal filtering and deletion trials (**Exact Unpumped Quiescence**). The site-summed condition is a separate bound on the triangle drift (**Conditional Triangle Drift**). It is not used to prove normalization or either information identity. Thus those identities hold independently of an additional recurrence or sparse-density estimate.

Q.E.D.

4.6.10 Type-Theoretic Validation via Lean 4 Core

Lean 4 Encoding of Move Disjointness and Addition Confluence

Type-theoretic certification of the frozen-proposal set operations used in **Emergent Dynamics** and specified by **Evolution Operator** proceeds via the following verification strategy:

1. **Move Grammar Encoding:** Edge subsets are represented as predicates over directed vertex pairs `Edge V -> Prop`. An addition proposal set A_{edges} satisfies `IsLegalAdditionSet E A_edges` if every proposed edge is absent from the existing topology E. A deletion set D satisfies `IsLegalDeletionSet E D` if every candidate deletion belongs to E.
2. **Move Disjointness and Race-Free Invariance:** The Lean theorem `dynamic_move_disjointness` formally proves that $A_{\text{edges}} \cap D = \emptyset$, ruling out conflicting update requests on identical edges. Theorem `dynamic_race_free_invariance` proves that newly added edges are guaranteed to survive deletions occurring within the identical tick.
3. **Step 3 Confluence Algebra:** The operator `merge_edge` accumulates additions into the intermediate graph. Theorems `parallel_addition_commutes` and `parallel_addition_idempotent` prove that concurrent additions commute in arbitrary order and fold duplicate proposals idempotently, ensuring deterministic state progression.

```
def Edge (V : Type) := V x V

def GraphEdges (V : Type) := Edge V -> Prop

def IsLegalAdditionSet {V : Type} (E A_edges : Edge V -> Prop) : Prop :=
  ? e, A_edges e -> ? (E e)

def IsLegalDeletionSet {V : Type} (E D : Edge V -> Prop) : Prop :=
  ? e, D e -> E e

/--
THEOREM 1: Dynamic Move Disjointness
Proves that the set of accepted additions and accepted deletions generated
within the same parallel tick are strictly disjoint: A_edges and D share no elements.
-/
theorem dynamic_move_disjointness {V : Type}
    (E A_edges D : Edge V -> Prop)
    (hA : IsLegalAdditionSet E A_edges)
    (hD : IsLegalDeletionSet E D) :
    ? e, ? (A_edges e ? D e) := by
  intro e <heA, heD>
  have h_not_E := hA e heA
  have h_in_E := hD e heD
  exact h_not_E h_in_E

/--
THEOREM 2: Dynamic Race-Free Invariance
Proves that newly added edges cannot be concurrently deleted within the same tick.
-/
theorem dynamic_race_free_invariance {V : Type}
    (E A_edges D : Edge V -> Prop)
    (hA : IsLegalAdditionSet E A_edges)
    (hD : IsLegalDeletionSet E D) :
    ? e, A_edges e -> (E e ? A_edges e) ? ? (D e) := by
```

```
  intro e heA
  refine <Or.inr heA, ?_>
  intro heD
  exact (dynamic_move_disjointness E A_edges D hA hD e) <heA, heD>

def merge_edge {V : Type} (E : GraphEdges V) (e : Edge V) : GraphEdges V :=
  fun x => E x ? x = e

/--
THEOREM 3: Parallel Addition Commutativity
Proves that pairwise additions commute: E ? {e1} ? {e2} = E ? {e2} ? {e1}.
-/
theorem parallel_addition_commutes {V : Type}
    (E : GraphEdges V) (e1 e2 : Edge V) :
    merge_edge (merge_edge E e1) e2 = merge_edge (merge_edge E e2) e1 := by
  funext x; dsimp [merge_edge]; apply propext
  constructor
  - intro h; rcases h with (hE | he1) | he2
    - exact Or.inl (Or.inl hE)
    - exact Or.inr he1
    - exact Or.inl (Or.inr he2)
  - intro h; rcases h with (hE | he2) | he1
    - exact Or.inl (Or.inl hE)
    - exact Or.inl (Or.inr he2)
    - exact Or.inr he1

/--
THEOREM 4: Parallel Addition Idempotence
Proves that redundant duplicate addition proposals fold idempotently.
-/
theorem parallel_addition_idempotent {V : Type}
    (E : GraphEdges V) (e : Edge V) :
    merge_edge (merge_edge E e) e = merge_edge E e := by
  funext x; dsimp [merge_edge]; apply propext
  constructor
  - intro h; rcases h with (hE | he1) | he2
    - exact Or.inl hE
    - exact Or.inr he1
    - exact Or.inr he2
  - intro h; rcases h with hE | he
    - exact Or.inl (Or.inl hE)
    - exact Or.inr he
```

Verification Summary: The formalization models the parallel move scheduler over arbitrary vertex types with zero postulated axioms and zero unverified dependencies. The constructive Lean theorem `dynamic_move_disjointness` certifies that accepted additions and accepted deletions generated within identical parallel ticks are strictly disjoint, while `dynamic_race_free_invariance` validates that newly created edges cannot be concurrently destroyed. Furthermore, theorems `parallel_addition_commutes` and `parallel_addition_idempotent` confirm that concurrent additions commute in arbitrary sequence and fold duplicate proposals idempotently. The Lean type-checker's acceptance of these machine-checked proofs validates the deterministic confluence and structural consistency of the discrete parallel scheduler under **Emergent Dynamics** .

4.6.Z Implications and Synthesis

Single Tick of Logical Time

The complete snapshot selection, merge and deletion law specifies one stochastic tick (**Evolution Operator**). Its realization probabilities include unsuccessful trials and the uniformly selected deletion edge. Summing all realizations with the same successor gives the normalized graph transition kernel (**Euclidean Transition Measure**).

The information conclusions concern trajectory entropy and common-kernel relative entropy (**Thermodynamic Arrow**). Snapshot Shannon entropy can decrease, and physical entropy production requires a separate forward/reverse and environmental accounting. The realization pushforward does not change those distinctions merely because it combines several outcomes into one graph.

These are the bounded conclusions of the revised synthesis (**Emergent Dynamics**). A global negative-drift estimate remains a separate obligation associated with the proposed dynamical bound (**Conditional Triangle Drift**). Stability of a chosen driven scalar closure does not by itself prove that graph-level estimate or persistent microscopic activity.

4.7 Finite-Kernel Quasi-Stationarity

Conditioning on remaining active changes the probability law of a finite chain. A substochastic transition matrix keeps only the active-to-active part of the tick. Its missing row mass is the chance of leaving the selected active set. By restricting the matrix powers to active states, the lost mass remains absent from all later terms. This permits survival probability and the conditioned state law to be read from the same calculation.

The conditioning problem asks when the distribution is preserved by an active transition followed by renormalization. Existence and lifetime depend on the spectrum of the chosen killed matrix, while convergence additionally depends on its cyclic structure. A primitive active class has a simple dominant mode with a strictly smaller subdominant spectral radius. The proof controls normalization separately because the unconditioned active mass can tend to zero.

The following statements take a finite killed class as a premise. They supply the spectral and lifetime implications for that class and separate them from the task of identifying a class for the physical dynamics. That identification requires a transition space on which the microscopic rule closes and on which communication can be proved. The finite-horizon bound supplies only the first time-limited control on the height variables.

4.7.1 Theorem: Finite-Kernel Quasi-Stationarity

Finite-Kernel Quasi-Stationarity via Conditional Survival and Convergence on a Finite Active Class

Let Q be the finite substochastic active-to-active matrix of a killed transition, irreducible with Perron root $\lambda \in (0,1)$. Its unique quasi-stationary probability ν has mean exit time $(1-\lambda)^{-1}$ when used as the initial law, and if Q is primitive the conditioned law converges to ν exponentially.

4.7.1.1 Commentary: Argument Outline

Structure of the Quasi-Stationarity Argument via Perron Normalization, Survival Summation, and Spectral Decay

The proof proceeds by construction of the normalized Perron vector, then derives the survival tail and the conditioned convergence estimate.

```
o 4.7.1 Theorem Finite-Kernel Quasi-Stationarity  [by construction]
|
+-- 4.7.2 Lemma: Perron Characterization of the QSD
|   +-- 4.7.2.1 Proof: Perron Characterization of the QSD
|   +-- 4.7.2.2 Commentary: Stationarity and Active Communication
|
+-- 4.7.3 Lemma: Geometric Lifetime from the QSD
|   +-- 4.7.3.1 Proof: Geometric Lifetime from the QSD
|   +-- 4.7.3.2 Commentary: Lifetime from the Conditioned Law
|
+-- 4.7.4 Lemma: Conditional Spectral Convergence
|   +-- 4.7.4.1 Proof: Conditional Spectral Convergence
|   +-- 4.7.4.2 Commentary: Spectral Remainders under Conditioning
|
+-- 4.7.5 Proof: Finite-Kernel Quasi-Stationarity
```

4.7.2 Lemma: Perron Characterization of the QSD

Perron Characterization of the QSD via Uniqueness of a Conditioned Stationary Probability

Let Q be a finite irreducible substochastic matrix with Perron root $\lambda > 0$. A distribution ν is quasi-stationary for killing on exit exactly when it is the normalized left Perron eigenvector of Q.

4.7.2.1 Proof: Perron Characterization of the QSD

Perron Characterization of the QSD via Normalization and the Positive Right Eigenvector

I. Domain and Objects

The killed matrix is substochastic on the finite active class (**Perron Characterization of the QSD**) (**Finite-Kernel Quasi-Stationarity**).

II. Construction

Perron-Frobenius gives positive vectors ℓ, r with $\ell Q = \lambda \ell$ and $Qr = \lambda r$. Set $\nu = \ell/(\ell \mathbf{1})$. Then

$$\nu Q \mathbf{1} = \lambda, \qquad \frac{\nu Q}{\nu Q \mathbf{1}} = \nu.$$

III. Uniqueness

For any quasi-stationary probability π, put $a = \pi Q \mathbf{1} > 0$, so $\pi Q = a\pi$. Associativity and $Qr = \lambda r$ give

$$a(\pi r) = (a\pi)r = (\pi Q)r = \pi(Qr) = \pi(\lambda r) = \lambda(\pi r).$$

In components, $\pi r = \sum_i \pi_i r_i \geq (\min_i r_i) \sum_i \pi_i = \min_i r_i > 0$. Dividing the displayed identity by this scalar gives $a = \lambda$. The left Perron eigenspace is one-dimensional, so normalization gives $\pi = \nu$.

Q.E.D.

4.7.2.2 Commentary: Stationarity and Active Communication

Stationarity and Active Communication

A quasi-stationary law is fixed after one transition conditioned on survival. Before normalization the update loses mass and preserves the direction of the probability vector. The Perron eigenvalue records the retained fraction of mass from that law. Irreducibility gives a strictly positive eigenvector and makes that normalized choice unique in the finite active class.

The active set specifies what counts as killing. Its matrix records only transitions that stay inside the set, so the choice of set matters for both the eigenvector and the retained mass. In an application to timestamped graph dynamics, the matrix must be justified for the actual state variables. Establishing communication on that class is a separate premise of the application. Positivity of the right eigenvector also excludes a distinct eigenvalue for a conditioned probability.

4.7.3 Lemma: Geometric Lifetime from the QSD

Geometric Lifetime from the QSD via Exit-Time Tail under a Quasi-Stationary Initial Law

Let $\nu Q = \lambda\nu$ with $\nu\mathbf{1} = 1$ and $0 < \lambda < 1$. The first exit time τ from the active set is geometric under ν, with $\Pr_\nu(\tau > t) = \lambda^t$ for $t \geq 0$ and $\mathbb{E}_\nu[\tau] = (1-\lambda)^{-1}$.

4.7.3.1 Proof: Geometric Lifetime from the QSD

Survival via Powers of the Killed Matrix

I. Domain and Objects

The initial law is the normalized left Perron vector (**Geometric Lifetime from the QSD**) (**Perron Characterization of the QSD**).

II. Survival Mass

The active unnormalized law after t steps is νQ^t. At $t = 0$, $\nu Q^0 = \nu = \lambda^0\nu$. If $\nu Q^t = \lambda^t\nu$, then

$$\nu Q^{t+1} = (\nu Q^t)Q = (\lambda^t\nu)Q = \lambda^t(\nu Q) = \lambda^{t+1}\nu.$$

Thus the survival mass at every tick is $\Pr_\nu(\tau > t) = \nu Q^t\mathbf{1} = \lambda^t(\nu\mathbf{1}) = \lambda^t$.

III. Mean Lifetime

For integer $m \geq 1$, $\min(\tau, m) = \sum_{t=0}^{m-1} \mathbf{1}_{\{\tau > t\}}$. The finite sum has expectation

$$\mathbb{E}_\nu[\min(\tau, m)] = \sum_{t=0}^{m-1} \Pr_\nu(\tau > t) = \sum_{t=0}^{m-1} \lambda^t = \frac{1-\lambda^m}{1-\lambda}.$$

As $m \to \infty$, the left side tends to $\mathbb{E}_\nu[\tau]$ by monotone convergence, and $\lambda^m \to 0$ because $0 < \lambda < 1$. Thus $\mathbb{E}_\nu[\tau] = 1/(1-\lambda)$.

Q.E.D.

4.7.3.2 Commentary: Lifetime from the Conditioned Law

Lifetime from the Conditioned Law

The geometric tail follows because the same law is recovered after every surviving step. Its surviving mass is multiplied by the Perron root each time. Summing this tail counts the expected number of ticks before the first exit, including the first attempted transition from the active set.

An arbitrary initial law can have a different short-time survival probability and a different lifetime expectation. The formula uses initialization in the quasi-stationary law and the specified killed matrix. If the conditioned law converges from another initial distribution, the same spectral data can describe its later survival behavior, with a prefactor determined by the initial law. That convergence uses the additional primitive-matrix hypothesis. For integer exit times the mean follows by summing the survival probabilities over all nonnegative ticks. For the triangle-active choice of the killed class, the initial graph is active and $\tau = \inf\{s \geq 1 : N_3(G_s) = 0\}$. Thus $\{\tau > t\} = \bigcap_{s=0}^{t} \{N_3(G_s) > 0\}$. This is the survival event used for the conditioned graph law in Chapter 5.

4.7.4 Lemma: Conditional Spectral Convergence

Conditional Spectral Convergence via Exponential Conditional Convergence for a Primitive Killed Matrix

Let Q be primitive and finite with Perron root $\lambda > 0$. For each initial law π on the active set and every $\eta \in (0, 1)$ larger than the ratio of the subdominant spectral radius to λ, the conditioned law satisfies $\|\pi Q^t/(\pi Q^t \mathbf{1}) - \nu\|_{\mathrm{TV}} \leq C_\eta(\pi)\eta^t$.

4.7.4.1 Proof: Conditional Spectral Convergence

Conditional Spectral Convergence via Spectral Projection and Control of the Conditioning Denominator

I. Domain and Objects

The positive Perron eigenvectors determine the limiting law (**Conditional Spectral Convergence**) (**Perron Characterization of the QSD**).

II. Perron Projection and Remainder

Normalize the positive right eigenvector by $\nu r = 1$. The Perron projection is $r\nu$. The remaining Jordan blocks of Q/λ have eigenvalues of modulus smaller than η. A block of size m with eigenvalue z contributes entries bounded by sums of $\binom{t}{j}|z|^{t-j}$ for $0 \leq j < m$. For $z \neq 0$, dividing by η^t bounds each term uniformly since $|z|/\eta < 1$. For $z = 0$, the block vanishes at $t \geq m$. Thus

$$\pi Q^t/\lambda^t = (\pi r)\nu + e_t, \qquad \|e_t\|_1 \leq A_\eta(\pi)\eta^t.$$

III. Normalization

Write $b = \pi r > 0$. Since $|e_t \mathbf{1}| \leq \|e_t\|_1$, for sufficiently large t the denominator $b + e_t \mathbf{1}$ is at least $b/2$. Subtracting ν gives

$$\frac{\pi Q^t}{\pi Q^t \mathbf{1}} - \nu = \frac{e_t - (e_t \mathbf{1})\nu}{b + e_t \mathbf{1}}.$$

The numerator has ℓ^1 norm at most $2\|e_t\|_1$. Total variation is half that norm after division, so the error is at most $2A_\eta(\pi)\eta^t/b$. Every active state has a positive outgoing active transition under the stated hypotheses. The survival denominators are therefore positive at the finitely many earlier times, which are covered by increasing the constant.

Q.E.D.

4.7.4.2 Commentary: Spectral Remainders under Conditioning

Spectral Remainders under Conditioning

The unnormalized active mass tends to zero when the Perron root is below one. Comparing it directly to a normalized probability obscures the rate of convergence. Factoring out the common leading power exposes a positive limiting denominator and a remainder controlled by the smaller spectral modes of the normalized matrix.

A defective subdominant eigenvalue can multiply its decay by a polynomial in the tick count. Choosing a decay ratio strictly above the subdominant ratio absorbs that factor. A nilpotent block instead vanishes after finitely many powers. This is a temporal mixing estimate for the finite primitive killed class, with the constants and decay rate tied to its actual spectrum. The normalization estimate uses the positive overlap between the initial probability and the right Perron eigenvector.

4.7.5 Proof: Finite-Kernel Quasi-Stationarity

Finite-Kernel Quasi-Stationarity via Assembly of the Conditioned Kernel Identities

I. Domain and Objects

The stated theorem assumes irreducibility and a Perron root in the open unit interval (**Finite-Kernel Quasi-Stationarity**) (**Finite-Horizon Height Bound**).

II. Conditioned Stationarity

Irreducibility gives a unique normalized positive left eigenvector, and the conditioned update fixes it (**Perron Characterization of the QSD**).

III. Exit Time and Convergence

The left eigenvector gives survival mass λ^t and its sum gives the mean lifetime (**Geometric Lifetime from the QSD**). If the matrix is primitive, its subdominant spectral radius is strictly below λ. A ratio η strictly between the normalized radius and one therefore gives exponential convergence (**Conditional Spectral Convergence**).

Q.E.D.

4.7.Z Implications and Synthesis

Conditioned Active Dynamics

A finite killed matrix with the stated irreducibility and Perron root has a unique quasi-stationary distribution (**Perron Characterization of the QSD**). The lifetime has a geometric survival tail (**Geometric Lifetime from the QSD**). Primitivity gives convergence of the conditioned law (**Conditional Spectral Convergence**).

The active snapshot includes its heights. A finite-horizon bound is available for the unpumped tick (**Finite-Horizon Height Bound**), and a finite relative-height quotient follows under a preserved span bound. The long-time active class and its communication structure require evidence from the transition rule.

A driven kernel uses a different outcome law. On a finite driven class, irreducibility gives a unique stationary law and aperiodicity gives convergence. Relating that law to a scalar mean-field equilibrium requires a separate microscopic correspondence. These distinctions enter the synthesis of the evolution model.

4.8 Formal Synthesis

End of Chapter 4

Life breathes into the static frame, assembling the complete runtime for the relational engine. The **Internal Causal Category** and **Historical Category** provide a rigorous syntax for evolution, ensuring that every update is a valid morphism that preserves causal structure. By equipping the graph with "Awareness" via the store comonad, the system gains the capacity to self-diagnose topological defects and guide its own growth without the need for an external observer.

Crucially, the iron laws of thermodynamics ground this engine. Deriving the critical temperature $T = \ln 2$ and the friction coefficient μ tunes the system to the precise edge of criticality where information creation is energetically neutral but structurally bounded. The **Evolution Operator** $\mathcal{U}$ functions as a biased pump, cycling through awareness, rewrite, projection, and sampling to drive the system forward. The thermodynamic arrow of time emerges here as the inevitable entropy production of the sampling step.

This runtime transforms the static tree into a living, breathing process. However, the question remains: what happens when this engine runs for billions of ticks? Does it produce a chaotic mess, or does it settle into a recognizable shape? We turn to **Chapter 5**, where the master equation supplies a conditional density model and the geometric limit has separate premises.

Table of Symbols

Symbol	Description	Context / First Used
$\mathbf{Caus}_t$	Internal Causal Category (Path Category)	Sec.4.1.1
Hist	Global Historical Category (Embeddings)	Sec.4.1.2
AnnCG	Category of Annotated Causal Graphs	Sec.4.3.1
R_T	Awareness Endofunctor	Sec.4.3.2
σ_G	Freshly computed syndrome map	Sec.4.3.2
ϵ	Counit (Context Extraction)	Sec.4.3.3
δ	Comultiplication (Meta-Check)	Sec.4.3.4
T_c, β_c	Vacuum Temperature / Base Conversion Modulus ($\ln 2$)	Sec.4.4.1
ΔS_{close}	Entropy of Closure ($\ln 2$)	Sec.4.4.3
ε_{geo}	Microscopic Channel Self-Energy ($\frac{\ln 2}{3} \approx 0.231049$)	Sec.4.4.4
α_{topo}	Macroscopic Topological Energy Scale ($\frac{\ln 2}{4} \approx 0.173205$)	Sec.7.4.3
Λ_{theory}	Simplicial Permittivity / Cosmological Vacuum Drive ($2^{-6} = 0.015625$)	Sec.4.4.5
λ_0	Catalytic Defect Relaxation Generator ($e - 1 \approx 1.7183$)	Sec.4.4.6
μ_0	Modular S-Duality Friction Coefficient ($1/\sqrt{2\pi} \approx 0.3989$)	Sec.4.4.7
τ_{homeo}	Operational Homeostatic Stopping Time	Sec.4.5.1
$\mathcal{R}$	Universal Constructor (Rewrite Rule)	Sec.4.5.1
$\chi(\sigma_e)$	Catalytic Tension Factor	Sec.4.5.2
P_{acc}	Addition Proposal Probability	Sec.4.5.3

Symbol	Description	Context / First Used
P_{del}	Deletion Proposal Probability	Sec.4.5.4
$\mathbb{P}_{\text{acc}}$	Acceptance Probability (Addition)	Sec.4.5.6
$\mathbb{P}_{\text{del}}$	Acceptance Probability (Deletion)	Sec.4.5.7
$\mathcal{U}$	Universal Evolution Operator ($\mathcal{D} \circ \mathcal{M} \circ \mathcal{P}_{\text{prop}} \circ \mathcal{A}$)	Sec.4.6.1
Σ_{valid}	State space of axiomatically compliant graphs	Sec.4.6.1
$\mathcal{A}$	Diagnostic Awareness Mapping	Sec.4.6.1
$\mathcal{P}_{\text{prop}}$	Stochastic Proposal Kernel	Sec.4.6.1
$\mathcal{M}$	Addition Merge Map	Sec.4.6.1
$\mathcal{D}$	Excision Deletion Operator	Sec.4.6.1
ρ	Probability measure over the state space	Sec.4.6.1
$\mathbb{P}(G \to G')$	Transition Probability	Sec.4.6.3

References

8. Awodey, S. (2010).

"Category Theory (2nd ed.)" * **Link:** https://global.oup.com/academic/product/category-theory-9780199237180

Overview: Awodey provides a systematic and accessible introduction to category theory, focusing on the structures and relations that unify different branches of mathematics. He covers functors, natural transformations, limits, colimits, and monads, emphasizing the structural perspective over set-theoretic foundations.

Relevance to QBD: Category theory is the formal language used to define the computational syntax of QBD in Chapter 1. By representing the substrate as a category where vertices are objects and edges are morphisms, we formalize the rewrite rules as functors. Awodey's text is the core reference for these category-theoretic structures, ensuring that the algebraic foundations of our model are carefully defined.

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72. Uustalu, T., & Vene, V. (2008).

"Comonadic notions of computation" * **Link:** https://www.sciencedirect.com/science/article/pii/S1571066108003435

Overview: Uustalu and Vene formulate a comonadic approach to describe context-dependent computations in computer science. They demonstrate that while monads are effective at modeling computations that produce effects, comonads are the natural category-theoretic tool to model computations that rely on surrounding context or historical execution states.

Relevance to QBD: This comonadic structure is the direct tool used to formalize the local update rules in Chapter 2. Because our rewrite rules rely on the surrounding context of neighboring vertices and edges, they are modeled comonadically. This construction provides the category-theoretic foundations required to define these context-dependent updates, ensuring algebraic consistency.

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